Can Religion Bridge Ethnic Divides?

Pastors, Chiefs, and a Field Experiment on Collective Action

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Abstract

Ethnic fragmentation is often associated with worse economic development. In one prominent mechanism, free-riding in heterogeneous societies undermines public good provision. This paper examines whether religious and ethnic leaders can sustain cooperation in a voluntary public-works program in an ethnically diverse city in the D.R. Congo. Residents are randomly assigned to ethnically homogeneous or mixed work teams that either include an ethnic chief, a pastor, or no leader at all. Chiefs increase work by 6.0 percentage points in homogeneous teams but not in mixed ones, while pastors increase work by 5.8 points in mixed teams but not in homogeneous ones. Heterogeneity by the strength of ethnic attachment explains this reversal. Chiefs still mobilize their coethnics on mixed teams, and on homogeneous teams their effect is concentrated among teams of residents who report a strong ethnic identity. Pastors do increase work on homogeneous teams whose members have weak ethnic attachments, and their effect is largest on mixed teams whose members hold strong but competing ethnic identities. A cross-randomized public disclosure intervention provides further evidence that the chief mobilizes labor contributions through social enforcement while the pastor uses internal incentives. The paper provides the first experimental evidence on how religious leaders can sustain voluntary public-goods provision in fragmented societies.

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1 Introduction

Ethnically diverse societies have slower economic growth and worse public goods than homogeneous ones (Easterly and Levine, 1997; Alesina and La Ferrara, 2005). One important channel, particularly in Africa, is the failure of voluntary provision: local public goods depend on contributions of labor, money, and materials made outside the formal tax system (Olken and Singhal, 2011). Voluntary contributions are lower in diverse communities because they struggle to sanction free-riders (Miguel and Gugerty, 2005; Habyarimana et al., 2007). In principle, local leaders could solve this enforcement problem (Ostrom, 1990; Kosfeld and Rustagi, 2015). But the authority of chiefs — the leaders best placed to do so — is also ethnically bounded (Acemoglu et al., 2014; Baldwin, 2016). Who, then, can bridge ethnic divides?

Religious leaders are natural candidates. Sub-Saharan Africa is now home to more Christians than any other region in the world, and religious leaders play important social roles — from conflict resolution to job training to service provision (Freeman 2012, WCD 2025). Religious leaders also differ from chiefs in two crucial ways. First, their scope of authority extends beyond the ethnic group (McCauley, 2014), and past work has documented a correlation between major religions and cooperation (Henrich et al., 2010). Second, rather than relying on social sanctions, religious leaders can promote internal incentives (e.g., fear of divine judgment), a source of enforcement that does not require an audience of peers (Shariff and Norenzayan, 2007). Yet there is little evidence on whether religious leaders can sustain cooperation across ethnic lines.

We present experimental evidence on the ability of ethnic and religious authorities to promote collective action in a citywide voluntary public-works program in the Democratic Republic of the Congo (DRC). As in many low-income settings, public good provision depends on voluntary labor that residents contribute through *salongo*, a community-work institution, to repair roads and drains, remove waste, and fight erosion. In villages, customary chiefs coordinate this work, while in cities, avenue chiefs play this role (Balán et al., 2022; Hoffmann et al., 2022). In both cases, chiefs' authority depends on ethnic ties, which poses a problem in large multi-ethnic cities like Kananga (pop. 1.8 million), the setting for this study. By contrast, Christian churches are ubiquitous in Kananga, and their congregations typically cut across ethnic cleavages. But can pastors promote costly salongo work in heterogeneous communities?

We partnered with the Kananga city government during a campaign to boost salongo participation. The campaign organized residents into work teams and tracked each resident's contributions every day. We embedded an experiment in this campaign. Across 28 neighborhoods, we randomly assigned 1,040 residents to 336 work teams. Each team included either the local avenue chief, a local pastor, or none of these leaders (without any special role for or mention of the status of the leader). We cross-randomized two features of work teams. First, teams were either ethnically homogeneous — and coethnics of the chief — or mixed, a variation we leverage to study the *scope of authority* of leaders. Second, we disclosed to teammates either each member's work contributions or only the team total, which allows us to study the *source of enforcement* on which leaders can draw. Because the same chief and pastor serve on several teams within a neighborhood,

the analysis can hold the leader fixed and vary the team setting in which they operate.

Theory gives a clear prediction for chiefs and an ambiguous one for pastors. Chiefs should be more effective in homogeneous teams, where shared ethnic ties make social sanctions more consequential (Miguel and Gugerty, 2005; Bhalotra et al., 2023). For pastors, one view is that their authority crosses ethnic lines, so they should be equally effective in both team types. Another view emphasizes that group composition shapes which identity — ethnic or religious — guides behavior (Akerlof and Kranton, 2000). In homogeneous teams, ethnicity may be so relevant for organizing cooperation that religious appeals fall flat. In mixed teams, religion may be the one identity the group shares, creating scope for the pastor's authority. Our preregistration reflects this: a positive chief effect in homogeneous teams and no signed prediction for pastors.

Our main result is that leaders do raise work contributions, but in different types of teams. Chiefs increase work by 6.0 percentage points in homogeneous teams ($p = 0.032$) and have no detectable effect in mixed teams (0.4 points, $p = 0.893$). Pastors increase work by 5.8 points in mixed teams ($p = 0.039$) and have no detectable effect in homogeneous teams (1.7 points, $p = 0.484$). This reversal is large and significant: the chief's advantage over the pastor is 9.8 points larger in homogeneous teams than in mixed teams ($p = 0.017$). Without a leader, work levels are the same in both team types: 58 percent of possible days in mixed teams and 57 percent in homogeneous teams ($p = 0.536$). The reversal therefore does not reflect baseline differences in cooperation. Ethnic cohesion alone does not raise contributions. Homogeneous teams may hold latent capacity for collective action, but it takes a chief to activate it. Instead, ethnic cohesion changes who can promote cooperation; while it enables chiefs, it notably prevents pastors from organizing collective action.

What explains the reversal in the effectiveness of leaders by group composition? Treatment effects track the *strength of ethnic attachment*. Start with the chief. In homogeneous teams, the chief effect is concentrated among groups with a strong ethnic identity: 9.7 points for them and indistinguishable from zero for others. In mixed teams, chiefs are still able to mobilize their coethnics by roughly as much as in homogeneous teams, though this estimate is imprecise. This suggests that the authority of the chief does not vanish in diverse teams. It is spread over fewer people who recognize him as their leader. Chief effects are also larger among residents who see the chief as able to sanction and reward, and among those who rate him favorably (such as coethnics). Coethnicity delineates where the chief has authority.

By contrast, the effect of pastors does not depend on residents' ethnic ties to the pastor. It depends on the ethnic cohesion of the team. In homogeneous teams, the average null masks offsetting effects. Pastors, in fact, boost work by 6.8 points in homogeneous teams with weak ethnic attachment. They fail only where strong ethnic attachments bind the whole team tightly. In mixed teams, strong ethnic attachments depress work contributions in the absence of a leader: such teams work 53% of possible days compared to 62% in mixed teams with weaker ethnic attachments. Yet these divided teams are where the pastor's effect is largest: 13.2 points, enough to close the gap. In short, where ethnicity divides most, religion has the most to bridge.

One concern is that these patterns reflect individual traits rather than team composition: perhaps residents

with strong attachments to the chief ethnicity are simply less religious. They are not. Moreover, the chief's coethnics barely respond to pastors in homogeneous teams but respond strongly in mixed ones. Put simply, the same type of resident responds differently to the pastor depending on who else is on the team. In sum, ethnic attachment is the source of the chief's authority and the main obstacle to the pastor's.

Having shown whom each leader can mobilize, we turn to their source of enforcement using the cross-randomized disclosure intervention. Although residents were part of small work teams, they were free to work at different times and sites, so teammates often did not know who had contributed. Disclosure varied this visibility gap. Chiefs are thought to rely on social sanctions through ethnic networks, so they should benefit when disclosure reveals who is shirking (Kosfeld and Rustagi, 2015). By contrast, pastors are thought to activate internal incentives, such as fear of divine judgment (Shariff and Norenzayan, 2007). Disclosure should not help them, and it could even hurt if social image concerns crowd out internal motivation.

Both predictions hold. Chiefs raise work by 10.6 points in homogeneous teams when individual contributions are disclosed ($p = 0.002$). Without disclosure, they have no significant effect, even in homogeneous teams, and they are ineffective in mixed teams either way. For pastors, by contrast, disclosure does not help. Their effect in mixed teams does not differ significantly with and without disclosure. The point estimate is larger without disclosure (9.0 points, $p = 0.019$), which offers suggestive evidence that social image concerns crowd out internal motivation. Yet, the stronger takeaway is that public disclosure matters more for chiefs, who rely on social sanctions, than for pastors, who appear to rely on internal incentives.

Three additional pieces of evidence support this interpretation. First, a treatment-blind text classifier sorts speeches into three different categories: calls for attendance and better group performance, requests for material support, and appeals linking divine oversight to solidarity and community work. Chiefs devote most of their speaking time to attendance and performance, emphasizing the social enforcement account of their role. Pastors devote theirs to divine oversight and community work, consistent with cultivating internal incentives. Second, the two leaders encourage work from different residents. If chiefs raise the reputational cost of shirking, their effects should be largest among prior nonparticipants when disclosure exposes them. That is what we find. Pastor effects are similar across prior participation, consistent with appeals that do not need peer enforcers. Third, pastor effects are concentrated among residents high in intrinsic religiosity (Allport and Ross, 1967), those for whom religion is a guiding principle and who should be most responsive to internal incentives.

Could leaders simply be trying harder in the teams where they succeed? We find little support for this explanation after triangulating evidence from several sources. Meeting transcripts show that leaders' discourse and effort are similar across team compositions. Detailed work records show that leaders do not devote unequal effort to bringing teammates to participate. Residents' reports show that differential contact outside meetings and workdays does not track leader effects. Leader influence appears to reflect how audiences respond, not how leaders behave.

Finally, we ask how leader-induced cooperation shaped residents' attitudes toward one another and toward their leaders. Mirroring the rise in voluntary work, residents on leader-led teams became more willing to help one another financially. This suggests that religion may help build informal insurance networks in multi-ethnic settings where the state provides little social protection (Townsend, 1994; Berman, 2000; Chen, 2010; Auriol et al., 2020). Successful pastors also gained standing in the eyes of their teammates. Successful chiefs did not, likely because they already enjoyed favorable evaluations in homogeneous teams. Religious leaders may thus be able to convert cross-ethnic mobilization into broader social and political influence (Chaney, 2013; Bazzi et al., 2020). Did sustained cooperation also shift attitudes beyond the team? We find no change in beliefs about social or divine accountability, the salience of religious or ethnic identity, or views of the state and other authorities. The effects of cooperation appear to be confined to relationships within the team.

To our knowledge, this is the first field experiment to illustrate the ethnic fault lines that determine which local leaders can mobilize voluntary work. Chiefs are effective when cooperation stays within an ethnic community and contributions are visible. Pastors are effective when cooperation crosses ethnic lines without relying on visibility. Religion can bridge ethnic divides, but struggles if ethnic ties already organize the group. Instead of impeding cooperation, we find that ethnic fragmentation changes who can govern collective action.

1.1 Related Literature

This paper contributes to three literatures. First, a large literature documents lower growth, civic participation, and public-good provision in more fragmented settings (Easterly and Levine, 1997; Alesina et al., 1999; Alesina and La Ferrara, 2000, 2005; Miguel and Gugerty, 2005; Habyarimana et al., 2007; Baldwin and Huber, 2010; Glennerster et al., 2013).[1] Research on governing the commons identifies shared rules, local monitoring, and informal sanctions as the foundations of cooperation outside formal enforcement (Olson, 1965; Ostrom, 1990, 2000). A central lesson from both literatures is that ethnic fragmentation weakens the enforcement of social norms that prevent free-riding in voluntary public good provision. We ask instead which leaders can sustain collective action when ethnic diversity complicates social enforcement. In doing so, we complement research showing that cross-cutting identities can mitigate ethnic divisions by shifting the social boundaries around which behavior is organized (Miguel, 2004; Posner, 2004; Clingingsmith et al., 2009; Dunning and Harrison, 2010; Sambanis and Shayo, 2013; Blouin and Mukand, 2019; Bazzi et al., 2019; Depetris-Chauvin et al., 2020).[2] Religion is one such identity. We show that its bridging power depends on whether ethnicity divides the group or unites it.

Second, the paper studies two sources of enforcement for cooperation. Social-image models argue that behavior responds to how others judge it (Bénabou and Tirole, 2006; Andreoni and Bernheim, 2009; DellaVigna

[1] Beyond public-good provision, another strand of research studies how ethnic and religious divisions shape discrimination and productivity within firms and organizations (Hjort, 2014; Marx et al., 2021; Ghosh, 2025).

[2] A related literature tests whether intergroup contact can weaken social divisions (Scacco and Warren, 2018; Rao, 2019; Mousa, 2020; Lowe, 2021; Ghosh et al., 2026). Findings in this literature emphasize that the effects of intergroup contact depend on how interaction is structured and often do not extend much beyond those directly encountered (Lowe, 2025).

et al., 2012; Schram and Charness, 2015; Bursztyn and Jensen, 2017; Breza et al., 2026). Models of identity and self-image argue that behavior responds to what one's actions reveal about oneself (Akerlof and Kranton, 2000; Bénabou and Tirole, 2003, 2006, 2011; Dal Bó and Terviö, 2013; Dal Bó and Dal Bó, 2014; Grossman, 2015; Bénabou and Tirole, 2016). Traditional authority is understood to enforce behavior through external sanctions (Kosfeld and Rustagi, 2015). Religious institutions both embed believers in communities that socially enforce prosocial conduct and invoke divine monitoring of private actions (Iannaccone, 1992; Berman, 2000; Auriol et al., 2020; Johnson, 2005; Shariff and Norenzayan, 2007; Norenzayan, 2013; Purzycki et al., 2016; Lang et al., 2019; Lowes et al., 2026a). Field experiments show that religious authority and messages affect behavior (Bryan et al., 2021; Buccione, 2023), but they do not separate the social and internal channels. Our design separates them by cross-randomizing leadership with the disclosure of individual participation. Chiefs' influence is complemented by public accountability, consistent with social enforcement. Disclosure does not help pastors. If anything, their point estimate is smaller under disclosure. This evidence aligns with prior evidence indicating that external control can crowd out internal motivation (Fehr and Rockenbach, 2003; Bénabou and Tirole, 2006; Falk and Kosfeld, 2006; Gneezy et al., 2012).

Third, we contribute to research on leadership and informal governance. Existing work studies how leaders coordinate, communicate, set examples, sanction free-riding, and shape expectations (Bianco and Bates, 1990; Hermalin, 1998, 2012; Caillaud and Tirole, 2007; Dewan and Myatt, 2008; Jack and Recalde, 2015; Kosfeld and Rustagi, 2015; Antonakis et al., 2022; Bhalotra et al., 2023; Boudreau et al., 2025). Related work examines who becomes a leader and how leader identity shapes political outcomes (Besley, 2005; Caselli and Morelli, 2004; Jones and Olken, 2005; Dal Bó et al., 2017). Research on informal governance studies how traditional authorities contribute to public-good provision and state capacity (Acemoglu et al., 2014; Baldwin, 2016; Baldwin et al., 2025; Balán et al., 2022; Henn and Tanutama, 2023; Lowes et al., 2026b). Less attention has been paid to whether a leader's effectiveness depends on fit with the audience.[3] Our design holds the public-good activity, the neighborhood, and the leader fixed, and varies the audience. We find that effective leadership depends on fit with the audience: ethnic cohesion determines which leader can mobilize a group.

2 Context

The Democratic Republic of Congo is one of the world's most populous countries and also one of the poorest. Kananga, the capital of Kasaï-Central Province and the setting for this study, is a city of roughly two million inhabitants. It is governed in a low-capacity fiscal environment. Roads are degraded, constant tropical storms cause severe erosion problems, and access to basic urban services such as electricity, piped water, and sanitation is low. The local administration is responsible for many of these services, but the lack of institutional capacity

[3]Social identity theories of leadership argue that leaders are more effective when they are perceived as members of the group. Evidence for these accounts is largely correlational or confined to lab outcomes (Hogg, 2001; Van Knippenberg and Hogg, 2003; Steffens et al., 2021; Roy and Houser, 2024).

prevents it from delivering them. The setting is one of weak fiscal capacity, underprovided public goods, and a heavy reliance on local authorities to mobilize residents directly (Balán et al., 2022).

Salongo, City Governance and Traditional Authority. The principal institution through which this direct mobilization occurs is *salongo*. In the 70s, salongo was formalized as obligatory civic labor and became part of a broader effort to mobilize local labor for public works (Callaghy, 1984; Robinson, 2023). Recent evidence shows that households are still regularly asked to contribute salongo to neighborhood upkeep projects throughout the country, with participation organized by local authorities and sustained through social pressure, civic obligation, and repeated local interaction (van den Boogaard et al., 2026). In Kananga, salongo remains part of everyday life, and many times it is the only solution residents have for waste management, road maintenance, public space cleaning, and, most urgently, anti-erosion works.[4]

Salongo operates through a local governance hierarchy that belongs to a broader political structure in which territorial origin shapes representation and expectations of political support (Hoffmann et al., 2022; Gobbers, 2016).[5] The *chef de quartier* is the formal neighborhood authority responsible for the administrative supervision of the population, hygiene, and public sanitation. Below the *chef de quartier*, the implementation of salongo relies on two further layers of authority. The first is the *point focal*, or *chef de localité*, which effectively serves as a focal point for the neighborhood chief.[6] *Points focaux* transmit instructions from the quartier administration, help identify feasible work sites, and organize the maintenance efforts across several avenue chiefs. They are implementation intermediaries. The second is the *chef d'avenue*, or avenue-level notable, who is the household-facing authority. He is closer to residents than the other chiefs.

The traditional-leader treatment focuses on the *chef d'avenue*, the level of chief authority that is closest to the resident population. Two features of their status matter for the experiment. The first is their ethno-territorial scope. The chief is better understood as a respected resident whose authority is anchored in a local ethno-territorial social order, so it should be strongest over residents who recognize him as speaking from within their own origin community. The second is their social source of enforcement. Given chiefs' standing and embeddedness in the local context, we expect their mobilization capacity to operate through reputational channels.[7] The treatment tests whether chiefly authority can make salongo participation socially consequential, and whether its force declines when the mobilized group falls outside their ethnic scope.

[4] The relevance of these tasks is visible in the World Bank's Kananga Emergency Urban Resilience Project (World Bank, 2022), which focuses on gully erosion, drainage, and climate-resilient urban planning. Reports emphasize that the constraint is partly organizational since neighborhood erosion-prevention committees have been established across Kananga; yet, community participation remains below target.

[5] Within Kasai, Ngalamulume (1995) traces the formation of geographically different identities among the Lulua around Kananga.

[6] The *localité* appeared as a lower subdivision under the 1982 territorial organization (République du Zaïre, 1982), but is absent from the current 2008–2010 framework, which organizes urban communes into quartiers and/or incorporated groupements and lists territoire, quartier, groupement, and village as deconcentrated entities (République Démocratique du Congo, 2008, 2010). In Kananga, the category remains a relevant de facto administrative scale.

[7] Balán et al. (2022) characterize Kananga city chiefs as local notables nominated by neighborhood elders, typically because they are longstanding and respected residents, and then ratified by the state. They report indefinite and often lifelong tenure, rare deposition, some family transmission of the position, and status as its main return. They also show that chiefs possess household-level information and that this information helps explain their effects on tax compliance.

Fragmentation. An initial assessment of Kananga's ethnic landscape may wrongly infer a homogeneous environment. The city is largely Tshiluba-speaking, and most residents are linked to the Lulua umbrella identity. However, this broad classification contains important internal divisions. It consolidated from smaller kin, territorial, and chiefly units in the Kasai during the colonial period. Belgian recognition of the Kalamba lineage as Lulua sovereign led to power struggles between central and peripheral clans that remain unresolved (Ngalamulume, 1995; Stearns, 2018). Because this umbrella identity would flatten most residents into a single group, we define ethno-territorial identity (henceforth, ethnicity) using the five administrative territories of Kasaï-Central (Demba, Dibaya, Dimbelenge, Kazumba, and Luiza), together with Kananga city[8]. The administrative territories help us identify residents' ties to different kin networks that structure political representation and expectations of mutual obligation. These ties matter for local governance because they shape the reach of the authority of chiefs. Given the informal nature of salongo mobilization by local chiefs, variations in neighborhood ethnic fragmentation should help explain why salongo may have weak support in certain places while succeeding in others.

Descriptive evidence collected in this study corroborates the hypothesis that fragmentation complicates local governance around salongo activities. Using participants' stated ethnicity, we build a fractionalization index for each neighborhood and investigate how it relates to the perceived importance of salongo among local residents. Figure A-1 reveals that fragmentation weakens perceived salongo importance in the neighborhood through two margins: (i) residents from the majority ethnicity in a given neighborhood see salongo as less important the more fragmented the neighborhood is; and (ii) at the average level of neighborhood fragmentation, residents from less represented ethnicities see salongo as less important than residents from the majority ethnicity. A key part of our account is that this governance problem is tied to the ethnic bounds of *the chef d'avenue's* authority. Consistent with this account, Appendix Table A-2 shows that chiefs' personal networks are disproportionately composed of people from their own ethnicity, while Appendix Table A-3 shows that residents who are coethnics with the chief rate the chief more favorably in several aspects than non-coethnics.

Conflict. Beyond historical fragmentation related to the consolidation of several groups under the broader Lulua identity, the Kamuina Nsapu conflict made these internal divisions especially salient. Beginning in 2016 as a dispute over customary authority in Dibaya, the conflict became a regional war after the state's refusal to recognize Jean-Prince Pandi as the successor to the Kamuina Nsapu chiefship.[9] The war not only hardened those divisions due to clashes occurring in rural areas but also created chaos in the city of Kananga, which became a stage for hostilities. Several killings took place in the city, while schools, health centers, homes, and

[8] Appendix Table A-1 locates the leaders' origins and religious denominations within the population from which experimental participants were recruited. Chiefs and pastors come from broadly the same origin territories as ordinary residents.

[9] Stearns (2018) interprets this episode as an instance of *dédoublement*, a broader strategy through which Congolese authorities weaken customary leaders viewed as sympathetic to the opposition by withholding recognition and supporting rival claimants, while locating this occurrence within the historical divisions of the Lulua umbrella ethnicity. The Kamuina Nsapu lineage heads the Bajila Kasanga, the dominant group within the peripheral Lulua clans, which has long contested the Kalamba lineage and the political center around Kananga.

public buildings were destroyed. It also became a new home for hundreds of thousands of displaced villagers who were unable or unwilling to return to their villages of origin.[10] Therefore, beyond the deaths, destruction, and tensions left by the war, Kananga recently absorbed a violent re-sorting of people from different ethnicity and customary ties (Mercy Corps, 2019). Many neighbors who now share a communal task to maintain a drainage channel or to address erosion risk are more likely to have arrived from different places of origin. The experiment focuses on the resulting mismatch between the chief's administrative neighborhood and the social networks through which his authority becomes relevant.

Religious Environment and Religious Authority. The DRC is overwhelmingly Christian, comprising 95% of the population, and is roughly divided between Catholics and broadly defined Protestants (Pew Research Center, 2011). The religious environment in Kananga roughly corresponds to these numbers, with the caveat that a broad Protestant classification hides the rapidly shifting religious landscape where autonomous *Églises du Réveil* are taking over the space previously filled by Catholics and traditional Protestant churches (Kapepula et al., 2022). The gap left by a weak state creates a wide space for churches to mobilize welfare and collective goods. Institutional churches, Catholic and historic Protestant networks, occupy the organizationally dense pole of this field through schools, clinics, development projects, and social assistance.

Entrant denominations perform a more decentralized and entrepreneurial role. They provide support through pastoral counseling, healing, deliverance, mutual aid, and informal insurance against illness, unemployment, and other risks. This decentralized model relies on local mobilization with few large bureaucratic assets.[11] Appendix Table A-1 locates leaders' religious denominations within the population from which experimental participants were recruited. Religious leaders are concentrated in Pentecostal/born-again churches (12 of 28), and the experiment also includes pastors from Kimbanguist (7), Neo-Apostolic (4), and Protestant/Presbyterian (2) denominations, while chiefs and residents are distributed more broadly across the Christian spectrum. The selection is informative in itself. Pastors whose authority depends on building a local following are exactly those most engaged in local mobilization for public goods. Catholic priests and other mission churches, by contrast, derive authority from their office within a larger hierarchy and need not build it through the same local involvement.[12]

Importantly, churches are spaces of ethnic mixing, where individuals with different kin networks meet and build lives unrelated to their origins. Baseline data show that more than 90 percent of identifiable congregations include members from multiple ethnicities (Appendix Table A-4). Congregants are no more likely to share an ethnicity within their churches than would be expected under random allocation within

[10] Mercy Corps reports more than 3,300 direct conflict-related deaths and 1.4 million displaced people in 2017; by early 2019, several hundred thousand displaced residents had still not returned because of livelihood collapse, insecurity, and ethnic tensions. The same report notes that more than 400 schools and 200 health centers, along with many government buildings, were damaged or destroyed (Mercy Corps, 2019).

[11] On institutional church provision in Congo, especially health and education under weak state capacity, see Seay (2013); Titeca et al. (2013). On healing churches and everyday social support in urban Congo, see Devisch (1996).

[12] Recruitment was open to leaders from all denominations.

their neighborhoods. Within-church pairs are co-ethnics in 48.2 percent of cases, compared with 48.8 percent under the no-sorting benchmark. Religious leaders are therefore already leading organizational constituencies that cross the ethnic boundaries on which chiefly authority rests.

Thus, the religious-leader treatment arm focuses on pastors from autonomous churches in a context where religion is an important institution for ethnic mixing. These leaders provide a contrast to avenue chiefs on the two dimensions central to the design. The first is cross-ethnic scope. Pastors address residents through a portable religious frame that cuts across the ethnic identities to which traditional leaders are confined.[13] The second is a different source of enforcement. Pastors can frame participation as conduct before God rather than as performance before their neighbors and local authorities.[14] The treatment asks whether this form of authority can sustain cooperation where the chief's ethnic authority is weak.

3 Experimental Design and Data

In this context of post-war recovery, where neighborhoods have recently integrated a massive wave of displaced residents, Kananga's city administration organized a city-led *salongo* campaign to coordinate basic neighborhood maintenance and incentivize citizens to join community activities. The campaign brought residents together to contribute labor and materials to neighborhood upkeeping tasks and used city evaluation and public recognition to reward performance, a common practice in the region.[15] We partnered with the administration in these efforts and embedded an experiment in the campaign.

Neighborhood selection. Before rollout, we identified neighborhoods with active *salongo*, identifiable boundaries, and sufficient scale for team implementation. Fifty neighborhoods met these criteria. We stratified them by commune and by whether the median predicted property value lay above or below the sample median, and randomly assigned 40 neighborhoods to receive the intervention rollout. The main estimands compare randomized teams within these intervention neighborhoods. Our main analysis covers 28 neighborhoods, 336 teams, and 1,040 non-leader participants due to a public health emergency that caused the program to stop.[16]

[13] On religious identity as an axis of mobilization that differs from ethno-linguistic identity in Africa, see McCauley (2014). On Pentecostal and charismatic Christianity as a portable religious form, see Robbins (2004).

[14] In Kananga's revival churches, concerns with witchcraft, protection, prosperity, and a worldview of hidden supernatural causality are translated into a Christian vocabulary of sin, blessing, and divine judgment. For an account of Congolese revival Christianity, see Pype (2012); Tonda (2005).

[15] Comparable civic-work and cleanliness contests elsewhere in the region include Kinshasa's Kin Bopeto, which promised prizes to the cleanest quartier in each commune (Actualite.cd, 2019); Rwanda's Umuganda, which links regular community work to infrastructure, environmental protection, and cleanliness (Rwanda Governance Board, 2017); Lomé's "salubrité challenge des quartiers," which had quartiers submit streets for jury evaluation (Vert Togo, 2019); and Cameroon's "Villes propres" competition, which began in 2019 as a Yaoundé "Quartiers propres" contest (Ministère de l'Habitat et du Développement Urbain, 2022).

[16] The study was planned as ten waves of four neighborhoods. Seven waves were completed under the standard protocol before the Ebola emergency disrupted the eighth wave and prevented the remaining rollout (World Health Organization, 2026). Appendix B.1 documents the stopping decision and reports robustness estimates incorporating the disrupted eighth wave.

Recruitment and short baseline. Before each wave, enumerators visited households in participating neighborhoods to introduce the campaign and register adults interested in taking part. This short baseline recorded residents' demographic characteristics, territorial origins (ethnicity), and willingness to contribute, and asked them to refer traditional and religious leaders who were especially committed to making *salongo* succeed. The resulting data defined the resident sampling frame, identified candidate leaders, and supplied the variables used in randomization.

3.1 Salongo Campaign

During each implementation wave, local officials organized daily maintenance activities and recorded residents' contributions for five weeks. In parallel, in the first week of the campaign, we invited volunteers to participate in coordination teams that would help advance salongo in the neighborhood. Each team met twice during the campaign in coordination meetings held separately from the daily work: first in the second week of activities and again in the third week. The meetings gave residents the opportunity to review their team's recent contributions, coordinate future work, and encourage each other to participate.

At the first meeting, participants got to know their assigned teammates, were exposed to their set of experimental conditions, and learned that the city would publicly recognize highly engaged teams at the end of the campaign. Because this was the first moment participants encountered their team and disclosure condition, the first meeting marked the onset of the randomized treatments. By the end of the fourth week of activities, the three teams with the highest engagement received public recognition from the city in a neighborhood ceremony attended by city officials. A fifth week of activities was organized after the recognition ceremony. All teams faced the same maintenance activities and recognition rules. The upper panel of Figure 1 summarizes the timeline of activities.

3.2 Randomization

The assignment proceeded in two stages. First, a screening stage selected participants from the registration sampling frame, oversampling those from the neighborhood's ethnic majority to facilitate the construction of homogeneous teams. Second, an assignment stage organized the selected residents into twelve teams of four in each neighborhood, composing a factorial design that crossed three leader conditions, two territorial-origin compositions, and two disclosure conditions. The design varies these conditions while keeping the neighborhood, campaign, team task, and recognition rules fixed. Figure 2 summarizes these assignments. Appendix B.2 presents the complete algorithm.

3.2.1 Leader Treatment

Each team counted on a fourth participant who would define the leader condition. Enumerators used the recruitment stage leader nominations to identify leaders who were available to join the campaign. They were

instructed to select the leader with the most mentions who was willing to participate. In the traditional leader condition (T), the same ethnic-majority traditional leader joined four teams in the neighborhood. The nominated religious leader did the same in the religious leader condition (R). In the no-leader condition (C), a placebo participant joined the fourth position.[17] We keep all non-repeated residents in the analysis sample while we exclude repeated leaders and placebo participants. All fourth participants followed the same meeting protocol. Leaders received no script or special instructions.[18]

Traditional leaders. The traditional leaders in the study are chiefs who are part of Kananga's customary system of territorial governance. We selected these leaders through residents' referrals collected during volunteer recruitment. These leaders are local notables whose main responsibilities include organizing communal work, mediating disputes, and representing residents before the state (Balán et al., 2022). Appendix Table A-2 shows that, relative to ordinary residents of the same neighborhood, the chiefs recruited into the study are twenty-two years older, have about 1.5 additional years of schooling, and have lived on their avenue for roughly thirteen years longer; they are also overwhelmingly male. Their institutional position is visible given the higher chances of holding a traditional head role, knowing local authorities, planning *salongo* activities, and evaluating higher chiefs' contributions to infrastructure, mediation, and representation favorably. Their personal networks are disproportionately composed of people from the same ethnicity, and they score higher than ordinary residents on an aggregate index of standard leadership traits, particularly conscientiousness. Appendix Table A-3 uses surveys administered before the first meeting to show that chiefs are known and recognized as authorities across ethnic lines, but their broader standing is concentrated among co-ethnics. Residents who share a chief's ethnic origin rate the chief as more motivating, morally exemplary, trustworthy, and hardworking, whereas residents from other origins recognize the chief's formal authority but do not evaluate the chief more favorably on these other dimensions. Coethnic and non-coethnic residents, nevertheless, expect chiefs to participate more in *salongo*, with especially strong expectations of their work effort. Therefore, chiefs enter the experiment as incumbent organizers of salongo whose authority, relationships, and reputations are centered on their ethnic origin.

Religious leaders. The religious leaders are local pastors drawn from Kananga's expanding landscape of Pentecostal, charismatic, and independent churches, institutions whose activities frequently extend beyond worship into community organization and public life (McClendon and Riedl, 2015, 2019; Kapepula et al., 2022; Ngoma Mavungu et al., 2025). We recruited religious leaders through the same resident-referral procedure we used to recruit chiefs. Appendix Table A-2 shows that pastors have a similar demographic profile to

[17] Beginning in Wave 3, the same randomly selected placebo participant joined all four no-leader teams, making team size and repeated participation symmetric across the three leader conditions. In Waves 1–2, each no-leader team contained a different fourth ordinary resident.

[18] The fixed placebo participant was selected from the majority ethnicity. Religious leaders were recruited through religious authority and did not always belong to the neighborhoods' majority. Eleven of the 28 assigned religious leaders differed from the traditional leader. We report a robustness check restricted to the 17 neighborhoods in which both leaders were from the majority.

chiefs. Relative to ordinary residents, pastors are thirteen years older, have approximately one and a half additional years of education, and have resided in the neighborhood for roughly thirteen years longer; they are even less likely to be female than chiefs. Their most significant difference rests on their participation in congregational life. Religious leaders are much more likely to head a congregation, attend Sunday services frequently, contribute to their church, and depend on networks centered on fellow church members. They are similar to ordinary residents on standard leadership-personality measures but report more positive and less negative religious coping, with weaker evidence of greater intrinsic religiosity. The institutional boundary is permeable: roughly 40 percent of pastors report holding a traditional role, and 29 percent of chiefs report a religious role. Nevertheless, the two leader types remain sufficiently different in their local roles, where chiefs are representatives of Kananga's territorial governance system, while pastors are more firmly embedded in congregational organization. Appendix Table A-3 shows how this distinction is reflected in residents' evaluation of leaders. Pastors are better known across ethnic lines and are rated as more motivating, morally exemplary, trustworthy, and hardworking, with similarly favorable assessments among co-ethnics *and* residents of a different ethnic origin. Residents also expect pastors to participate in *salongo* and contribute financially, although the expectations of their efforts are weaker than those for chiefs. Pastors, thus, enter the experiment with less incumbent responsibility for *salongo* but with a favorable standing that extends across the territorial boundaries organizing customary authority.

3.2.2 Cross-randomizations

Team composition. We filled each of the three remaining seats of the neighborhoods' twelve teams in random order. For the six homogeneous teams, all three residents were drawn from the pool that shared the traditional leader's territorial origin. For the six mixed teams, we assigned residents from territories not yet represented in the team, repeating an origin only when no additional territory was available. This procedure ensured that the three ordinary residents in mixed teams came from different territories of origin. Within each leader condition, two teams were homogeneous and two were mixed.

Disclosure. Residents could contribute at different times and work sites; therefore, they could not fully observe their teammates' contributions. Leveraging the incomplete information setup, we varied the information shown at the meeting. Under group disclosure, participants saw only their team's aggregate prior *salongo* contribution (private condition). Under individual disclosure, they also saw each member's identifiable contribution record, increasing external reputational incentives (public condition). Within every leader-by-composition cell, one team was assigned to each disclosure condition.

3.2.3 Rerandomization and Replacements

For each wave, the assignment program generated 48 complete candidate assignments[19] and retained the one that minimized an imbalance score across 15 pre-assignment characteristics in the cumulative implemented sample. During the invitation period, some selected residents reported that they would not be able to attend their meeting slot. These residents were replaced before treatment using randomly ordered replacement lists of respondents from the same territory of origin. Replacements inherited the original resident's team and experimental conditions, and all availability decisions occurred before residents met their teammates or the disclosure condition.

3.3 Data

Beyond the short survey used to construct the experimental assignments, the analysis combines administrative records, survey data collected before and after treatment, and recordings of the team meetings. The bottom of Figure 1 shows when each data collection instrument was administered.

Administrative data. The administrative records contain individual daily measures of labor, in-kind contributions such as food or materials, and monetary contributions. Local staff recorded contributions using a digital tracking system created for the campaign. The registry contains the neighborhood, date, and activity site, along with the geolocation of each salongo location; classifies the type of maintenance activity; and collects photographs of the sites before and after work. Arrival and departure time stamps measured contributions of labor for each participant, along with participant ratings of effort. In-kind and monetary contributions were recorded with quantities and estimated values. The records were merged with registered participants and dates, producing high-frequency, individual-level data on actual contributions to neighborhood maintenance.

Survey data. We collected survey data at baseline, two midlines, and endline. The baseline survey was administered before meeting invitations and measured participants' demographic and socioeconomic background, religious practices and beliefs, local social embeddedness, perceived salongo norms, trust in local authorities, and expectations of social and divine reciprocity. The first midline was conducted around the first team meeting. Right before the team meeting, it measured participants' proximity to each other and shared attributes such as church affiliation or kinship. Immediately after the meeting, it measured meeting impressions and immediate expectations of future peer participation. The second midline took place around the follow-up meeting and measured interactions during the inter-meeting period, including communication among team members, encouragement to contribute, praise for contributors, and sanctioning of free-riders. The endline questionnaire was administered in the week after the recognition ceremony and repeated the main baseline modules.

[19] Following Banerjee et al. (2020), we set the rerandomization budget to $K = \min\{N, 100\}$, where N is the number of experimental units. Each wave covered four neighborhoods, each with 12 teams, for a total of $N = 4 \times 12 = 48$ units.

Meeting recordings. We recorded team meeting discussions with individual consent and transcribed the content to conduct a textual analysis of the leader's mobilization strategy. Because leaders received no script, these recordings show what they endogenously contributed to the discussion. We specifically investigate whether leaders activated the authority channels specified in the design.

3.4 Implementation Checks

Appendix Table B-4 displays covariate balance across experimental arms. The characteristics measured before assignment are similar across experimental arms. Participation during the six days after assignment and before the first meeting is imbalanced. At that point, residents had not learned their assigned leader or reporting rule, so the difference is unlikely to reflect a response to treatment. We nevertheless examine whether anticipation effects can account for the results in the alternative-explanations subsection. The remaining implementation checks are reassuring. Although 22.2 percent of residents entered through replacement queues, nearly 95 percent of groups still had the territorial composition assigned by the design. Homogeneous groups remained overwhelmingly composed of residents from the chief's territory, while mixed groups drew residents from several territories (Appendix Table B-5). Treatment also reached the intended groups. Residents attended first meetings 95.0 percent of the time, assigned leaders attended 99.6 percent of treated meetings, and delivery of the meeting instructions and reporting rules was verified in at least 99.4 percent of groups (Appendix Table B-6). Finally, administrative participation is observed for every resident over the same six-day outcome window, and survey follow-up remains high (Appendix Table B-7). Group formation, meeting delivery, and measurement of the main outcome therefore proceeded as planned.

4 Scope of Authority Results

Traditional and religious leaders are notably different from each other and from ordinary citizens, but how do the different profiles translate into the capacity to lead different groups toward a collective goal? We call a leader's scope of authority the range of people and social settings in which the leader can influence behavior toward such a goal. A leader's scope may depend both on whether individuals recognize the leader's authority and on whether the relationships among group members give relevance to the obligations that authority invokes. Group composition can change both.

For chiefs, sharing the same origin provides one channel for individual recognition of authority, as residents who share the chief's ethnic origin should be more responsive to an ingroup authority (Miguel and Gugerty, 2005; Bhalotra et al., 2023; Roy and Houser, 2024). This account predicts greater effectiveness of chiefs in homogeneous groups, where all participants share the chief's origin. But it does not necessarily imply that chiefs lose influence over coethnics when they are placed in mixed groups, nor that their authority is necessarily bounded by ethnicity. Ethnic proximity is only one reason for recognizing the authority of the

chief. Participants from other ethnic groups may also regard the chief as a legitimate authority and therefore respond to his appeals, even if they are, on average, less closely connected to him than the coethnics who compose homogeneous groups. Hence, chiefs' scope should follow their relationships with particular residents, even when ethnic identity is not shared by the group as a whole.

For pastors, an open question is whether religious motives can broadly influence believers across and within ethnic divides, or whether the social setting can interfere with how much individuals respond to religious leaders' appeals. If belief in divine judgment is sufficient for religious leaders' speeches to influence behavior, pastors may be effective among believers regardless of group composition. An alternative account builds on the distinction between systems of enforcement that operate through kin and community sanctions versus those that rely on internalized moral obligations (Enke, 2019). These systems may coexist within individuals, with social context shaping which identities and associated obligations become salient and guide behavior (Akerlof and Kranton, 2000). In homogeneous groups, shared ethnic identity may give socially enforced ethnic obligations precedence in organizing collective action. In mixed groups, where ethnicity provides no common basis across all participants, it is when religion becomes the common identity denominator that pastors can leverage as an alternative basis for encouraging cooperation. Under this interpretation, even residents who recognize religious authority may respond differently to a pastor depending on whom they are asked to cooperate with.

We next use our design to examine how group composition shapes each leader's scope of authority.

4.1 Research Design

Within each neighborhood, we compare groups assigned a traditional or religious leader with groups assigned no leader. Let i index participants, $g(i)$ their accountability group, and $n(i)$ their neighborhood. The variables $T_{g(i)}$ and $R_{g(i)} \in \{0,1\}$ indicate whether participant i's group was assigned a traditional or religious leader, respectively; no leader is the omitted category. Group composition is $c(i) \in \{H, M\}$, where H denotes groups with shared territorial origins and M denotes groups with participants from mixed origins. Finally, $q(i) \in \{0,1\}$ indicates whether individual participation was disclosed.

The estimation sample contains 1,040 participants in 336 groups across 28 neighborhoods.[20] The outcome Y_i is average salongo participation, defined as contributing labor, money, or materials to communal public works, measured in percentage points over the six-day administrative window immediately following the first accountability meeting. All specifications control for baseline participation, Y_{i0}, defined as average participation over the six-day window preceding any accountability meeting.[21] For the scope analysis, let

[20] The sample includes the three ordinary residents assigned to each group and 32 additional ordinary residents who filled the fourth position in no-leader groups during the first two waves. Repeated fourth participants are excluded.

[21] We use a common pre-meeting window to ensure that the baseline measure is not affected by potential spillovers from earlier meetings. The main specification controls for potential spillover exposure, the construction of which is detailed in Appendix Section C.1.

$e(i) = c(i)$ and $\mathcal{E} = \{H, M\}$, pooling leader effects across disclosure conditions. We estimate:

$$Y_i = \lambda_{n(i),q(i),c(i)} + \rho Y_{i0} + \sum_{e \in \mathcal{E}} \left(\tau_e^T T_{g(i)} + \tau_e^R R_{g(i)}\right) \mathbf{1}\{e(i) = e\} + X_i'\beta + \varepsilon_i. \tag{1}$$

The vector X_i contains the 25-meter spillover indicator and 11 LASSO-selected baseline controls used in the main figures and the third table specification. Appendix Section C.2 describes the selection. The neighborhood-by-disclosure-by-composition fixed effects $\lambda_{n(i),q(i),c(i)}$ ensure that each leader group is compared with no-leader groups in the same neighborhood and experimental environment. Hence, for each environment e, τ_e^T, and τ_e^R are the intention-to-treat effects of assignment to a group with a traditional or religious leader, respectively, relative to no leader under the same conditions.

4.2 Main Results

The lower panel of Figure 3 plots the average predicted levels of groups without a leader and shows similar participation across no-leader groups ($p = 0.536$), establishing our first finding: in the absence of a leader, ethnic cohesion alone does not improve participation. This finding aligns with previous work on how informal institutions govern the commons, which highlights the role of leaders in establishing sanctions to prevent free-riding (Ostrom, 2000; Miguel and Gugerty, 2005; Kosfeld and Rustagi, 2015).

Figure 3 upper panel then reports the paper's main finding: ethnic fragmentation changes which leader can mobilize collective action. Assigning an urban chief to a group of participants who share the leader's territorial origin increases participation by 6.0 percentage points (SE 2.8; $p = 0.032$), a 10.5 percent increase over the 56.8 percent no-leader mean. In contrast, the chief's effect in mixed groups is only 0.4 points (SE 2.7; $p = 0.893$). Perhaps surprisingly, pastors display the *opposite* pattern. The presence of a religious leader increases participation by 5.8 points in mixed groups (SE 2.8; $p = 0.039$), or 9.9 percent relative to the 58.3 percent no-leader mean, and by *just* 1.7 points in homogeneous groups (SE 2.4; $p = 0.484$). The traditional-minus-religious contrast is 4.3 points in homogeneous groups (SE 2.7; $p = 0.113$) and -5.4 points in mixed groups (SE 3.0; $p = 0.067$); the difference between these gaps is $\widehat{\Delta}_C = 9.8$ percentage points (SE 4.1; $p = 0.017$).

We interpret these results as consistent with chiefs and pastors having different scopes of authority. Leaders are effective when participants recognize their right to ask them to contribute. Chiefs derive this standing from kinship, ethnic ties, and their customary responsibility for *salongo*. Thus, when chiefs share origins with participants in the group, their presence strengthens the pressure to meet a demand that participants already recognize as legitimate. In contrast, chiefs may lack the same standing to demand contributions from residents of other origins in mixed teams. Pastors derive their authority from a shared Christian worldview that crosses territorial boundaries. Because mixed teams lack a common customary basis for participation, pastors can frame community work as an obligation before God and provide a shared reason to contribute.

The same appeal has little impact in homogeneous teams, where participation is already recognized as an ethnically enforced obligation. The force of this social obligation, however, appears to depend on the presence of the chief who is entitled to invoke and enforce it. Pastors' religious appeals to God's judgment seem to matter only where ethnic obligations do not organize behavior.

This interpretation echoes other studies on how religious commitments coexist with ethnic obligations and how their relative importance depends on the social context. Gluckman (1940) describes how kinship obligations remain binding across religious divisions in certain contexts in Zululand, while Laitin (1986) shows how ancestral-town affiliations take precedence over Christian–Muslim divisions among the Yoruba. In the Congolese context, Placide Tempels' *Bantu Philosophy* (1959) links clan membership to communal duties and chiefs' enforcement authority while emphasizing the persistence of inherited understandings among Christian converts. These accounts distinguish religious affiliation from the obligations that govern a particular interaction. They complement our interpretation by establishing a precedent where socially enforced ethnic obligations coexist with internally enforced religious obligations (Enke, 2019), and where the relevance of those obligations depends on the setting an individual faces (Akerlof and Kranton, 2000). In our setting, group composition shapes the obligations that govern communal work; and where territorial ties organize social relationships, ethnic duties take precedence and reduce the relevance of religious appeals, even among committed Christians.

Composition changes which leader can sustain collective action. A shared territorial origin places *salongo* within the chief's recognized authority, while mixed origins create room for authority that crosses territorial boundaries. This is our *scope of authority* result. But before moving into complementary evidence for the proposed account, we discuss the results' robustness and unlikely alternative explanations.

The response is concentrated in labor. Across alternative outcome definitions, the effects primarily appear in labor participation. We estimate Equation (1) using alternative measures of labor participation: total days, total hours in levels, inverse-hyperbolic sine transformation of total hours, and ratings of effort by city staff. Across the same 1,040 participants and main outcome window, the Δ_C pattern corresponds to 0.56 additional days of participation. Total hours contributed and rated effort also move in the same direction. Adjusted p-values across the four labor outcomes are at most 0.047 for Δ_C (Appendix Table C-1), and the asinh-hours specification yields the same conclusion (Appendix Table C-1). Contrastingly, cash and material contributions show no comparable pattern (Appendix Table C-2).

Robustness. The main pattern is stable after accounting for exposure to potential spillovers from neighbors that were treated and other baseline controls (Appendix Table C-3 and Appendix C.1). Randomization inference, which repeats the original group-formation procedure across 5,000 assignments while holding the realized sample fixed, reinforces our conclusions (Appendix Table C-4). The main contrasts also remain significant when allowing errors to be correlated across groups assigned to the same leader.

Alternative explanation I: endogenous leader behavior. The main alternative explanation for the composition pattern is that leaders changed their behavior in the groups where they succeeded. We follow the sequence from what leaders say in the first meeting, to arriving together at *salongo*, and finally to group participants' reports of follow-up contact.

i. *Meeting effort.* We first compare speaking time and topic shares of leaders across compositions, using the three topics we discuss in more detail in Section 5.3 and leader fixed effects (Panels A–B of Appendix Table C-5).[22] Chiefs could have put in more effort in mobilization discourse among their own coethnics, or pastors could have changed their speeches depending on the composition of the group. We find little evidence of leaders behaving differently across meeting environments, but we find significant differences between the behaviors of pastors and chiefs. Pastors speak more and place emphasis on religiously related speeches, while chiefs speak for less time and emphasize speeches about social accountability. Leaders are better understood as delivering consistent treatments across environments while differing substantially in what is delivered.

ii. *Salongo coordination.* Leaders could, nevertheless, differ in how they coordinate participation after the meeting. Chiefs could actively target coethnics more often because of their social proximity, or pastors could exert more effort to bring mixed teams' participants to the activity because they know they would be constrained by the chief's jurisdiction when with members of homogeneous groups. We approach this potential explanation with behavioral and reported measures. If mobilization efforts occur outside of the meeting and leaders bring participants to the activity with them, we should observe this pattern arising in arrival times at the worksite. Leaders and other team members would arrive at the same time. We measure co-arrival in two ways: (i) whether the leader's and participant's recorded arrival times are no more than one or two minutes apart; and (ii) whether they belong to the same arrival group, defined as a sequence of check-ins with no more than one minute between consecutive check-ins (thirty seconds in the alternative definition).

   The pattern we find mirrors the speech evidence. Behavior differs across leader types, while differences across compositions are imprecisely estimated. Chiefs arrive with participants more often; participants of a group with a chief arrive with the leader on 8.2% of days in mixed groups and 10.5% in homogeneous groups (Panels A–B of Appendix Table C-6). With pastors, the same levels are around 5–6% of days. These rates use all six post-meeting days, counting nonattendance as zero. Pooling compositions, the chief co-arrival rate is 9.4%, about 14% higher than the 8.2% rate per ordinary groupmate; the pastor rate is 5.5%, about 40% lower than the corresponding 9.2% rate. The pastor rate is lower than both the chief rate (difference -3.8 percentage points; $p = 0.009$) and the ordinary-groupmate rate (-3.7 points; $p < 0.001$).

[22] For each leader type, we estimate $y_{\ell g} = \alpha_\ell + \beta H_g + \varepsilon_{\ell g}$, where $y_{\ell g}$ is leader ℓ's speaking time or topic share in group g's first meeting, H_g indicates a homogeneous group, and α_ℓ is a leader fixed effect. We pool disclosure conditions and cluster standard errors by neighborhood.

iii. *Effort outside meetings and salongo.* Finally, follow-up contact may have occurred and may not necessarily be related to arrival times. Leaders could be devoting time outside of activities to exert sensitization efforts. We use the group members' reports collected at the start of the second meeting about contact with the assigned leader during the preceding week. Panels A–B of Appendix Table C-7 compare the rate of these responses between homogeneous and mixed groups. We ask whether participants had interacted with the group's leader, whether they discussed salongo, whether the leader explicitly asked the respondent to participate in salongo during this conversation, and if they mentioned social or divine consequences associated with showing up. About 41–50% of participants report contact with the leader during the work week, and 11–14% report being explicitly asked to participate. Explicit mention of social or divine consequences is far rarer, occurring in fewer than 4% of cases. There is no statistically significant evidence that leaders adjust these behaviors based on the composition of the group.

Taken together, these checks provide no consistent evidence that leaders intensify their activity or tailor their appeals in the groups where they succeed; they leave room for differences in effort and in the quality of interactions.

Alternative explanation II: belonging to the majority ethnicity. The composition pattern could instead reflect a belonging to the neighborhood's majority ethnicity. One could argue for an account in which pastors do not work with homogeneous groups because coethnics with the chief would only respond to the chief. In this account, it is not the group context that matters (context-dependent) but the trait of belonging to the ethnicity that composes homogeneous groups, the ethnic majority of the neighborhood (trait-dependent). As we discuss in Section 4.3, the regression in Appendix Table C-8 shows a positive pastor effect among majority residents assigned to mixed groups, compared with a small effect that is statistically indistinguishable from zero in homogeneous groups. Although the direct difference is imprecise, this pattern suggests that belonging to majority ethnicity alone does not explain pastors' limited reach in homogeneous teams.

Alternative explanation III: pastors' null is not due to ethnic mismatch. Pastors may fail in homogeneous groups because they do not always share the territorial origins that link chiefs to those participants. In this account, pastors would fail to sustain participation in homogeneous groups not because of ethnic homogeneity, but because they are from a different territory. Out of the twenty-eight localities and twenty-eight recruited religious leaders, seventeen share their origins with the chief and the homogeneous groups in the neighborhood, while eleven do not. We leverage these seventeen religious leaders to verify whether our conclusion holds in a restricted sample where the religious leader also shares their origin with the homogeneous group.

Appendix Table C-9 restricts pastors' comparisons to neighborhoods where the assigned chief and pastor have the same territorial relationship with homogeneous-group participants. Pastors with the relevant terri-

torial connection do not become more effective in homogeneous groups, and their effect remains concentrated among groups composed of diverse origins. The pastor result, therefore, reflects the limited reach of religious leadership within ethnically cohesive groups, rather than a simpler account in which an outsider leader would have a smaller impact.

Alternative explanation IV: leadership by example. Rather than projecting authority onto different residents, one could argue that leaders would incentivize contributions by serving as an example for the group, and the different effects by composition would be explained by whether residents consider that specific leader example to be followed. Leadership by example would predict larger effects under leaders who contributed more before the experiment (Jack and Recalde, 2015). Appendix Table C-10 compares leaders above and below their type-specific median participation in the common pre-meeting week. The higher-minus-lower difference is -5.8 percentage points for chiefs in homogeneous groups (SE 5.2; $p = 0.262$) and 3.2 points for pastors in mixed groups (SE 5.2; $p = 0.542$). These estimates provide little support for differences in leaders' earlier participation as an explanation of the composition pattern.

Alternative explanation V: No evidence of anticipation or mean reversion. The pre-assignment characteristics used to form groups are balanced. However, administrative participation, measured after assignment and before the first accountability meeting, differs across the realized cells (Appendix Table B-4). Participants received their invitation to participate in volunteer teams during this week, but had not yet learned about their teammates or the disclosure condition during this period. Nevertheless, two concerns arise: (i) participants may have learned that a leader was assigned to their team before the first meeting, creating anticipation effects, and (ii) even without anticipation, the estimates could be driven by mean reversion in the imbalanced cells.

We address the first concern in Appendix Table C-11, which examines the six daily pretreatment observations. The daily data show no systematic increase in participation before the first meeting, although some pretreatment paths differ. We address the mean reversion concern in Appendix Table C-12, allowing the relationship between pretreatment and subsequent participation to differ across all 12 assignment cells. All estimated coefficients remain essentially the same. Together, these checks provide little support for the imbalance as an explanation of the leader effects that emerge after the meeting.

4.3 Heterogeneous Treatment Effects: Who Can Mobilize Whom?

Our key result shows that leaders have distinct scopes of authority. Chiefs are effective when placed in teams where they share origins with residents. Pastors, on the other hand, are effective precisely in mixed teams, where chiefs have little influence. These results point to an account where ethnic cohesion has opposing impacts on the ability of leaders to organize collective action. While it enables chiefs by placing them in

a context where their standing in an origin-based governance system[23] matters for enforcing contributions, it limits pastors by grounding cooperation in socially enforced ethnic obligations that reduce the relevance of a religious framing of participation. But what determines the boundaries of each leader's authority? Do broader patterns in who responds to each leader support this interpretation?

We examine this question using groups' average ethnic identification, residents' relationships with their assigned leader, and their religious orientation. We also ask whether recognition of the chief's authority and personal receptiveness to religion predict participation. We estimate heterogeneity across these characteristics using an extension of the main specification. Let Z_{ig} indicate the higher category of the characteristic under study. Retaining the neighborhood-by-composition-by-disclosure fixed effects from Equation (1), we estimate

$$\begin{aligned} Y_i = \lambda_{n(i),q(i),c(i)} + \rho Y_{i0} + \sum_{c\in\{H,M\}} \eta_c Z_{ig} \mathbf{1}\{c(i)=c\} \\ + \sum_{L\in\{T,R\}} \sum_{c\in\{H,M\}} \left(\tau_c^L + \gamma_c^L Z_{ig}\right) \mathbf{1}\{L_{g(i)}=L, c(i)=c\} + X_i'\beta + \varepsilon_i. \end{aligned} \tag{2}$$

Here L_g denotes the assigned leader type. Both leader effects and the higher-minus-lower differences, γ_c^L, are pooled across disclosure. In some specifications, Z_{ig} is defined only for participants assigned a leader (e.g., leader ratings). In those cases, we omit the η_c terms, and the comparison is with the average participant in a no-leader group within the corresponding experimental stratum defined by $\lambda_{n(i),q(i),c(i)}$. We also estimate heterogeneous effects according to whether residents belong to the neighborhood's ethnic majority. Since homogeneous groups contain only majority residents by design, homogeneous-group effects remain pooled.

Figures 4 and 5 show chief and pastor effects, respectively. The main figures include the pretreatment outcome, the design fixed effects, the 25-meter spillover indicator, and 11 LASSO-selected baseline controls. Appendix Tables C-13–C-18 report both leader types, with and without spillover and LASSO controls. Appendix Section C.5 describes the variables. Standard errors are clustered by accountability group.

Chiefs' authority rests on ethnic ties with residents. Our account of chiefs emphasizes how their ethnic proximity to homogeneous teams enables them to influence contributions. Within these teams, all residents share the chief's origin. Therefore, we ask whether chiefs are more effective in groups whose members are more strongly identified with their ethnicity.

Before being introduced to their groups, respondents ranked six identities by their importance. We score identification with one's own ethnicity from least important to most important and average individual scores. We divide groups at the median of group ethnic identification, retaining all 1,040 residents in 336 groups. Figure 4A and Appendix Table C-13 compare groups with higher and lower ethnic identification. In homogeneous groups, chiefs raise participation by 9.7 points in groups with higher identification (SE 4.0), compared with 3.4 points in those with lower identification (SE 3.9). The 6.3-point difference is positive but

[23] See Section 2.

imprecise (SE 5.7; $p = 0.275$). In mixed groups, the corresponding effects are 5.2 points (SE 5.3) and -2.5 points (SE 3.5), a positive but imprecise 7.7-point difference (SE 6.8; $p = 0.260$). Among these four chief effects, only the effect in homogeneous groups with higher ethnic identification is statistically significant at the 5 percent level. The direction in homogeneous groups is consistent with stronger ethnic identification making a shared ethnic tie more consequential for chiefs' enforcement. Appendix Table C-14 shows that the positive chief estimate in mixed groups with high ethnic identification is concentrated among his coethnics (15.1 points, SE 10.5), compared with 1.2 points among other residents (SE 6.5), although the 13.9-point difference is imprecise ($p = 0.284$).

Chiefs can reach residents in mixed groups. Residents' recognition of chiefs' standing provides complementary evidence on their ability to enforce contributions, including in mixed groups. Figure 4B and Appendix Table C-15 compare residents above and below the median answer to a Likert question on whether the leader of their group is able to sanction and reward residents. The panel shows effects in mixed and homogeneous groups, with pooled no-leader references below, since the assessment is undefined without an assigned leader. Estimates show that chiefs are able to influence the participation of residents in mixed groups who recognize their authority. If the resident is above the cutpoint, chiefs lead to 6.8 percentage points (SE 4.0) higher participation, compared with -2.1 points among those below it (SE 3.3). The 8.9-point difference is suggestive (SE 5.0; $p = 0.075$).

An index that bundles together more evaluations produces a clearer difference. We asked each resident to rate the leader in their team in terms of trustworthiness, hard work, moral stance, and how much the leader inspires them. Appendix Table C-16 shows a 6.8-point chief effect in mixed groups among residents who evaluate him more favorably (SE 3.6), compared with -3.5 points among those who do not (SE 3.4). The difference is 10.3 points (SE 4.5; $p = 0.023$). The corresponding differences for pastors are smaller and imprecisely estimated. These comparisons suggest that a chief can mobilize outside the homogeneous group when residents recognize his standing or view him favorably.

Territorial ties point in the same direction, although the comparison is less precise. Appendix Table C-8 compares residents of mixed teams with the same or different territorial origins as the neighborhood chief. Since, by design, all participants of homogeneous groups share their territorial origin with the chief, homogeneous group effects are pooled. Estimates show that the chief effect is 5.3 points among residents who share his territory (SE 6.7) and -1.3 points among those who come from a different origin (SE 3.3). The 6.5-point difference is imprecise (SE 8.0; $p = 0.414$). We interpret these results with caution since they draw on relatively few residents from mixed groups who share the chief's origin (34 groups/44 participants in chiefs' groups and 35 groups/42 respondents in the no leader condition). Together, these comparisons are consistent with ethnic ties grounding chiefs' authority, while recognition of their standing can also sustain their influence beyond those ties.

Pastors' reach depends on the social context. In our favored account, pastors failure in homogeneous groups concerns how members collectively understand the activity. The more the group sees the activity as an ethnic undertaking, the less the pastor should be able to influence behavior, and their appeals may even become counterproductive if they displace a strong ethnic basis for cooperation with a weaker religious one. To investigate those patterns, we follow the same split on group average identification with ethnicity presented for chiefs in Figure 4A, and we present the heterogeneous treatment effects of pastors across these categories in Figure 5A and Appendix Table C-13.

Our findings show that the null average effect of pastors in homogeneous groups masks opposing effects across the distribution of group ethnic identification. In homogeneous groups at or below the median, pastors raise participation by 6.8 points (SE 3.3), compared with −4.4 points above it (SE 4.6). The higher-minus-lower difference is −11.1 points (SE 6.4; $p = 0.081$); the negative effect above the median is imprecise. In mixed groups, where greater ethnic identification means identification with different origins and therefore more division, we observe the opposite pattern: the pastor effect increases with ethnic identification, from 0.1 points (SE 3.8) to 13.2 points (SE 4.2), mainly because it recovers cooperation in highly divided groups, which, without a leader, have considerably lower participation compared to mixed groups that are less attached to their divisive ethnic identities. The 13.0-point difference is positive (SE 5.8; $p = 0.025$). The homogeneous-minus-mixed difference in these gradients is −24.2 points (SE 8.6; $p = 0.005$). This pattern is consistent with shared ethnic identification giving ethnic obligations precedence over religious appeals in homogeneous groups. In mixed groups, stronger identification with different territories instead leaves more room for a religious framing to organize cooperation.

Could these patterns reflect individual receptiveness to religion rather than the group context? Embeddedness in the chief's ethnic network of reciprocity might weaken residents' responsiveness to religious appeals. We investigate this possibility first by verifying the correlation between ethnic identification and religiosity in homogeneous groups and find little support that individuals who are attached to chiefs' ethnic identity are less religious. Furthermore, by design, all residents in homogeneous teams come from the neighborhoods' ethnic majority, whereas mixed teams can include residents from that origin alongside others. Comparing pastors' influence on the chief's coethnics assigned to homogeneous groups with their influence on coethnics assigned to mixed groups helps distinguish a limit rooted in residents' ethnic ties to the chief from one created by team composition. Appendix Table C-8 shows that pastors raise participation by 12.5 points among the chief's coethnics placed in mixed groups (SE 6.7; $p = 0.063$), while their effect in homogeneous groups is 1.7 points (SE 2.4). The 10.8-point difference is positive but imprecise (SE 7.1; $p = 0.129$). Since the estimates indicate that pastors can influence majority residents when they are part of mixed groups, we interpret these results as evidence consistent with a group context in which shared ethnic origins limit the effectiveness of religious appeals.

Pastors reach intrinsically religious residents. Religiosity, nevertheless, shapes who can be influenced by pastors. Following Allport and Ross (1967), we distinguish between intrinsic religiosity, where faith is internalized as a guide to life, and extrinsic religiosity, where religion serves other personal or social needs. Extrinsic religiosity includes personal reliance, such as seeking comfort or support from religion, and social reliance, such as using the church for social contact. If pastors work through a religious frame of obligation that resonates with the audience listening to the appeal, we should expect greater influence among residents for whom religion guides personal conduct. We average the intrinsic and extrinsic items separately and divide residents into lower and higher categories.

Figure 5B and Appendix Table C-17 present the results. In mixed groups, pastors raise participation by 9.9 points among residents with higher intrinsic religiosity (SE 4.3), compared with 0.7 points among those with lower intrinsic religiosity (SE 5.0). The 9.2-point difference is positive but imprecise (SE 7.4; $p = 0.213$). The corresponding difference for chiefs is smaller and imprecise (Appendix Table C-17). This pattern supports the interpretation that pastors' influence depends on residents' intrinsic religiosity. Extrinsic religiosity provides a useful comparison. Appendix Table C-18 shows pastor effects in mixed groups of 7.7 points among residents with lower extrinsic religiosity (SE 4.3) and 3.6 points among those with higher extrinsic religiosity (SE 3.9). The -4.1-point difference is imprecise (SE 6.0; $p = 0.498$). The pooled estimates therefore provide no clear evidence that responsiveness to pastors increases with extrinsic religiosity.

Denominational ties may extend religious authority. Collected data also allow us to establish whether residents share the neighborhood pastor's religious denomination. This matters for whether mobilization needs to occur through a familiar set of dogmas or can hold through a broad religious framing of obligation. Appendix Table C-19 shows that pastors raise participation by 7.6 points among mixed-group residents who do not share their denomination (SE 3.1; $p = 0.015$). Within-denomination links are therefore not necessary for this effect. In homogeneous groups, the pastor effect among residents who share his denomination is positive, at 12.3 points, but imprecise (SE 8.3; $p = 0.142$); its difference from the effect among other residents is also imprecise ($p = 0.155$). Only 42 religious-arm groups contain a resident who shares the pastor's denomination, so there is suggestive evidence that parochial ties extend pastors' reach into homogeneous groups.

Which traits make leaders effective? These comparisons suggest that chiefs' scope depends on ethnic ties that give their standing force with individual residents, while pastors' scope depends on the social context and residents' receptiveness to religious obligation. Appendix Section C.6 complements this evidence with leaders' own characteristics. Chiefs' effects are suggestively stronger among those with deeper local roots in homogeneous groups, while pastors' effects are stronger among those with greater conviction in divine moral enforcement in mixed groups. These results corroborate the interpretation that chiefs' influence operates through established relationships in the neighborhood, while pastors depend on religious claims

regarding divine judgment.

5 Source of Enforcement Results

After investigating *whose* participation a leader can enforce, a question that naturally follows is *how* participation is enforced. The preceding results suggest that leaders make participation consequential through different channels. Yet the evidence is not enough for a conclusive interpretation. Do residents contribute because they expect others to judge their behavior? Or do they contribute even when others are not watching because of an internalized obligation?

For chiefs, their greater influence over residents who recognize their capacity to sanction and reward is consistent with their role as *external*, social enforcers of behavior (Kosfeld and Rustagi, 2015). If this account is true, increased public accountability should strengthen their capacity to enforce behavior. For pastors, their larger influence over residents who score higher in intrinsic religiosity points toward internal religious motivations (Shariff and Norenzayan, 2007). These internal motivations do not require others to observe participation, so public accountability should not strengthen pastors' influence. Pastors' effect on participants could even decline if disclosure shifts attention toward social image concerns that provide weaker incentives in ethnically mixed groups. Yet the intrinsic religiosity account does not rule out a religious version of social enforcement, where costly behavior may signal religious commitment and enable future cooperation (Iannaccone, 1992; Levy and Razin, 2012). In this alternative account, where a religious frame influences social expectations, public accountability should instead strengthen their impact.

We next use our experimental variation in public accountability to disentangle these sources of enforcement.

5.1 Research Design

To distinguish between internal and external sources of enforcement, we randomized whether meetings disclosed individual information about peers' contributions. In teams with disclosure, each member's contribution record was public at every meeting, while teams without disclosure could only see the total group contribution. This experimental arm changed how much participants knew about their peers' participation. Public accountability can strengthen external enforcement by making behavior visible to others, but it should not strengthen internal enforcement channels because individuals already observe their own actions.

We now set $e(i) = (q(i), c(i))$ and $\mathcal{E} = \{(0, H), (1, H), (0, M), (1, M)\}$, producing leader effects in each composition-by-disclosure environment while retaining the sample, baseline control, and fixed effects from Equation (1). This specification asks whether the two leaders' effectiveness also depends on the visibility of individual participation. We expected chiefs to be most effective in homogeneous groups with disclosure, where the group falls within their authority and they can evaluate individual behavior. Pastors should rely less on disclosure because their authority can also make participation binding through internal accountability.

5.2 Main Results

Figure 6 and Appendix Table D-1 show how leader effects across homogeneous and mixed teams vary under the two disclosure conditions. In homogeneous groups with disclosure, assigning a traditional leader increases participation by 10.6 percentage points (SE 3.5; $p = 0.002$), while the religious-leader effect is 1.7 points (SE 3.1; $p = 0.579$); the direct traditional-over-religious gap is 8.9 points (SE 3.7; $p = 0.016$). In homogeneous groups without disclosure, neither leader has a statistically significant effect relative to no leader. Pastors show the opposite pattern in mixed groups. In mixed groups with disclosure, neither leader has a statistically significant effect. The pastor's effect in mixed groups is concentrated in teams without disclosure. In mixed groups without public disclosure of individual participation, religious leadership increases participation by 9.0 percentage points (SE 3.8; $p = 0.019$) relative to having no leader. The traditional-leader effect in this environment is -1.1 points (SE 3.5; $p = 0.751$). The traditional-minus-religious gap in mixed groups without public accountability is -10.1 points (SE 4.2; $p = 0.016$). The cross-environment contrast summarizes how the relative effectiveness of chiefs and pastors changes between these two environments. It equals $\widehat{\Delta}_E = 8.9 - (-10.1) = 19.0$ percentage points (SE 5.6; $p < 0.001$).

These results reinforce the interpretation that religious leaders are weak mobilizers of homogeneous groups and that chiefs are weak mobilizers of mixed groups due to their mismatched scope. They also provide evidence that chiefs and pastors mobilize through distinct enforcement channels. This second reversal, in which pastors hold an advantage in private disclosure and chiefs in public disclosure, is difficult to reconcile with a common social mechanism that operates more strongly for one leader than the other. Chiefs in same-origin groups appear to rely on external enforcement, with residents responding to the chief's judgment of their behavior when participation is public. In contrast, pastors appear to work through internal religious accountability, using a shared Christian framework that leads residents to evaluate their own behavior before God, which becomes most relevant when their actions are private. When actions are public, pastors appeals may become less effective because behavior is more strongly influenced by the existing social norm and religious motivations become secondary. This is our *source of enforcement* result.

The response remains concentrated in labor. The environment gradient also appears across the alternative labor measures introduced in Section 4.2. It corresponds to 1.10 additional days of participation, with adjusted p-values below 0.004 across the four labor outcomes (Appendix Table D-2). Other definitions of labor contributions, including effort ratings, move in the same direction. Yet cash and material contributions are less frequent and do not show the same pattern (Appendix Table D-3).

Chiefs may become counterproductive when individual records are private. In homogeneous groups without disclosure, the chief effect falls from near zero in the main window to -7.4 percentage points in the final week (SE 4.1; Appendix Figure D-2 and Table D-5). Although imprecisely estimated, this

negative effect suggests that assigning a chief may eventually reduce participation. One possible interpretation is free riding: if residents expect the chief to contribute more to the group's work, they may reduce their own effort when individual participation is not publicly disclosed.

Alternative explanations: endogeneous leader responses. Most alternative explanations that were discussed as potential accounts for earlier results could also explain these patterns. Rather than an account in which teammates respond to the reputational exposure to chiefs and intrinsic incentives under pastors, one could argue, for example, that the chiefs focus their efforts on sensitizing participants under public disclosure, while pastors would employ more effort to sensitize participants when records are not public. Hence, we revisit the earlier outcomes on the behavior of leaders to further investigate those possibilities. Panels D of Appendix Tables C-5- C-7 present the results.

We find some evidence for different responses of leaders depending on the disclosure environment, but they cannot fully explain the patterns or compete with our favored interpretation. When it comes to behavior during debriefing meetings, we find that the only significant difference arises among pastors. Although they do not change the topics they raise in their speeches on average, we find that in mixed groups, pastors speak for half a minute less in a public disclosure environment compared to the average of 3.75 minutes spoken in a private disclosure environment (-0.56 minutes; $p = 0.047$). This result helps us understand the more muted impact of pastors under public disclosure, but it cannot fully account for the entire pattern of results.

Our analysis of salongo co-arrival outcomes provides little support for a leader-response account. Estimates are small and imprecise. In homogeneous groups with public disclosure, the roughly 10 percentage point increase in participation induced by chief presence includes an estimated 2.33 percentage point increase in participation accompanied by co-arrival with the chief when we compare it to how chiefs behave in the same homogeneous composition without disclosure. This provides weak evidence consistent with chiefs using public disclosure to identify and bring teammates to salongo, although co-arrival alone does not establish active mobilization. Nevertheless, the estimated increase in co-arrival is too small to account for the full participation effect.

Lastly, our analysis of the survey reports about leader contact also yields little support for a leader-driven account of participation outcomes. The only statistically significant estimate is a 14.06 percentage point ($p = 0.023$) increase in the probability of being asked by the chief to participate in mixed groups with public disclosure, relative to the 7 percent request rate in mixed groups with a chief but no disclosure. Therefore, if anything, the evidence we find is consistent with chiefs trying to compensate for their lack of influence in mixed groups when they are able to see how much each member is contributing. The pattern of contact rates does not suggest that the additional effect is due to a leader-effort account, but rather to how participants respond to leaders' presence under distinct disclosure environments.

Alternative explanations: other accounts. The extended heterogeneous treatment effects specification that compares the influence of pastors on majority-ethnicity residents still supports the conclusion that pastors' null in homogeneous groups is about group composition rather than an individual trait explanation (Appendix Table D-6). The restriction to pastors belonging to the majority ethnicity again yields no detectable pastor effect in homogeneous groups. If anything, it reveals that the null pastor effect in mixed groups with public disclosure is not robust, as estimates in the restricted sample show similar pastor effects in mixed groups under public and private disclosure (Appendix Table D-7). Leadership by example again receives little support (Appendix Table D-8). The pretreatment data show no systematic increase before the first meeting, and allowing cell-specific baseline adjustment leaves the estimates essentially unchanged (Appendix Tables D-9 and C-12). Overall, we find no robust evidence that alternative explanations are driving our results. We now present complementary evidence on chiefs' and pastors' different sources of enforcement.

5.3 What Leaders Say: Pastors Talk About God, Chiefs About Attendance

The public accountability variation suggests that chiefs and pastors draw on different sources of enforcement. If this is indeed the case, we should observe different leader contributions and approaches to coordination meetings. Do chiefs emphasize attendance and social accountability while pastors invoke obligations before God? Recordings of the first group meetings allow us to study these appeals before the subsequent participation outcomes are realized.

If leaders sustain collective action through different enforcement channels, these differences should be reflected in leaders' discourse. To test this interpretation, we use a topic model to analyze transcripts from the first group meetings (Blei et al., 2003). The model groups words that often appear together into latent topics and is blind to every other aspect of the intervention, including group structure, treatment condition, and leader role. Because our passages are short, we follow the approach of Quinn et al. (2010) and assign each passage to one of three categories. We name the topics only after estimation. We present this simplest specification and compare it with more flexible topic models and embedding-based methods in Appendix Section D.4.[24] The conclusions are unchanged across models. Appendix Table D-10 reports translated passages that help interpret each topic.

Latent topics. The first passages in Appendix Table D-10 highlight a discourse that creates social pressure around participation. We labeled these as attendance-and-performance speeches. The excerpts display leaders setting clear targets for hours and regular participation, calling out those residents who did not show up and praising groups that meet expectations. This fits the interpretation we developed for the chiefs' source of enforcement, where participation is induced by stronger social accountability over the activity.

The last block of the table establishes a different enforcer. The examples tell volunteers to look to God

[24]For the use of text as data and the measurement and inference issues it raises, see Gentzkow et al. (2019) and Dell and Rambachan (2026).

rather than wages for reward, state that "God rewards each person according to their deeds", and connect collective work to patience, peace, and mutual aid. This language frames participation as a decision subject to divine judgment and calls for solidarity between peers. It is the speech counterpart to the strategy attributed to pastors, whose appeal can bind residents through self-evaluation before God, even when individual contribution records are not disclosed.

The third topic concerns operational constraints. A recurrent topic of discussion was the coordination of who would provide the materials for neighborhood improvement. Some examples concern within-group coordination, where peers would discuss and agree on who would provide the necessary tools for the work to take place. Yet, the passages also show substantial reliance on the centralized provision of these materials, with many examples praising the municipality for providing the tools for the work. Although it may be useful material for program improvement and future research, we interpret this topic as neutral concerning our main goal. The distinction between mobilization strategies comes from the relative emphasis on attendance and performance or on divine reward and solidarity.

Who talks about what, and when? After assigning each passage to its most likely topic, we sum the passage duration to measure how many minutes each speaker spends on attendance and performance, material support, and divine rewards and solidarity. We aggregate the classified passages to the leader-by-meeting level. Figure 7 shows the average share of speaking time devoted to each topic, pooling the four composition-by-disclosure environments. Blue bars report traditional leaders and gold bars religious leaders. Appendix Table C-5 reports total speaking time and topic shares separately by environment.

The figure shows that leaders' discourse differs across leader types. Aligned with our interpretation, chiefs devote 44.1 percent of their speech to attendance and performance, compared with 26.0 percent among pastors. In contrast, pastors devote 48.3 percent to divine reward and solidarity, compared with 27.4 percent among chiefs. The differences remain significant after multiple testing adjustments ($q = 0.009$), while material-support shares are similar (28.5 vs. 25.7 percent). Appendix Figure D-3 shows the same distinction across the full set of models. Appendix Table C-5 shows that these differences persist across experimental conditions. Pastors speak for 3.43 minutes on average, compared with 2.55 minutes for chiefs ($p = 0.067$). Within each leader type, speaking time and topic shares are fairly stable across composition and disclosure conditions. All tests of equality in topic shares across environments have $p > 0.30$, while the speaking-time tests have $p = 0.701$ for chiefs and $p = 0.139$ for pastors. Thus, what participants hear differs mainly by leader type, not by the experimental environment. Appendix Figure D-4 reports the corresponding levels across models in each environment.

Although what leaders say is similar across experimental conditions, their influence on the discussion varies. Appendix Figure D-5 shows estimates of Equation (1) using participants' time spent on each topic. Chiefs do not consistently shift participants' speech toward their characteristic appeals, while pastors tilt the discussion

toward divine reward and solidarity. Pooling disclosure conditions, this shift accompanies additional speaking time in homogeneous groups. In mixed groups, divine speech instead rises as attendance-and-performance talk falls, with little change in total speaking time. Appendix Figure D-6 reports participant levels, including the no-leader control, across all models and shows the same broad direction of effects. Pastors' influence on religious speech extends beyond the environment where they raise participation. The meetings therefore show that pastors can steer the discussion toward their characteristic appeals, although these changes do not always translate into greater participation.

5.4 Heterogeneous Treatment Effects: Who Responds To Revealed Records?

The scope results show that chiefs' influence is stronger among residents who recognize their standing, while pastors' influence is stronger among residents for whom religion guides personal conduct. Meeting recordings reveal the different appeals each leader makes. We now examine a behavioral implication of these two sources of enforcement by asking who responds when contribution records are revealed.

Participation histories reveal distinct sources of enforcement. First meetings under the public information condition affect participants in two ways: (i) it establishes what the reporting conditions for future meetings will be, and (ii) it also provides information on who shirks and who contributes to everyone who attended the meeting. The treatment provides a signal that influences the social image of every group participant based on how much they have contributed up to that point. Shirkers will be the ones who suffer the largest damage to their standing since the group will find out they did not contribute. If chiefs operate through social judgment, the chiefs' effect should be strongest on exactly these shirkers, where the social standing of the participant is worse.

Under our interpretation, pastors' enforcement operates differently. Intrinsically religious residents will change their behavior because the pastor will remind them of their internal religious motivations. What that motivation is, whether it is a fear of divine punishment or moral convictions associated with religious practice, matters less than where the audience judging residents' behavior is located. In pastors' enforcement, the relevant audience is the self (or God), who always knows whether the resident cooperated or not, regardless of the disclosure condition. Therefore, there is no reason to believe that the pastor effect will also be concentrated among shirkers; it may very well extend across initial participation histories.

We test these predictions by extending Equation (2) from composition c to the full environment $e = (q, c)$, allowing both the category-specific no-leader levels and each leader effect to vary with residents' pretreatment participation in every composition-by-disclosure environment. Figure 8 and Appendix Table D-11 show the results for the two compositions where each leader is effective: chiefs in homogeneous groups and pastors in mixed groups. The top row plots each leader's effect relative to no leader at lower and higher levels of pretreatment participation. The bottom row shows the corresponding no-leader means.

Exposed shirkers drive the chief effect, pastors influence everyone. The left column of Figure 8 displays what one would expect from external social enforcement. With disclosure, chiefs raised participation by 19.0 percentage points (SE 6.1) among residents who participated one day or fewer in the first week, compared with 1.7 points (SE 4.2) among those with higher participation. The difference is −17.3 points (SE 7.9; $p = 0.030$). Without disclosure, the corresponding effects are −5.8 and 7.9 points, and the 13.7-point difference is positive but imprecise ($p = 0.145$). Pastors' effects display no comparable difference based on prior participation. The positive chief effect is concentrated among prior nonparticipants and infrequent participants. The weak visible record gives the chief something to judge and the resident something to repair, while residents with stronger records have less room to improve. Appendix Table **??**, which displays robustness specifications, yields the same reversal.

The religious leader effect displays a completely different pattern of results. Pastor effects are positive in both baseline-participation categories when groups are mixed and individual records are not disclosed, at 6.3 points among residents with weaker prior participation (SE 5.9) and 9.4 points among those with stronger records (SE 5.8). Each subgroup estimate is imprecise, while the pooled pastor effect is positive and statistically significant in the main specification. The 3.1-point higher-minus-lower difference is also imprecise (SE 8.9; $p = 0.727$). This result strengthens the interpretation that they operate through internal incentives, but it does not explain why the pastor effect is weakened by public disclosure.

Unpacking the null: public accountability substitutes for the pastor effect. A possible explanation can be seen in the bottom panel of Figure 8. The aggregate no-leader comparison conceals differences between initial shirkers and stronger contributors. In no-leader groups, the implied participation of higher contributors is 74.4 percent without disclosure and 80.6 percent with disclosure, while the corresponding prediction falls for lower contributors. These are standardized no-leader predictions for the residents supporting each leader effect. The implied no-leader difference for higher contributors is 6.3 points (SE 5.0; $p = 0.211$), while the pastor effect falls by 11.7 points across disclosure conditions (SE 7.4; $p = 0.116$; Appendix Table D-11). These differences suggest that public accountability can substitute for internal accountability among stronger contributors, although the pastor-effect contrast is imprecise. The lower predicted participation among weaker contributors is also consistent with a less favorable response under disclosure.

5.5 Policy Implications: Hidden Levers of Fiscal Capacity

The exposition of results in terms of participation shares facilitates their interpretation; however, it underplays the magnitude of the program in terms of state capacity. As highlighted in Olken and Singhal (2011), standard public finance accounts miss an important source of developing states' fiscal capacity when they omit informal taxation. Our collaboration allows us to partially account for this source of capacity in the studied sample, as described in Appendix D.6.

Our study revealed a surprising demand among citizens for the arrangements facilitated by the city program. Once the city provided basic tools and organizational capacity for neighborhood improvement, residents volunteered their labor. Across the 28 campaign neighborhoods, residents contributed 70,250 hours and completed an estimated 832 worksites. The labor amounts to about 8,800 eight-hour workdays, valued at $26,344 using the historical STEP II public-works rate of $3 per day (Agence Congolaise de Presse, 2021). Including recorded cash and in-kind contributions brings the total to $27,890—30.5% of Kasaï-Central's reported 2025 property-tax receipts (Direction Générale des Recettes du Kasaï Central, 2026).

Incorporating our findings into a rough counterfactual that assigns chiefs with public disclosure to homogeneous groups and pastors without disclosure to mixed groups, the program could have mobilized 7,251 additional hours among the 1,040 experimental residents relative to no leaders under the same disclosure conditions. Assuming the six-day effects persist throughout the campaign, this is equivalent to about 75 average worksite workloads. This exercise highlights both the scale of hidden capacity through mobilizing residents and the scope for improvement when leadership and disclosure conditions align with the enforcement channels uncovered by this experiment.

6 Beyond Participation: What Else Changes?

Our results establish when leaders can promote collective action and we develop an interpretation of why they succeed. What are the consequences of this success? The main behavioral outcomes are observed during a short window of time, but sustained cooperation may reshape the social and political views of participants, as well as their relationships. We turn to investigating these possibilities by examining beliefs about cooperation and accountability, the salience of religious and social identities, views of government and other local institutions, and opinions about other members of the group. For the first set of outcomes measured at the individual level, we use the specification in Equation (1) to assess leader treatment effects in different group environments.

More participation does not change beliefs or political attitudes. The first set of outcomes captures beliefs about social and divine accountability and the identities residents consider most important. The second covers government services and accountability, institutional performance, trust in state and non-state actors, and evaluations of the locality chief. Across these outcomes, we find no robust patterns. Appendix Tables E-1, E-2, and E-3 report the full results. Additional participation therefore changed behavior without broadly changing residents' beliefs, identities, or political attitudes.

Sustained cooperation precedes greater willingness to help one another. The clearest changes concern relationships within the group. We compare residents' evaluations of ordinary peers and assigned leaders before the first discussion and at the start of the second meeting. The outcome is the

standardized change in these evaluations. To distinguish how leader presence affects evaluations of ordinary peers from evaluations of the leader, we extend equation (1) as follows. Let j index the person being evaluated and let $J_{jg} = 1$ indicate that this person is the assigned leader:

$$Y_{ij} = \lambda_{n(i),q(i),c(i)} + \sum_{e \in \mathcal{E}} \Big[\left(\tau_e^T + \kappa_e^T J_{jg}\right) T_{g(i)} + \left(\tau_e^R + \kappa_e^R J_{jg}\right) R_{g(i)} \Big] \mathbf{1}\{e(i) = e\} + X_i'\beta + \varepsilon_{ij}. \tag{3}$$

Here Y_{ij} is the standardized change in an individual i's evaluation of person j. For leader type $L \in \{T, R\}$, τ_e^L captures the effect of leader assignment on evaluations of ordinary peers in environment e, while $\tau_e^L + \kappa_e^L$ captures the effect on evaluations of the assigned leader. Appendix Tables E-4 and E-5 report all six outcomes across the four experimental environments. Figure 9 focuses on willingness to help, showing separately the environment in which each leader increased participation and pooling the other three environments.

The effects are sizable where leaders sustain participation. In homogeneous groups with disclosure, assigning a chief increases residents' willingness to help an ordinary participant by 0.19 standard deviations. At the average baseline answer of about 4,600 Congolese francs, this effect corresponds to roughly 1,400 francs more willingness to help, or 30 percent of the baseline mean. In mixed groups without disclosure, assigning a pastor increases the willingness to help an ordinary participant by 0.35 standard deviations and the willingness to help the pastor by 0.34 standard deviations. These effects correspond to about 2,760 and 2,680 francs, or 60 and 58 percent of the baseline means. Other evaluations of peers do not change systematically. Sustained cooperation therefore appears to strengthen the willingness to assist others within the group without improving opinions of peers more generally.

Pastors gain support when they sustain cooperation. The pattern is different for peers and leaders. Chief assignment mainly increases the willingness to help ordinary peers, while evaluations of the chief change little. Pastor assignment increases the willingness to help both ordinary participants and the pastor himself by about the same amount. This difference fits their initial positions. Chiefs already hold established authority among coethnics, while pastors enter mixed groups without the same standing and can gain it when they successfully sustain cooperation. Religious leaders can therefore leave divided groups more cohesive while also gaining greater support from residents.

7 Conclusion

What can unite ethnically divided societies around collective goals when social enforcement breaks down? In Kananga, we have found that religious authorities succeed in promoting collective action in ethnically heterogeneous groups where traditional authorities have failed. In a large-scale field experiment that involved more than a thousand volunteers and hundreds of small neighborhood improvement projects, they not only succeeded in promoting the participation of fragmented teams; they were especially likely to succeed when information about participation was unavailable and social enforcement was the hardest. Our analysis, therefore, points to two reasons that explain their success: the authority they invoke can work across ethnic relationships, and enforcement occurs internally rather than being dependent on social sanctions.

Perhaps more surprising is the result that religious leaders *fail* to sustain high levels of cooperation precisely in settings where traditional leaders succeed. When the audience is ethnically cohesive *or* public accountability shapes incentives, religious leaders are less effective in inducing cooperation. Exploratory evidence suggests that ethnic cohesion limits the influence of religious leaders because ethnic identity organizes cooperation as an ethnically enforced obligation, limiting the scope for framing the activity as a religious duty. Public accountability, in turn, appears to limit the influence of the self-accountability activated by religious moral appeals. Unlike traditional leaders, who display patterns consistent with an authority that depends on social monitoring and is rooted in ties to each resident that may or may not be ethnic, the authority of religious leaders appears to rely on preexisting beliefs and on a setting that allows those beliefs to influence behavior.

These findings expose two dimensions of authority that are rarely studied in tandem and show why future considerations of these dimensions should matter for understanding how collective action operates. Its *scope* determines whom a leader can mobilize; its *source* determines what makes participation incentive compatible—evaluation by others, obligations individuals apply to themselves, or both. Collective action depends on the fit between the leader in charge, the composition of the audience, and whether the enforcement technology is appropriate for the setting. Although extending these considerations to other leaders and institutions requires further research, the implications of this theory of change could apply far beyond religion, customs, and the DRC. More broadly, the same logic becomes relevant where a narrow identity sustained by social accountability coexists with a more encompassing identity whose obligations are internally enforced.

Therefore, beyond the policy implications of the studied framework, which can be applied to any development policy that uses local leaders as intermediaries in program implementation, the paper opens important research avenues by revisiting the relationship between ethnic fragmentation and collective action. Indeed, our paper confirms that fragmentation is a corrosive force for the *social enforcement* of collective action. Yet, by corroding social enforcement, something else also happens: internal motives begin to enforce conduct. Fragmentation does not simply corrode collective action; it changes which institutions can govern behavior. Religion is one widespread institution that enables cooperation in divided settings, potentially among many. What else can govern cooperation when social enforcement breaks down remains the subject of future research.

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Figures

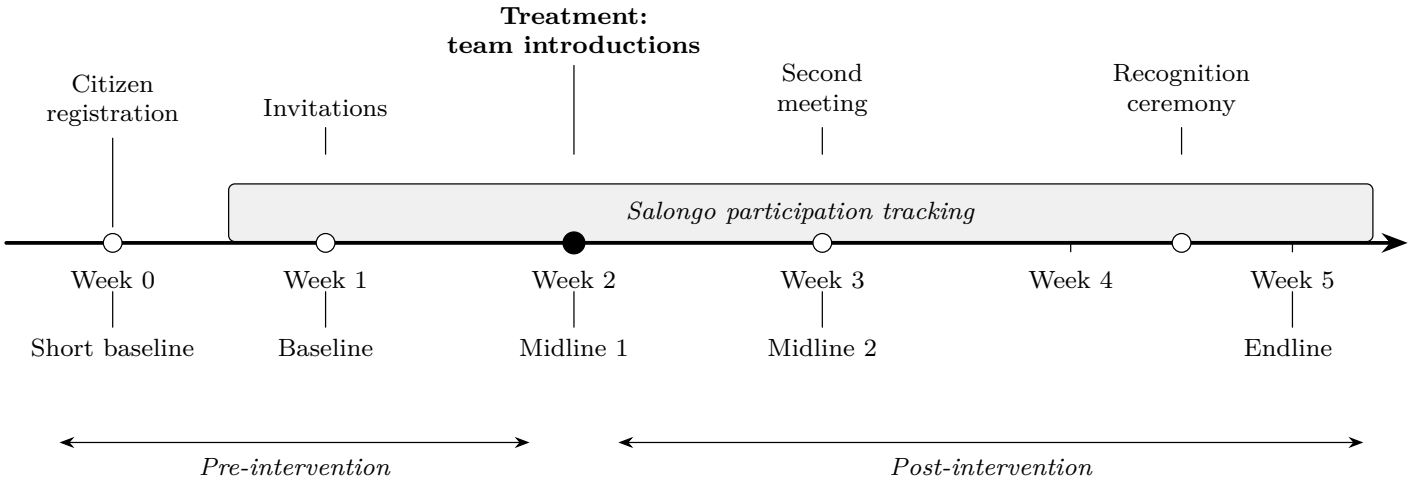
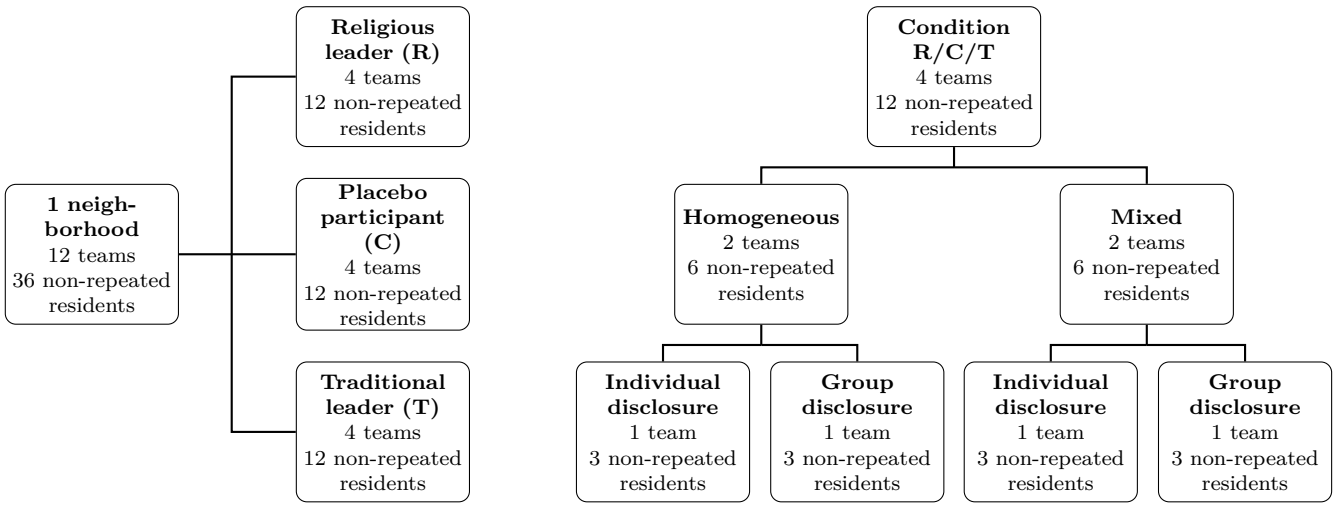


Figure 1: INTERVENTION AND DATA COLLECTION TIMELINE

Notes: The figure shows the sequence of activities during one five-week campaign wave. Residents registered before the city began tracking daily *salongo* participation. Invitations and the baseline survey followed registration. At the first meeting, residents met their assigned teammates and learned the team's leader assignment and whether individual contributions would be disclosed, marking the start of the intervention. The same team met again one week later. The city held the recognition ceremony near the end of the tracking period, and the endline followed. Survey timing appears below the timeline. The intervention is described in Section 3; Appendix Figure D-1 shows the outcome windows used in the analysis.



(a) Leader experiment **(b) Within-leader variation**

Figure 2: EXPERIMENTAL DESIGN

Notes: Each neighborhood contains twelve teams. Panel (a) assigns four teams to a traditional leader, four to a religious leader, and four to no leader. Panel (b) divides the four teams in each condition evenly between homogeneous and mixed groups and between individual and group disclosure. The diagram shows the standard four-person team used from Wave 3 onward. The slightly different no-leader roster used in Waves 1–2 is explained in Appendix Tables B-5–B-7. The figure is discussed in Section 3; balance and implementation checks appear in Appendix Tables B-4–B-7.

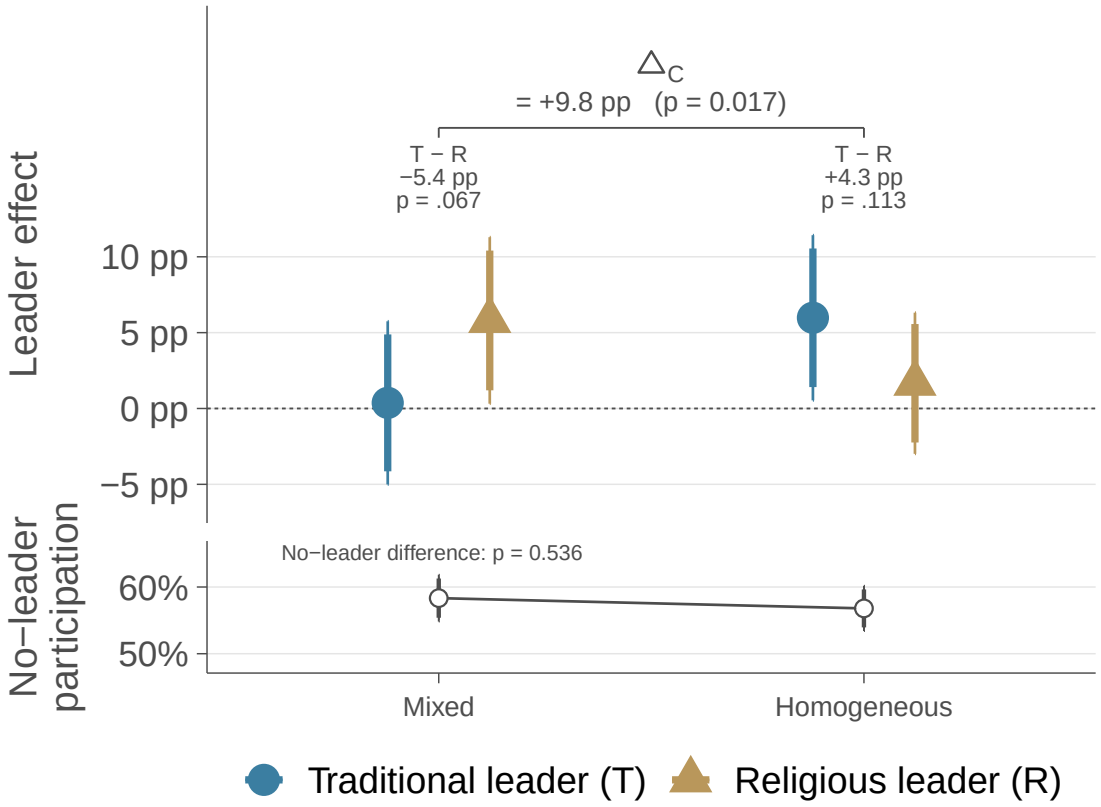


Figure 3: LEADER EFFECTS BY GROUP COMPOSITION

Notes: Blue circles and gold triangles show how assigning a traditional leader or religious leader changes participation relative to no leader, pooling across disclosure conditions. The upper panel compares mixed and homogeneous groups; T–R labels compare the leaders directly, and Δ_C measures the change in this gap from mixed to homogeneous groups. Open circles below show no-leader participation. Thick and thin bars are 90 and 95 percent confidence intervals. Estimates adjust for pretreatment participation, 25-meter spillover exposure, and 11 LASSO-selected baseline controls and compare groups within the same neighborhood, composition, and disclosure condition. Section 4.2 discusses the figure; Appendix Table C-3 reports the estimates and specification checks.

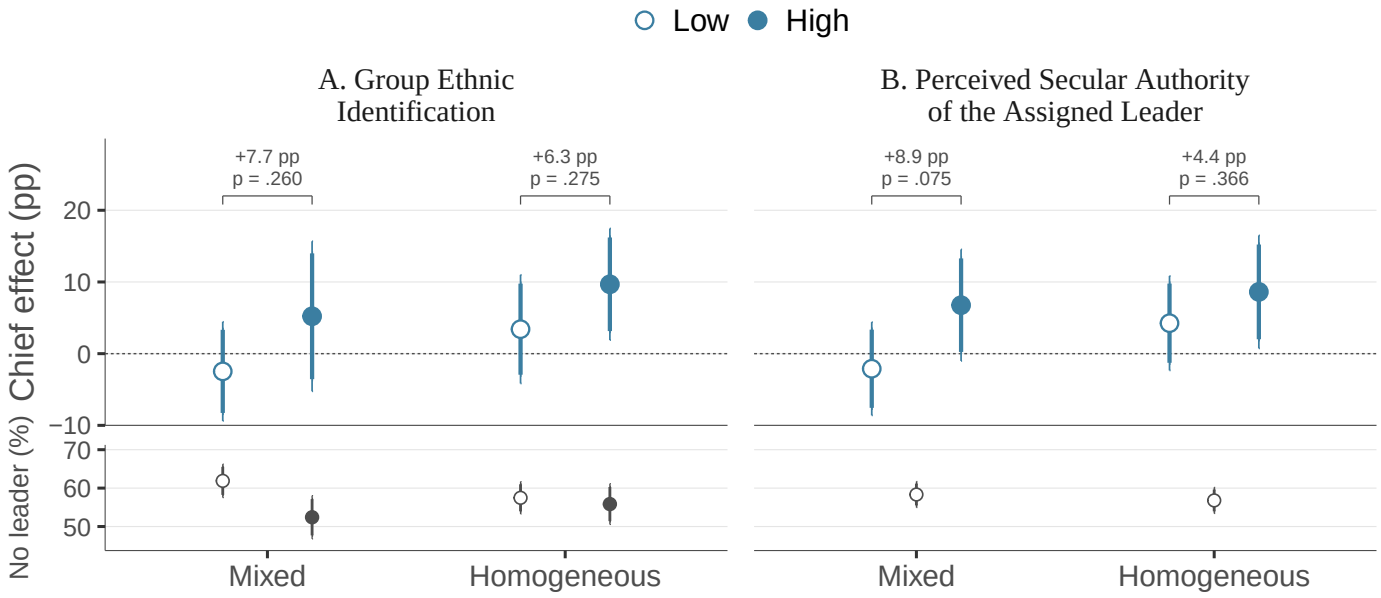


Figure 4: ETHNIC IDENTIFICATION, RECOGNIZED AUTHORITY, AND CHIEF EFFECTIVENESS

Notes: Panel A compares groups above and at or below the group median of ethnic identification, 10/3. Before treatment, residents ranked their identities; own-territory scores run from 1–6, with higher scores indicating greater importance. We average the available scores of ordinary members, excluding leaders: three in 330 groups and two in six (1,040 residents). Panel B compares residents at or below the secular-authority cutpoint (4) with those above it. The assessment concerns the assigned leader's capacity to sanction and reward and precedes the first meeting discussion. Missing values for 33 residents are filled with their group's observed mean; three residents without an observed group score are excluded. Open and filled markers show the lower and higher categories. Lower plots average no-leader predictions within each observed identity category in Panel A and pool them within composition in Panel B, where assessments are undefined without a leader. Blue circles report traditional leader effects relative to no leader, pooling disclosure. Thick and thin bars are 90 and 95 percent group-clustered confidence intervals. Estimates include baseline participation, 25-meter spillover exposure, 11 LASSO-selected baseline controls, and neighborhood-by-composition-by-disclosure fixed effects. Annotations above pairs report higher-minus-lower differences. Section 4.3 discusses the figure; Appendix Tables C-13 and C-15 report both leader types and both specifications for Panels A and B, respectively.

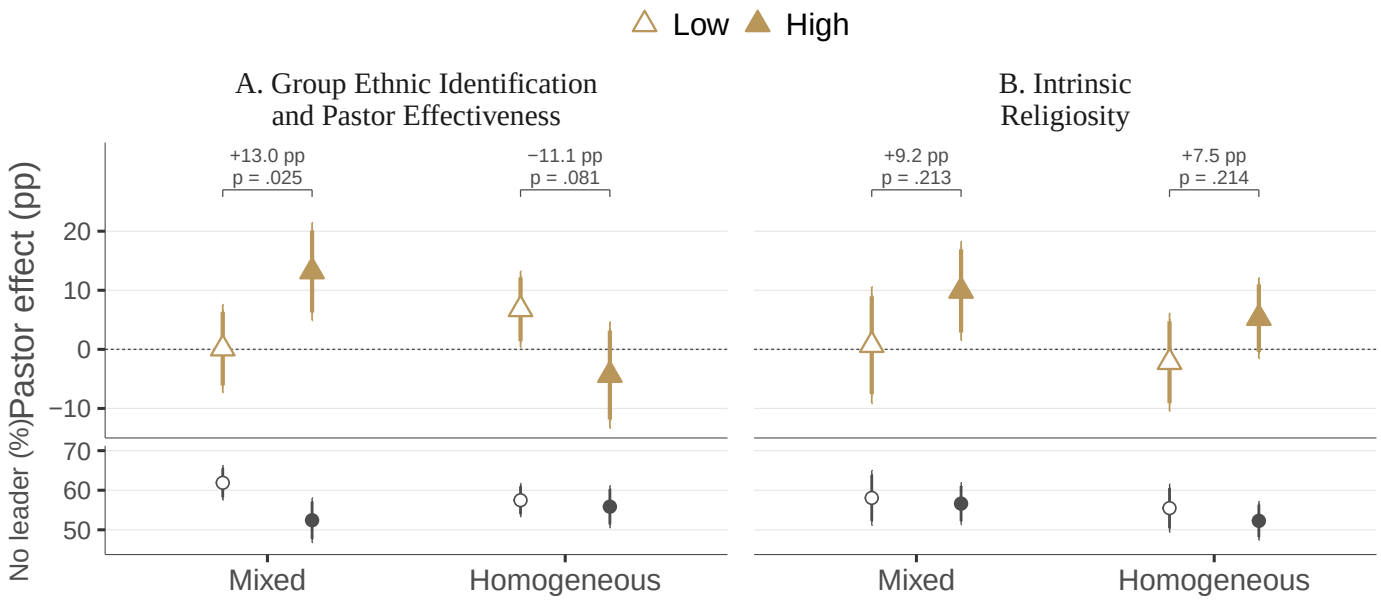


Figure 5: ETHNIC IDENTIFICATION, INTRINSIC RELIGIOSITY, AND PASTOR EFFECTIVENESS

Notes: Panel A compares groups above and at or below the group median of ethnic identification, 10/3. Before treatment, residents ranked their identities; own-territory scores run from 1–6, with higher scores indicating greater importance. We average the available scores of ordinary members, excluding leaders: three in 330 groups and two in six (1,040 residents). Panel B compares residents above and at or below 3.2 on intrinsic religiosity, the cutpoint that best balances category sizes (498 lower, 536 higher). We measure intrinsic religiosity from five questions, asked at baseline, about faith as a guide to life. After reversing two items, we take the average for residents with at least three valid answers, leaving 1,034 residents. For each composition, the lower category is on the left with open markers, and the higher category is on the right with filled markers. Panel A's lower plot averages fitted predictions over actual no-leader residents within each observed composition and identity category; Panel B uses common no-leader predictions. Gold triangles report religious leader effects relative to no leader, pooling disclosure. Thick and thin bars are 90 and 95 percent group-clustered confidence intervals. Estimates include baseline participation, 25-meter spillover exposure, 11 LASSO-selected baseline controls, and neighborhood-by-composition-by-disclosure fixed effects. The numbers above paired markers subtract the lower-category effect from the higher-category effect. We discuss these comparisons in Section 4.3. For both leader types and specifications, see Appendix Table C-13 for Panel A and Table C-17 for Panel B.

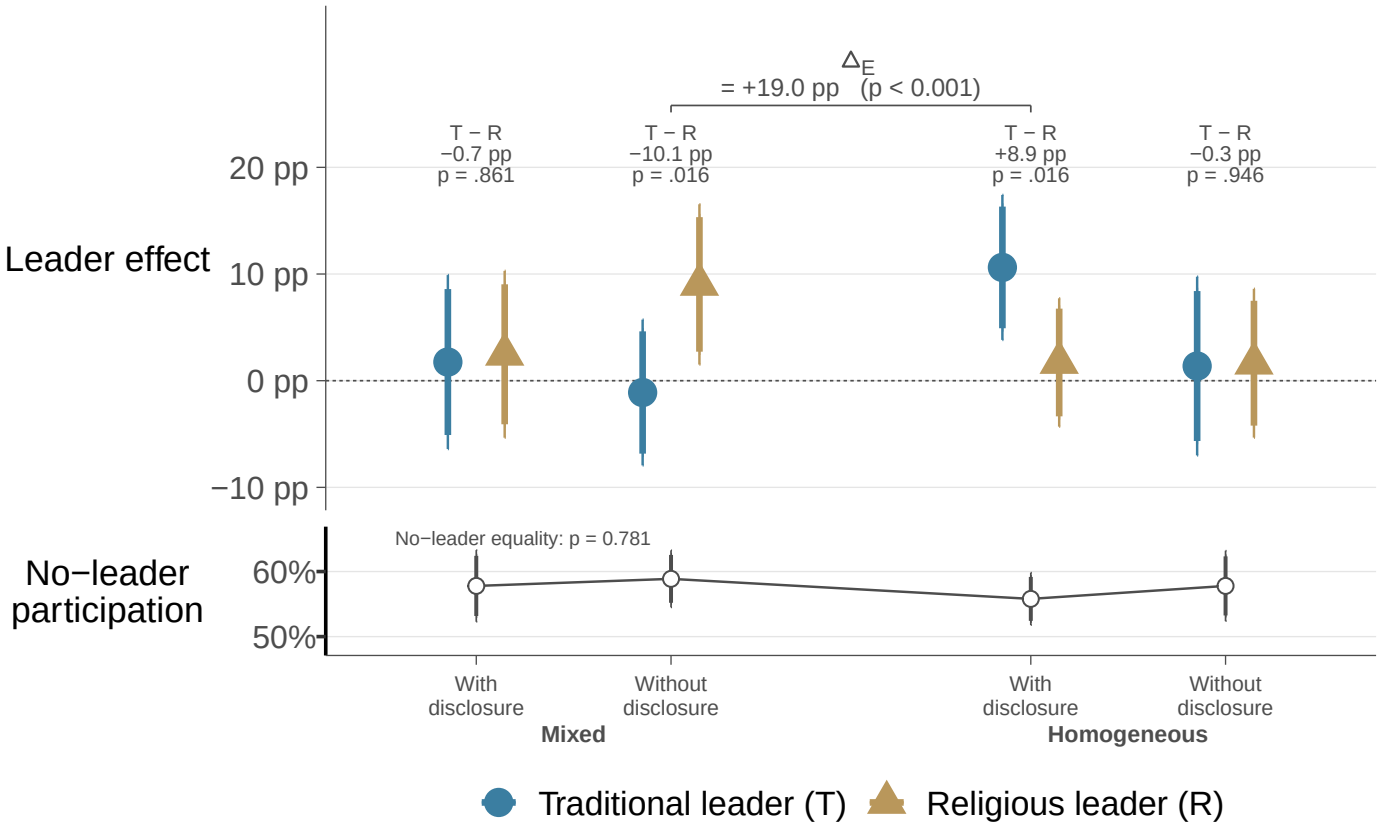


Figure 6: LEADER EFFECTS BY COMPOSITION AND DISCLOSURE

Notes: Blue circles and gold triangles show traditional leader and religious leader effects relative to no leader in each composition-by-disclosure environment. Traditional leaders raise participation most in homogeneous groups with disclosure, and religious leaders in mixed groups without disclosure. T–R labels compare the leaders directly; Δ_E is the difference between their gaps in these two environments. Open circles below show no-leader participation. Thick and thin bars are 90 and 95 percent confidence intervals. Estimates use the same sample, pretreatment participation control, spillover exposure, LASSO-selected baseline controls, and neighborhood-by-composition-by-disclosure fixed effects as Figure 3. Section 5 discusses the figure; Appendix Table D-1 reports the estimates and specification checks.

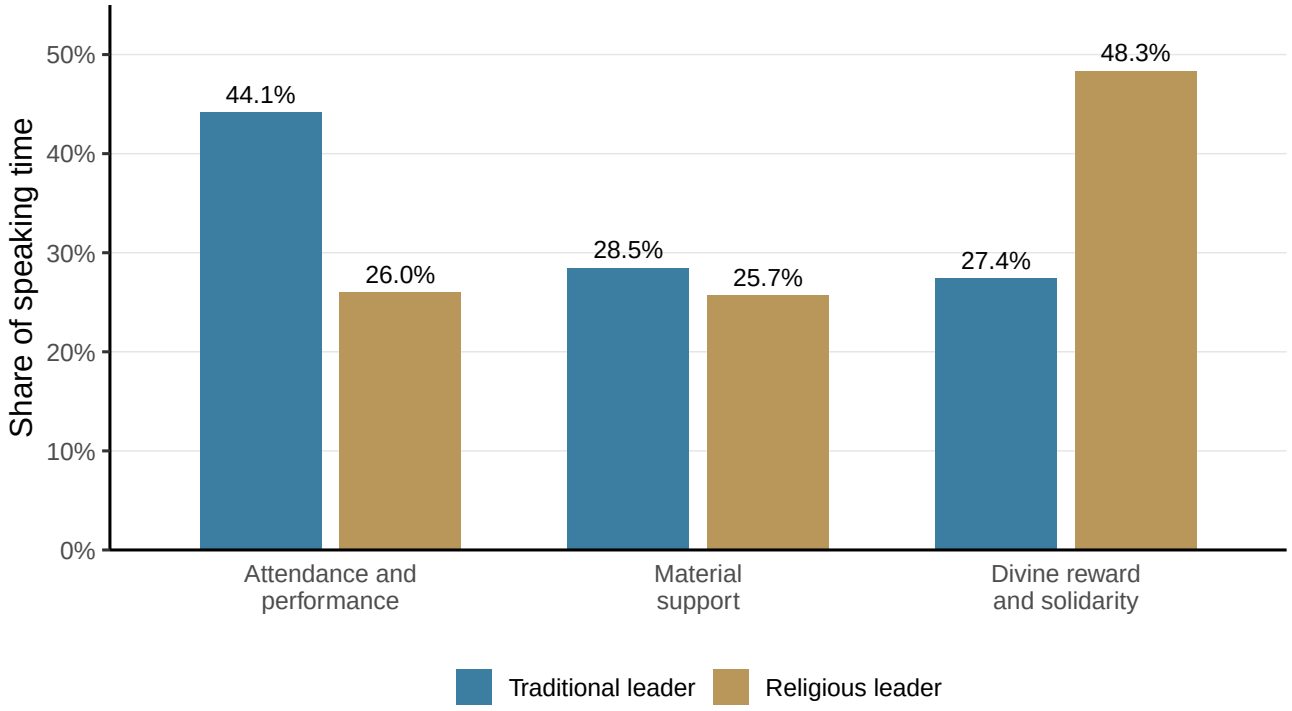


Figure 7: TRADITIONAL LEADERS STRESS PERFORMANCE; RELIGIOUS LEADERS STRESS DIVINE REWARD

Notes: Bars show the mean share of speaking time devoted to each topic during the first recorded meeting, by leader type and pooling the four experimental environments. Each classified passage is assigned to one topic and weighted by its duration. Shares are calculated within each leader-meeting observation and then averaged, giving each observation equal weight. The three shares sum to 100 percent within leader type. Blue bars report traditional leaders and gold bars religious leaders. Section 5.3 discusses the figure. Appendix Table C-5 reports speaking time and topic shares separately by environment; Appendix Table D-10 gives characteristic words and translated examples.

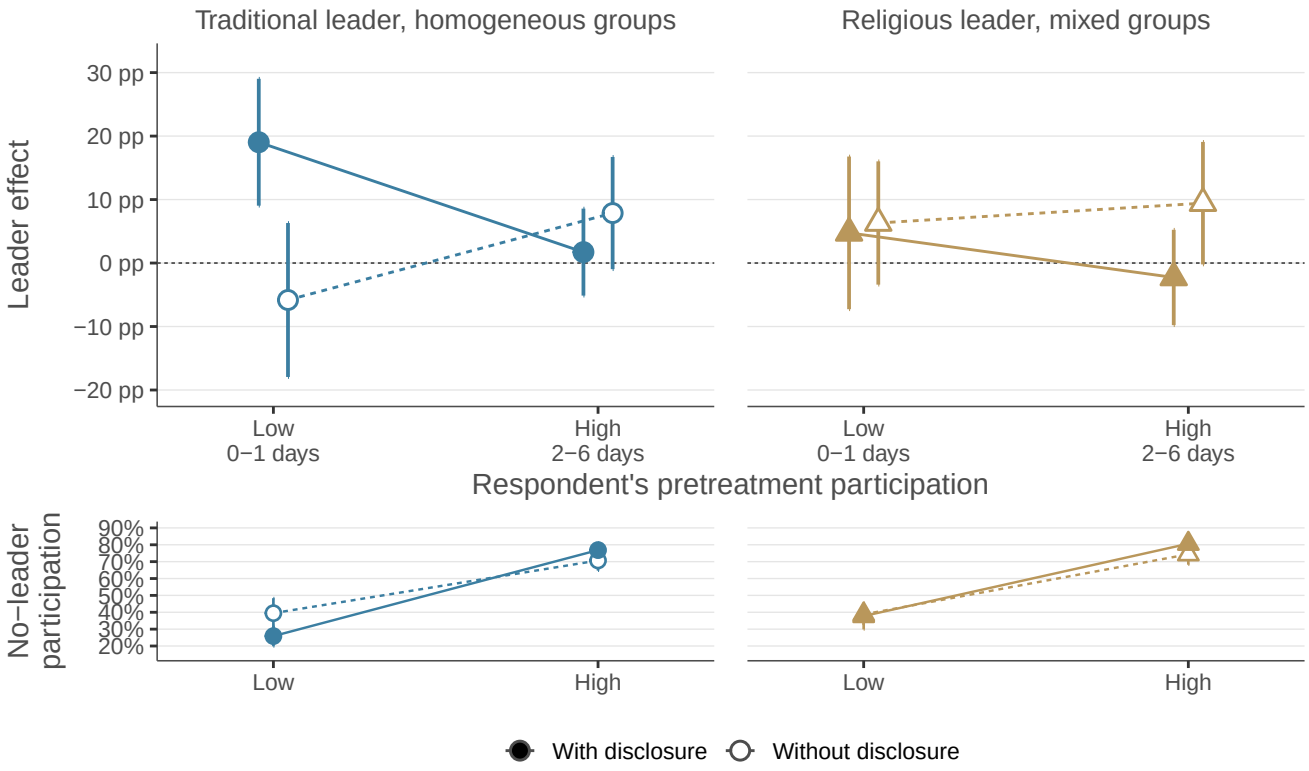


Figure 8: TRADITIONAL LEADERS DRAW IN PRIOR NONPARTICIPANTS UNDER DISCLOSURE; RELIGIOUS LEADER EFFECTS SPAN PARTICIPATION HISTORIES

Notes: Read each column from top to bottom. The upper panels compare leader effects among residents with lower and higher pretreatment participation. Below them, we show predicted participation without a leader for the same types of residents. The left column reports traditional leaders in homogeneous groups and the right column religious leaders in mixed groups; blue circles and gold triangles distinguish leader types. Filled markers and solid lines indicate disclosure; open markers and dashed lines indicate no disclosure. Lower participation means zero or one of the six eligible pre-meeting days, and higher participation means two to six days. With disclosure, the traditional leader effect is concentrated among prior nonparticipants and infrequent participants. Without disclosure, the religious leader effect extends across both participation groups. The no-leader means help distinguish differences in leader influence from differences in participation without a leader. Estimates include baseline participation, 25-meter spillover exposure, 11 LASSO-selected baseline controls, and neighborhood-by-composition-by-disclosure fixed effects. Lines are 90 percent confidence intervals. The figure is discussed in Section 5.4; Appendix Table D-11 reports the exact plotted specification and direct disclosure contrasts.

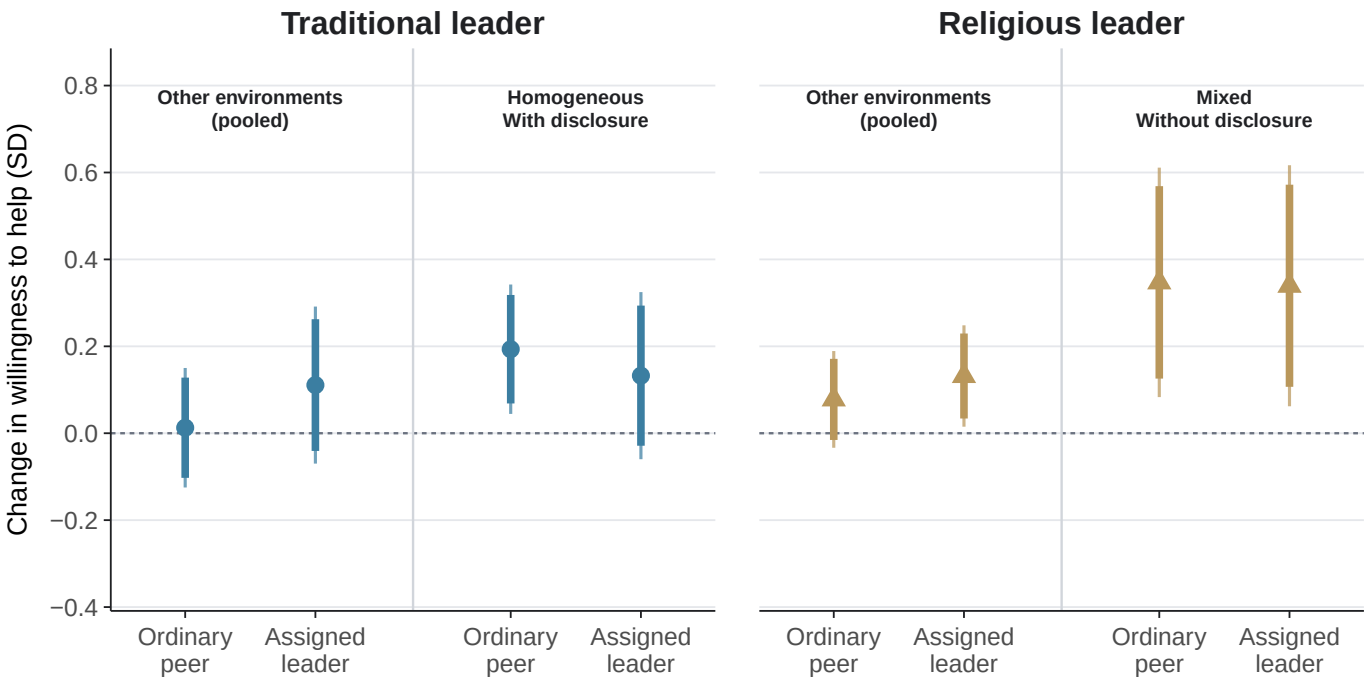


Figure 9: SUCCESSFUL MOBILIZATION STRENGTHENS DIFFERENT RELATIONSHIPS WITHIN THE GROUP

Notes: Before the first discussion and again at the start of Meeting 2, residents reported how much they would help a named group member. The figure shows how leader assignment changes willingness to help over time. The left panel reports traditional leaders and the right panel religious leaders. Within each panel, the right block isolates the environment where that leader raises participation; the left block pools the other three environments. The named person is either an ordinary peer or the assigned leader. Blue circles report traditional leader effects and gold triangles religious leader effects. Effects are measured in baseline standard deviations of the help measure; thick and thin lines are 90 and 95 percent confidence intervals. Willingness-to-help estimates include the 25-meter spillover indicator; outcome-specific LASSO selected no additional baseline controls. Section 6 discusses the figure. Appendix Tables E-4 and E-5 report all outcomes and environments. Panel B of those tables estimates each environment separately, whereas this figure pools the other three environments separately for each leader type; the resulting coefficients therefore differ.

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A Context and Leader Characteristics

Table A-1 – LEADER COUNTS AND RECRUITMENT POPULATION SHARES

	Leader counts		Population
	Religious leaders	Traditional leaders	Share (%)
Panel A. Origin territories			
Dibaya	12 (42.9)	14 (50.0)	36.8
Demba	9 (32.1)	7 (25.0)	25.9
Kazumba	2 (7.1)	5 (17.9)	18.2
Dimbelenge	3 (10.7)	1 (3.6)	5.7
Kananga	1 (3.6)	1 (3.6)	5.3
Other / Not Reported	1 (3.6)	0 (0.0)	8.2
Total	28 (100.0)	28 (100.0)	100.0
Panel B. Main denominations			
Pentecostal, autonomous	5 (17.9)	6 (21.4)	22.8
Pentecostal, Branham	3 (10.7)	3 (10.7)	17.7
Catholic	0 (0.0)	4 (14.3)	11.8
Neo-Apostolic	4 (14.3)	4 (14.3)	10.8
Pentecostal, network	4 (14.3)	0 (0.0)	9.1
Kimbanguist	7 (25.0)	2 (7.1)	6.7
Traditional religion	1 (3.6)	1 (3.6)	5.3
Adventist	0 (0.0)	2 (7.1)	4.1
Protestant/Presbyterian	2 (7.1)	2 (7.1)	4.0
Other / not reported	2 (7.1)	4 (14.3)	7.8
Total	28 (100.0)	28 (100.0)	100.0

Notes: How similar are the leaders to residents recruited in the same neighborhoods? We compare individual records in the full recruitment sample, restricting neighborhoods to final assignment Waves 1–7. The leader columns count each assigned person once and show within-column percentages in parentheses. In the final assignment files, these are the people for whom `leader_religious` or `leader_traditional` equals one. Population percentages use the $N = 2,795$ recruited residents with usable neighborhood information. Territory is their reported origin territory; denomination comes from the coded religion/denomination item, rather than the open-text church name. Other / not reported combines rows with less than 4 percent of the recruitment population.

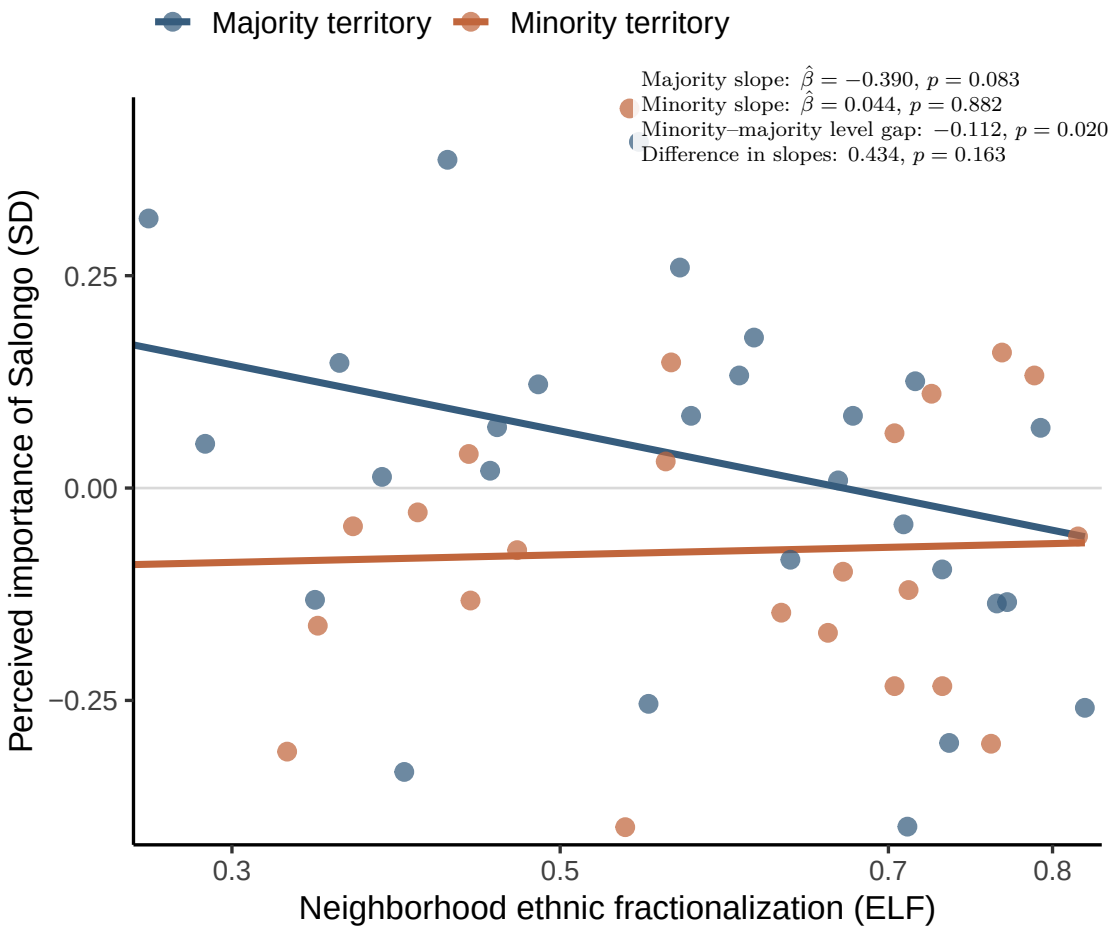


Figure A-1: Neighborhood ethnic fractionalization and perceived importance of *salongo*

Notes: The sample contains 2,791 recruitment-survey respondents in the 28 nonpilot neighborhoods. Neighborhood ethnic fractionalization is the leave-one-out territory-based ELF, $1 - \sum_g s_g^2$, computed from all respondents in the recruitment roster with nonmissing neighborhood and territory. The outcome is the respondent's perceived importance of *salongo*, standardized once over the full estimation sample. The question reads, "How important do people in your community think it is to participate in Salongo?" Responses use a five-point ordered scale anchored at 1, "They think it is not important at all," and 5, "They think it is very important"; values 2–4 are unlabeled intermediate points. Minority status indicates that the respondent's territory is not the modal territory in the neighborhood. The fitted lines come from one individual-level regression of standardized *salongo* importance on mean-centered ELF, minority status, and their interaction, with no additional controls. Standard errors and the displayed tests are clustered by neighborhood. Points are neighborhood-by-status outcome means; the lines are fitted values from the individual-level model. The level gap is evaluated at mean ELF.

Table A-2 – LEADER SELECTION AND INSTITUTIONAL EMBEDDEDNESS

	Observations (1)	Citizen mean (2)	Coeff. on religious leader (3)	Coeff. on traditional leader (4)	p-value of diff., columns 3–4 (5)
Panel A. Demographic and economic selection					
Female	2,681	0.603	**-0.590**	**-0.448**	**0.043**
			[**0.000**]	[**0.000**]	
Age	2,681	34.098	**13.289**	**22.253**	**0.035**
			[**0.000**]	[**0.000**]	
Education (yrs)	2,681	9.531	**1.765**	**1.408**	0.662
			[**0.001**]	[**0.020**]	
Non-clergy salaried job	2,681	0.037	0.033	0.105	0.424
			[0.519]	[0.126]	
Government job	2,681	0.006	0.065	0.030	0.573
			[0.198]	[0.416]	
Log income	1,354	11.558	0.106	**0.417**	0.499
			[0.805]	[**0.007**]	
Avenue tenure	1,327	15.032	**12.880**	**12.487**	0.937
			[**0.002**]	[**0.002**]	
Panel B. Institutional embeddedness					
Traditional authority and local embeddedness					
Any traditional role	1,362	0.047	**0.346**	**0.953**	**0.000**
			[**0.001**]	[**0.000**]	
Traditional head role	1,362	0.026	**0.260**	**0.617**	**0.005**
			[**0.006**]	[**0.000**]	
Secrets network: same territory	1,362	0.295	0.073	**0.163**	0.459
			[0.360]	[**0.039**]	
Material network: same territory	1,362	0.425	0.037	**0.134**	0.311
			[0.632]	[**0.043**]	
Planned salongo hours	1,350	16.807	**6.837**	**10.051**	0.296
			[**0.036**]	[**0.000**]	
Religious authority and church embeddedness					
Any religious role	1,362	0.139	**0.862**	0.147	**0.000**
			[**0.000**]	[0.090]	
Religious head role	1,362	0.053	**0.662**	**0.197**	**0.001**
			[**0.000**]	[**0.024**]	
Secrets network: same church	1,362	0.177	0.049	-0.004	0.601
			[0.471]	[0.956]	
Material network: same church	1,362	0.223	0.091	**-0.128**	**0.008**
			[0.126]	[**0.006**]	
Sunday attendance	1,362	3.466	**0.319**	0.176	0.381
			[**0.034**]	[0.113]	
Weekday attendance	1,362	1.590	**0.877**	**1.055**	0.742
			[**0.030**]	[**0.002**]	
Log religious contribution	1,317	6.244	**1.482**	0.882	0.296
			[**0.000**]	[0.089]	
Knowledge of local authorities					
Knows authorities	1,362	0.292	**0.315**	**0.434**	0.087
			[**0.000**]	[**0.000**]	
Knows quartier chief	1,362	0.309	**0.298**	**0.477**	0.057
			[**0.001**]	[**0.000**]	
Knows locality chief	1,362	0.352	**0.256**	**0.470**	0.083
			[**0.008**]	[**0.000**]	
Knows chief phone	1,362	0.071	**0.322**	**0.358**	0.802
			[**0.001**]	[**0.001**]	
Chief help index	1,362	1.499	0.170	**0.705**	0.067
			[0.491]	[**0.003**]	
Panel C. Psychological traits					
Standard psychological measures					
Big Five Inventory (main index)	1,362	-0.005	0.019	**0.226**	0.086
			[0.838]	[**0.025**]	
Extraversion	1,358	3.210	-0.013	0.112	0.346
			[0.884]	[0.298]	
Agreeableness	1,361	3.844	0.030	0.120	0.593
			[0.790]	[0.255]	
Conscientiousness	1,360	3.905	0.023	**0.273**	0.085
			[0.846]	[**0.004**]	
Neuroticism	1,362	2.872	-0.104	-0.175	0.659
			[0.464]	[0.116]	
Openness	1,362	3.192	-0.085	0.058	0.476
			[0.497]	[0.733]	
Emotional intelligence	1,362	4.296	-0.014	0.007	0.803
			[0.870]	[0.936]	
Locus of control	1,362	4.069	-0.140	0.324	0.091
			[0.612]	[0.091]	
Social desirability	1,362	0.560	**0.055**	0.025	0.322
			[**0.022**]	[0.278]	
Religious psychological measures					
Positive RCOPE	1,362	3.794	**0.106**	-0.073	0.073
			[**0.012**]	[0.448]	
Negative RCOPE	1,362	3.540	**-0.605**	-0.162	0.115
			[**0.005**]	[0.249]	
Intrinsic religiosity	1,362	3.326	0.109	0.009	0.297
			[0.077]	[0.888]	
Extrinsic religiosity: personal	1,358	4.416	-0.060	-0.024	0.839
			[0.586]	[0.863]	
Extrinsic religiosity: social	1,362	2.678	-0.320	0.133	0.079
			[0.094]	[0.491]	

Notes: We compare each leader type with ordinary residents of the same locality. Each row reports a separate respondent-level regression of the listed characteristic on religious- and traditional-leader indicators and locality fixed effects. Locality-clustered p-values are reported in square brackets. Column 5 reports the p-value for equality of the two leader coefficients. We identify the leaders selected to join the experimental teams from the selected meeting/panel leader IDs. The first five demographic and employment rows use the recruitment survey (N = 2,681). The remaining rows combine the study panel with the capped sample of additional baseline-control respondents, giving up to 1,362 observations depending on data availability. Network shares code missing network counts as zero and code shares as zero for respondents with a zero-size network.

Table A-3 – SOCIAL PROXIMITY AND PERCEIVED AUTHORITY

Outcome	N	Non-leader		Traditional leader		Religious leader	
		Different ethnicity mean	Same-ethnicity premium	Different ethnicity	Same ethnicity	Different ethnicity	Same ethnicity
Panel A. Initial Social Cohesion and Pre-Existing Ties							
Know							
Know other person	2,894	2.938	**0.131**	**0.383**	**0.466**	**0.294**	**0.491**
			[0.042]	[0.000]	[0.000]	[0.000]	[0.000]
Pre-existing ties							
Same church	2,894	0.024	0.019	0.003	0.039	0.036	0.041
			[0.061]	[0.882]	[0.071]	[0.056]	[0.064]
Same groupement	2,894	0.000	**0.057**	-0.003	**0.087**	-0.001	**0.048**
			[0.000]	[0.402]	[0.000]	[0.831]	[0.005]
Same village	2,894	0.000	**0.046**	-0.003	**0.060**	0.001	**0.048**
			[0.000]	[0.321]	[0.002]	[0.760]	[0.005]
Close family	2,894	0.010	0.013	-0.002	0.024	0.002	0.013
			[0.076]	[0.806]	[0.176]	[0.799]	[0.323]
Extended family	2,894	0.017	0.020	0.008	**0.050**	0.001	0.033
			[0.060]	[0.608]	[0.024]	[0.924]	[0.093]
Reciprocity							
Mutual sickness support	2,894	1.157	**0.168**	**0.309**	**0.310**	0.163	**0.242**
			[0.042]	[0.005]	[0.001]	[0.111]	[0.038]
Willing to help them	2,894	1.268	**0.168**	0.179	**0.241**	0.103	0.141
			[0.033]	[0.077]	[0.017]	[0.304]	[0.209]
Expected help from them	2,894	1.056	0.168	**0.438**	**0.379**	0.224	**0.343**
			[0.094]	[0.002]	[0.003]	[0.058]	[0.011]
Panel B. Opinions and Authority							
Motivates respondent	2,894	4.041	0.112	0.173	**0.393**	**0.239**	**0.329**
			[0.057]	[0.051]	[0.000]	[0.001]	[0.000]
Moral example	2,894	4.119	**0.160**	0.167	**0.352**	**0.243**	**0.293**
			[0.005]	[0.056]	[0.000]	[0.000]	[0.000]
Trustworthy	2,894	3.845	0.091	0.118	**0.344**	**0.254**	**0.321**
			[0.138]	[0.158]	[0.000]	[0.000]	[0.000]
Hardworking	2,894	4.043	0.046	0.119	**0.340**	**0.301**	**0.268**
			[0.415]	[0.215]	[0.000]	[0.000]	[0.001]
Seen as authority	2,894	3.586	0.084	**0.212**	**0.253**	0.117	**0.210**
			[0.134]	[0.002]	[0.000]	[0.092]	[0.016]
Panel C. Participation and Material Expectations							
Trust they will join	2,894	3.160	0.098	**0.224**	**0.330**	**0.240**	**0.246**
			[0.067]	[0.004]	[0.000]	[0.000]	[0.004]
Expected money contribution	2,485	0.109	-0.054	**1.056**	**0.698**	**0.939**	**1.209**
			[0.807]	[0.002]	[0.014]	[0.001]	[0.001]
Expected work days	2,691	3.253	0.071	**0.485**	**0.580**	0.271	0.349
			[0.537]	[0.002]	[0.000]	[0.055]	[0.059]

Notes: Each respondent in this table is an ordinary resident evaluating a particular teammate. We estimate a separate regression for each outcome to compare these evaluations by the teammate's leader status and ethnic ties to the respondent. An evaluation of an ordinary teammate from a different ethnicity is the omitted category. The first data column reports its mean, and coefficients in the remaining columns measure differences from this group. Standard errors allow correlation both within respondent and within accountability group in the same neighborhood. Brackets give two-sided p-values, and coefficients are bold when $p < 0.05$.

For familiarity ("Know other person") and trust in participation, we show comparison means on the original 1–4 scales, while Panel B uses the original 1–5 scales. The coefficients for these ratings are in standard deviations of the relevant outcome. For pre-existing ties, which are binary, they are in probability units. We transform monetary answers in CDF using the inverse hyperbolic sine and estimate coefficients on that scale. To express comparison means in USD, we take the hyperbolic sine of mean transformed CDF and divide by 2250. The conversion is $\sinh(\overline{\text{asinh}(\text{CDF})})/2250$. To measure mutual sickness support, we take the mean of the two help amounts after transforming them, then use this mean for estimation and back-transformation. Expected work is in days; we convert earlier answers given in hours or minutes using six hours for each day.

All regressions include neighborhood fixed effects. We select controls separately for each outcome, after removing neighborhood means, using post-double-selection LASSO. A control is kept if it is selected for the outcome or for the coethnicity and leader regressors. We let LASSO choose from the respondent's baseline social-desirability index, sex, age, education, wealth stratum, territory, and religion; indicators for whether respondent and teammate match in location, sex, age, education, and wealth; and agreement between the respondent's report of the teammate's religion and the teammate's recorded religion.

Table A-4 – TERRITORY-OF-ORIGIN COMPOSITION OF RECOVERED CHURCH CONGREGATIONS

	Full recruitment roster		Randomized study sample	
	Exact normalized	Strict fuzzy	Exact normalized	Strict fuzzy
Recovered church clusters	66	71	10	12
Observed congregants in recovered clusters	513	555	67	79
Church clusters spanning multiple origin territories	60	64	10	12
Share of church clusters spanning multiple territories	90.9%	90.1%	100.0%	100.0%
Share of congregants in those church clusters	91.6%	91.4%	100.0%	100.0%
Same-territory pairs within church clusters	48.2%	48.4%	53.2%	55.4%
Expected same-territory pairs under no sorting	48.8%	48.5%	53.6%	55.2%
Observed minus expected same-territory pairs	-0.6%	-0.1%	-0.4%	0.2%
Permutation p-value: positive territorial sorting	0.696	0.570	0.566	0.485

Notes: Following the experimental design, ethnic identity is measured using respondents' reported territory of origin; two respondents are coethnic when their normalized territories match exactly. The analysis uses Christian respondents with a valid territory, neighborhood, and free-text church name in the cumulative waves 1–7 recruitment roster. To identify congregations, we compare church names only among residents of the same neighborhood and denomination. We first match identical sets of name tokens after lowercasing, removing accents and punctuation, standardizing saint abbreviations, and dropping generic church and denomination words. We exclude names that provide no information for identifying a congregation. We also allow close spelling matches to reference names ranked by frequency, using Jaro–Winkler similarity of at least 0.97 (the maximum over original and sorted token order). Single-token names must both have at least five characters and similarity of at least 0.99; multi-token names with different token counts also require token Jaccard similarity of at least 0.80. Names with conflicting numeric identifiers cannot match. A match must link directly to the reference name, without passing through intermediate names. The table retains clusters with at least five observed congregants. The no-sorting benchmark reallocates territories within blocks while preserving recovered church sizes; the one-sided permutation test uses 999 draws. Cluster size is the number observed in recruitment, not total congregation size. Neighborhood blocking can split congregations drawing members from multiple neighborhoods.

B Experimental Design and Data

B.1 Fieldwork continuation after Wave 8

On Friday, May 15, 2026, the Democratic Republic of the Congo declared a Bundibugyo Ebola outbreak after laboratory confirmation in Ituri Province; on May 17, the World Health Organization (WHO) determined that the event constituted a Public Health Emergency of International Concern. Wave 8 was underway when the outbreak happened. Waves 9 and 10 were scheduled for the upcoming months of June, July, and August and would have required the repeated convening of new groups for collective work and program meetings. The institutional response to the outbreak underscored the risks of initiating such activities. WHO recommended considering the postponement of mass gatherings; UN staff in Bunia were instructed to work from home and avoid physical contact and crowded areas; the U.S. Department of State temporarily suspended all visa services at its embassy in Kinshasa; and the African Development Bank introduced remote participation for its Annual Meetings in neighboring Congo while explicitly monitoring risks from the outbreak, despite no confirmed or suspected cases in the host country. Initiating additional waves under these conditions was unwarranted and posed severe ethical risks. We therefore decided to stop the study on May 30, 2026, interrupting the implementation of the upcoming cohorts.[25] The emergency overlapped Wave 8 implementation. At the time of the declaration, Wave 8 was only 30.4 percent complete.

We therefore use Waves 1–7 for the main analysis and report estimates that add the disrupted Wave 8 here. Stopping fieldwork raises two questions. Were the neighborhoods reached before the emergency different from those scheduled later? And how much precision did we lose? We first compare predetermined neighborhood characteristics (Section B.1.1), then reassess power under the planned ten-wave design and the seven completed waves (Section B.1.2). We finally compare the estimates from Waves 1–7 and Waves 1–8.

B.1.1 Predetermined neighborhood characteristics

The ten-wave rollout order was randomly assigned before implementation. Table B-1 compares property values, building size and counts, and commune composition for neighborhoods assigned to Waves 1–7 and Waves 8–10.

None of the 12 characteristic-specific tests rejects at the 5 percent level. The exact max-statistic test across all 12 characteristics yields $p = 0.787$. We find no systematic differences in the observed characteristics of early- and late-wave neighborhoods. These tests have limited power, however, and cannot rule out differences in characteristics we did not measure. Within each realized neighborhood, treatment remains randomized, and the specifications absorb neighborhood × disclosure × composition fixed effects.

B.1.2 Power under ten planned waves and seven pre-declaration waves

Figure B-1 reassesses expected precision for the planned ten-wave design and for Waves 1–7, the sample completed before the national declaration. The calibration data come from the municipal field pilot conducted before July 1, 2025, well before the current experiment. The sample contains 251 participants from four pilot neighborhoods with a six-day baseline week followed by a separate six-day outcome week. Ngdilayi, an earlier pilot neighborhood, is excluded because its calendar lacks this consecutive-week structure. The preregistration set a minimum detectable effect (MDE) of 0.20 outcome standard deviations and planned 40 neighborhoods and 480 groups. We retain that effect-size threshold and use the group sizes implemented in each wave: no-leader groups contain four ordinary participants in Waves 1–2 and three thereafter, while traditional- and religious-leader groups contain three ordinary participants in every wave. The seven-wave design therefore has 1,040 participant observations in 336 groups, and the ten-wave design has 1,472 observations in 480 groups. Every simulation estimates the paper's model with neighborhood × disclosure × composition fixed effects and baseline participation; standard errors are clustered by group within neighborhood.

At the preregistered MDE of 0.20 outcome standard deviations (7.95 percentage points), power for the T_H and R_M leader effects is 0.902 under ten waves and ranges from 0.791 to 0.793 under seven waves. For the T_{H1} and R_{M0} effects, the corresponding ranges are 0.646 to 0.647 and 0.514 to 0.518. Power for the composition contrast Δ_C is 0.620 under ten waves and 0.478 under seven waves. Power for the environment contrast Δ_E is 0.364 and 0.276 under the two designs. Across the four leader effects and two delta contrasts, moving from ten to seven waves raises simulated standard errors by 17–19 percent. The ten-wave plan would

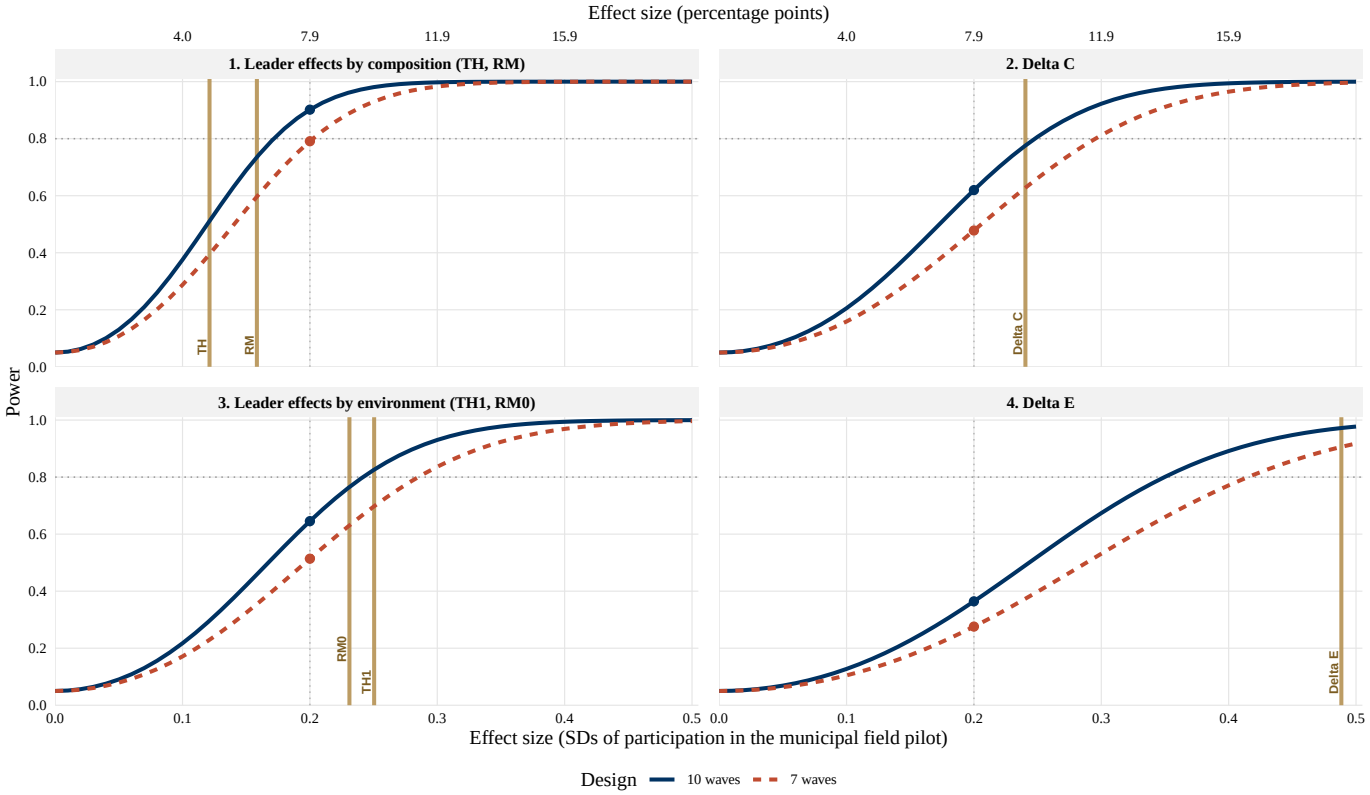
[25] WHO, Disease Outbreak News, May 16, 2026; WHO, PHEIC statement, May 17, 2026; Associated Press, May 18, 2026; U.S. Department of State, May 18, 2026; African Development Bank, May 21, 2026; and WHO, Disease Outbreak News, July 3, 2026.

Table B-1 – PREDETERMINED NEIGHBORHOOD CHARACTERISTICS BY PLANNED WAVE

	Waves 1–7	Waves 8–10	Difference	Std. diff.	RI p
Mean predicted property value (USD)	2612.09	2183.94	-428.15	-0.47	0.358
Median predicted property value (USD)	1359.32	1165.25	-194.07	-0.66	0.187
Mean building area	39.70	38.13	-1.57	-0.34	0.518
Median building area	34.59	33.50	-1.09	-0.35	0.496
Total built area	19261.66	18624.36	-637.30	-0.12	0.810
Number of main properties	313.86	322.00	8.14	0.15	0.769
Number of buildings	484.82	473.17	-11.65	-0.12	0.803
Commune: Kananga (share)	0.11	0.25	0.14	0.50	0.429
Commune: Katoka (share)	0.21	0.08	-0.13	-0.39	0.500
Commune: Lukonga (share)	0.36	0.42	0.06	0.13	1.000
Commune: Ndesha (share)	0.07	0.08	0.01	0.06	1.000
Commune: Nganza (share)	0.25	0.17	-0.08	-0.20	0.868
Joint max-statistic test			Maximum |standardized difference| = 0.66		0.787

Notes: The sample contains the 40 treated neighborhoods assigned ex ante to ten rollout waves: 28 neighborhoods in Waves 1–7 and 12 in Waves 8–10. Each row reports the early-wave mean, late-wave mean, their difference (late minus early), the difference standardized by the across-pair standard deviation, and a two-sided exact randomization-inference p-value. Wave assignment occurred at the neighborhood-pair level. Commune rows are neighborhood shares. The final row reports the randomization p-value for the maximum absolute standardized difference across all 12 predetermined characteristics.

have improved precision. Power at the 0.20-SD effect size would remain below 80 percent for T_{H1}, R_{M0}, Δ_C, and Δ_E. The seven-wave curves correspond to the Waves 1–7 main analysis. Below, we report the same effects after adding all Wave 8 observations. Gold bars show the magnitudes of the Waves 1–7 estimates for scale.



Dotted line: preregistered MDE benchmark (0.20 SD); gold bars: magnitudes of the labeled Waves 1–7 estimates

Figure B-1 – POWER FOR LEADER EFFECTS AND THE PAPER'S COMPOSITION AND ENVIRONMENT CONTRASTS

Notes: T_H is the traditional-leader effect in homogeneous groups, and R_M is the religious-leader effect in mixed groups. T_{H1} is the traditional-leader effect in homogeneous groups under disclosure, and R_{M0} is the religious-leader effect in mixed groups without disclosure. The four panels report power for (1) T_H and R_M, (2) $\Delta_C = (\tau_H^T - \tau_M^T) - (\tau_H^R - \tau_M^R)$, (3) T_{H1} and R_{M0}, and (4) $\Delta_E = (\tau_{H1}^T - \tau_{M0}^T) - (\tau_{H1}^R - \tau_{M0}^R)$. Panels 1 and 3 plot the lower-power curve across their two stated cells; Panels 2 and 4 use the full clustered covariance matrix for the stated linear contrast. The bottom axis reports standardized effects; the top axis translates them into participation percentage points using the outcome standard deviation from the municipal field pilot. The horizontal dotted line marks 80% power, and the vertical dotted line marks the preregistered 0.20-standard-deviation MDE benchmark (7.95 percentage points). Gold bars mark the magnitudes of the Waves 1–7 T_H and R_M estimates in Panel 1, Δ_C in Panel 2, T_{H1} and R_{M0} in Panel 3, and Δ_E in Panel 4. The ten- and seven-wave SEs are, respectively, 0.061 SD (2.42–2.43 percentage points) and 0.072 SD (2.85–2.86 percentage points) for T_H and R_M; 0.088 SD (3.51 percentage points) and 0.105 SD (4.17 percentage points) for Δ_C; 0.086 SD (3.41 percentage points) and 0.101 SD (4.00–4.02 percentage points) for T_{H1} and R_{M0}; and 0.124 SD (4.94 percentage points) and 0.148 SD (5.86 percentage points) for Δ_E. The outcome and pretreatment covariate are separate consecutive six-day participation weeks drawn from the municipal field pilot conducted before July 1, 2025. The simulation applies the group sizes implemented in the randomization scripts and uses 1,000 bootstrap designs; the largest Monte Carlo standard error among the reported power calculations is 0.003. Every simulated ANCOVA absorbs neighborhood × disclosure × composition fixed effects, identifies leader effects within the corresponding fixed-effect cell, and controls for baseline-week participation. Standard errors cluster by group within neighborhood using the paper's small-sample correction.

B.1.3 Comparison of Waves 1–7 and Waves 1–8

Table B-2 – ENVIRONMENT-SPECIFIC ESTIMATES IN WAVES 1–7 AND WAVES 1–8

	Waves 1–7			Waves 1–8		
	Baseline adjustment	+ Spillover	+ LASSO	Baseline adjustment	+ Spillover	+ LASSO
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed, without disclosure	-0.29	-0.31	-1.11	0.45	0.45	0.07
	(3.48)	(3.48)	(3.48)	(3.39)	(3.39)	(3.38)
Mixed, with disclosure	1.75	1.67	1.75	5.53	5.35	5.68
	(4.53)	(4.52)	(4.16)	(4.32)	(4.32)	(4.08)
Homogeneous, without disclosure	-0.31	-0.34	1.37	-3.47	-3.65	-2.30
	(4.46)	(4.45)	(4.27)	(4.43)	(4.44)	(4.37)
Homogeneous, with disclosure	9.95***	10.13***	10.62***	7.63**	7.89**	8.41**
	(3.63)	(3.63)	(3.47)	(3.73)	(3.72)	(3.54)
Religious leader						
Mixed, without disclosure	9.18**	9.16**	9.02**	6.80*	6.88*	7.03*
	(3.97)	(3.98)	(3.83)	(3.78)	(3.79)	(3.65)
Mixed, with disclosure	3.41	3.40	2.48	5.57	5.51	4.77
	(4.30)	(4.32)	(3.99)	(4.01)	(4.04)	(3.80)
Homogeneous, without disclosure	1.66	1.67	1.64	3.29	3.21	3.22
	(3.81)	(3.80)	(3.56)	(3.75)	(3.74)	(3.58)
Homogeneous, with disclosure	0.01	-0.01	1.71	-0.72	-0.72	0.70
	(3.27)	(3.26)	(3.07)	(3.13)	(3.13)	(2.95)
Panel B. Treatment-effect contrasts						
Mixed, with: T − R	-1.66	-1.73	-0.73	-0.04	-0.16	0.90
	(4.40)	(4.41)	(4.15)	(4.15)	(4.17)	(4.02)
Mixed, without: T − R	-9.47**	-9.47**	-10.13**	-6.34*	-6.43*	-6.96*
	(3.98)	(4.01)	(4.17)	(3.84)	(3.87)	(4.00)
Homogeneous, with: T − R	9.94**	10.13**	8.91**	8.35**	8.60**	7.71**
	(4.03)	(4.00)	(3.67)	(3.98)	(3.96)	(3.63)
Homogeneous, without: T − R	-1.97	-2.01	-0.27	-6.77	-6.86	-5.52
	(4.13)	(4.12)	(3.90)	(4.30)	(4.28)	(4.21)
Δ_E: H, with − M, without	19.41***	19.60***	19.04***	14.69***	15.04***	14.67***
	(5.67)	(5.69)	(5.57)	(5.55)	(5.56)	(5.41)
Observations	1,040	1,040	1,040	1,184	1,184	1,184
Group-neighborhood clusters	336	336	336	384	384	384
Neighborhoods	28	28	28	32	32	32
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes	Yes	Yes	Yes
Pretreatment participation control	Yes	Yes	Yes	Yes	Yes	Yes
25m spillover control	No	Yes	Yes	No	Yes	Yes
LASSO-selected baseline controls	No	No	Yes	No	No	Yes

Notes: Panels A and B reproduce the environment-specific half of the main results table for Waves 1–7 and the pooled Waves 1–8 sample. Panel C, on the following page, gives no-leader means for all six specifications. Each observation is a respondent's average salongo participation, in percentage points, over the six eligible days after Meeting 1 through the day of Meeting 2 within the assigned treatment cell. Panel A reports leader effects relative to no leader; Panel B reports traditional-minus-religious contrasts and Δ_E. All columns include the ANCOVA pretreatment-participation control and neighborhood × disclosure × composition fixed effects. The spillover columns add the 25m exposure indicator. The columns labeled + LASSO add 11 LASSO-selected baseline controls in both wave windows. Standard errors are clustered by group within neighborhood. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B-2 – ENVIRONMENT-SPECIFIC ESTIMATES IN WAVES 1–7 AND WAVES 1–8 (continued)

	Waves 1–7			Waves 1–8		
	Baseline adjustment	+ Spillover	+ LASSO	Baseline adjustment	+ Spillover	+ LASSO
Panel C. No-leader counterfactual means						
Mixed, without disclosure	58.88	58.88	58.88	57.53	57.53	57.53
	(2.46)	(2.44)	(2.24)	(2.36)	(2.34)	(2.15)
Mixed, with disclosure	57.79	57.79	57.79	54.65	54.65	54.65
	(3.14)	(3.14)	(2.81)	(2.97)	(2.97)	(2.73)
Homogeneous, without disclosure	57.79	57.79	57.79	57.69	57.69	57.69
	(2.91)	(2.91)	(2.76)	(2.74)	(2.75)	(2.64)
Homogeneous, with disclosure	55.80	55.80	55.80	56.09	56.09	56.09
	(1.96)	(1.98)	(2.05)	(1.97)	(1.99)	(2.03)

Notes: These are the model-implied no-leader participation means, in percentage points, for the six specifications on the preceding page. Predictions are averaged over the no-leader observations in each environment in the corresponding column's estimation sample. Parentheses report group-clustered standard errors.

Table B-2 compares Waves 1–7 with all eight collected waves while holding the estimator fixed. The columns labeled + LASSO use the same 11 LASSO-selected baseline controls in both samples, selected in Waves 1–7 and held fixed when adding Wave 8. The Waves 1–8 columns retain the complete scheduled Wave 8 six-day outcome window, including Friday, May 15, and Saturday, May 16.

Across the three specifications, Δ_E is 19.04–19.60 percentage points in Waves 1–7 and 14.67–15.04 percentage points in Waves 1–8 (SE 5.41–5.56; p-values 0.007–0.008). Adding Wave 8 therefore reduces the point estimate by roughly 4.4–4.7 percentage points. The Waves 1–8 estimate remains positive, and its confidence interval excludes zero in every specification. The two component leader effects retain their signs. The traditional-leader effect in homogeneous groups under disclosure, T_{H1}, is 7.63–8.41 percentage points (p-values 0.018–0.041). The religious-leader effect in mixed groups without disclosure, R_{M0}, is 6.80–7.03 percentage points (p-values 0.055–0.073).

B.2 Randomization and Team-Construction Protocol

Assignment proceeded in waves of four neighborhoods, with twelve four-person accountability teams in each neighborhood. The twelve teams covered all cells of the $3 \times 2 \times 2$ design: a traditional leader, religious leader, or no leader; homogeneous or mixed composition; and individual disclosure or no individual disclosure. We first constructed the eligible roster and sampled residents within sex-by-wealth strata. After forming homogeneous and mixed teams, we chose an assignment under the prespecified rerandomization rule, attached the information and scheduling assignments, and replaced unavailable residents from pre-generated queues. These are the six steps described below. All random draws were made by the assignment program and saved with the assignment files for each wave.

B.2.1 Stage 1: Eligible rosters and fixed participants

The assignment was run separately by wave using the recruitment roster that had been completed before that wave's cutoff date. We retained consenting residents with the information needed to invite them and form their teams. We excluded other identified religious and traditional leaders from the ordinary-resident pool so that no-leader teams would not receive a leader through that pool.

The same chief and pastor appeared across teams within a neighborhood. We identified one traditional and one religious leader before assigning teams, and each joined the teams assigned to his leader condition. From Wave 3, a randomly selected ordinary resident from the chief's territorial origin occupied the fourth seat across no-leader teams. We call this resident the fixed placebo participant. Having that person appear again in each no-leader environment, without any leadership role, gave the no-leader condition the same team size and number of repeated participants as the leader conditions.

B.2.2 Stage 2: Sampling ordinary residents

Within each neighborhood, ordinary residents were divided into four sampling strata defined by sex and predicted housing wealth: female–lower wealth, female–higher wealth, male–lower wealth, and male–higher wealth. Higher and lower wealth were defined relative to the sex-specific median within the neighborhood; residents with missing property values entered the lower-wealth stratum.

The program sampled up to 48 ordinary residents per neighborhood, prioritizing the traditional leader's territorial origin for homogeneous teams and drawing across territories for mixed teams. Homogeneous teams were formed first. Mixed teams drew from the remaining sampled residents, including those sharing the traditional leader's territorial origin, while prioritizing territorial diversity. Thirty-six ordinary residents filled the twelve teams; remaining sampled residents entered the baseline-only control group.

The initial wave targeted equal representation of the four sex-by-wealth strata. In subsequent waves, the quotas were adjusted using the cumulative distribution of assignments in completed waves: an overrepresented stratum received fewer positions in the next wave and an underrepresented stratum received more. The adjusted totals were then distributed across the four neighborhoods entering that wave, with each neighborhood receiving the same total number of positions.

When a requested sex-by-wealth cell did not contain enough eligible residents, the program used a fixed priority order: first the opposite sex within the same wealth category, then the same sex in the other wealth category, and finally any remaining eligible resident in the relevant composition pool. These fallbacks preserved team size while keeping the assignment as close as possible to the target margins.

B.2.3 Stage 3: Forming homogeneous and mixed teams

Each of the twelve group identifiers corresponded to one experimental cell, as shown in Table B-3. The order in which group identifiers were filled was randomized. Each group received three ordinary residents. The appropriate traditional leader, religious leader, or fixed placebo participant was then added as the fourth member.

Homogeneous teams. The program filled groups 1–6 from the pool associated with the traditional leader's territorial origin. It assigned residents to teams in randomized group order and one position at a time, which prevented any group identifier from systematically receiving the first available residents. When a territorial-origin or sampling-stratum constraint was binding, the program used the prespecified fallback order described above to preserve team size.

Table B-3 – Mapping from group identifiers to experimental cells

Group identifiers	Composition	Fourth team member	Individual disclosure
1, 2	Homogeneous	Fixed placebo participant	No, Yes
3, 4	Homogeneous	Traditional leader	No, Yes
5, 6	Homogeneous	Religious leader	No, Yes
7, 8	Mixed	Fixed placebo participant	No, Yes
9, 10	Mixed	Traditional leader	No, Yes
11, 12	Mixed	Religious leader	No, Yes

Notes: Within each row, the odd-numbered group received no individual disclosure and the even-numbered group received individual disclosure. In Waves 1–2, no fixed placebo participant was used; the no-leader teams instead contained four independently sampled ordinary residents.

Mixed teams. The program filled groups 7–12 in randomized group order and one position at a time. For every new position, it first selected from territories not already represented in that team. If no such resident remained, it selected from the remaining eligible mixed-team pool. Filling one position at a time kept mixed teams as territorially diverse as the available roster allowed, while preserving the sex-by-wealth targets.

Repeated participants. The traditional leader appeared in four teams in the same neighborhood: two homogeneous and two mixed, split evenly across information conditions. The religious leader and, from Wave 3 onward, the fixed placebo participant were used in the same way. Leaders participated as ordinary team members and received no script or formal power. The estimation sample excludes the traditional leader, religious leader, and fixed placebo participant.

B.2.4 Stage 4: Cumulative rerandomization

The program generated 48 candidate assignments per wave and selected the assignment with the best cumulative covariate balance, including assignments from earlier waves. Balance was measured by summing joint F-statistics for the leader and composition indicators and their interactions, controlling for neighborhood fixed effects, across the 15 balancing covariates.[26]

B.2.5 Stage 5: Information and time slots

The two group identifiers within every leader-by-composition cell were assigned opposite information conditions. Odd-numbered groups received the status-quo information regime, in which individual contributions were not disclosed; even-numbered groups received individual disclosure. Random allocation across group identifiers produced one team in each information condition for every leader-by-composition cell in every neighborhood.

Finally, the twelve group identifiers were mapped without replacement to the twelve available weekday-by-daypart meeting slots. A group identifier had the same slot across the four neighborhoods in a given wave. In the second half of fieldwork, the slot ordering mirrored the first-half ordering, reversing both day and daypart so that leader and information cells were not systematically concentrated in earlier or later slots.

B.2.6 Stage 6: Unavailable residents and replacements

Before field implementation, all eligible residents not selected for the initial roster received a random replacement order within neighborhood and territorial-origin category. If a selected ordinary resident was unavailable, the program assigned the first available person in the corresponding queue, prioritizing the same neighborhood and territorial-origin category before relaxing the match within neighborhood. For mixed teams, the fallback continued to prioritize residents outside the traditional leader's territorial origin. Replacement residents inherited the unavailable resident's group, experimental condition, and meeting slot. Leader replacements were handled separately and preserved leader type.

[26] Waves 1–2 followed an earlier version of the protocol, with minor differences in sampling and the balance criterion. No-leader teams contained four independently sampled residents rather than three residents and a fixed placebo participant.

B.2.7 Implementation refinements across waves

The assignment logic was stable across the main sample, with two implementation refinements:

1. In Waves 1–2, no-leader teams contained four independently sampled ordinary residents. Beginning in Wave 3, one fixed placebo participant occupied the fourth seat in all four no-leader teams in a neighborhood, leaving three non-repeated residents in every experimental team.
2. Beginning in Wave 3, the sampling quotas explicitly offset cumulative sex-by-wealth differences across group identifiers from completed waves. Candidate assignments had already been evaluated cumulatively beginning in Wave 2.

These changes standardized the fourth team position and improved cumulative balance. They did not alter the mapping between group identifiers and the leader, composition, or information treatments.

B.2.8 Pseudocode

Input: eligible recruitment roster for four neighborhoods in wave w; fixed traditional and religious leaders; pretreatment covariates; assignments from waves $1, \ldots, w-1$.

For each neighborhood:
  identify the traditional leader's territorial origin;
  select the fixed placebo participant when $w \geq 3$;
  divide ordinary residents into sex $\times$ wealth strata;
  create candidate pools for homogeneous and mixed teams;
  set stratum quotas, offsetting cumulative imbalances from earlier waves.

For candidate seed $s = 1, \ldots, S_w$**:**
  sample residents within composition pools and strata;
  generate complementary three-person stratum patterns across neighborhoods;
  randomly order homogeneous group identifiers 1-6;
  fill each homogeneous team slot-by-slot, prioritizing the leader's origin;
  randomly order mixed group identifiers 7-12;
  fill each mixed team slot-by-slot, prioritizing a new territory;
  add the traditional leader to groups 3, 4, 9, and 10;
  add the religious leader to groups 5, 6, 11, and 12;
  add the placebo participant to groups 1, 2, 7, and 8 when $w \geq 3$;
  append the candidate to completed-wave assignments;
  score candidate s using the prespecified rule in Stage 4 above.

Select the seed with the lowest prespecified assignment score.
Assign no disclosure to odd group identifiers and individual disclosure to even group identifiers.
Attach the wave's without-replacement time-slot permutation.
Export the assignment roster, seed, assignment diagnostics, and ordered replacement queues.

$S_w = 48$ under the final protocol.

B.3 Implementation Checks

Table B-4 – ASSIGNMENT BALANCE AND PRETREATMENT PARTICIPATION

	No leader	Leader − no leader, within environment										
Characteristic	$M0$ mean	$M1 - M0$	$H0 - M0$	$H1 - M0$	$T - M0$	$R - M0$	$T - M1$	$R - M1$	$T - H0$	$R - H0$	$T - H1$	$R - H1$	N
Panel A. Characteristics used in rerandomization													
Female	0.565	-0.043	-0.033	-0.022	-0.027	0.021	0.076	-0.007	-0.006	0.017	0.007	-0.005	1,040
	(0.057)	(0.071)	(0.080)	(0.072)	(0.066)	(0.077)	(0.073)	(0.064)	(0.082)	(0.078)	(0.065)	(0.072)	
Predicted property value (USD per cubic meter)	2444.612	-167.307	-75.311	113.918	26.979	-214.419	64.212	624.430	120.720	324.920	-293.164	-290.367	968
	(280.312)	(313.857)	(317.962)	(377.632)	(350.713)	(278.422)	(375.699)	(403.680)	(332.820)	(517.079)	(438.059)	(433.837)	
Community role	0.783	-0.011	0.000	0.054	0.038	0.002	-0.046	-0.058	-0.010	-0.022	-0.040	-0.076	1,040
	(0.049)	(0.058)	(0.055)	(0.053)	(0.056)	(0.059)	(0.062)	(0.066)	(0.052)	(0.051)	(0.046)	(0.054)	
Age	34.554	-1.185	-0.293	-0.815	-0.354	-0.711	1.105	0.081	0.499	2.464	1.581	-0.193	1,040
	(1.770)	(1.877)	(1.956)	(2.006)	(2.334)	(2.238)	(2.100)	(1.794)	(2.069)	(1.760)	(2.163)	(1.771)	
Education (years)	9.511	0.130	-0.130	0.402	0.638	0.007	-0.135	0.532	0.352	0.507	-0.157	-0.538	1,040
	(0.353)	(0.480)	(0.494)	(0.514)	(0.556)	(0.506)	(0.491)	(0.445)	(0.457)	(0.462)	(0.482)	(0.455)	
Government job	0.000	0.011	0.011	0.000	0.012	0.000	-0.011	-0.011	-0.011	0.001	0.012	0.000	1,040
	(0.000)	(0.010)	(0.010)	(0.003)	(0.011)	(0.003)	(0.011)	(0.011)	(0.010)	(0.015)	(0.011)	(0.003)	
Salaried job	0.043	0.000	0.000	0.000	-0.006	-0.018	-0.006	0.030	0.006	0.053	0.006	0.030	1,040
	(0.021)	(0.032)	(0.030)	(0.028)	(0.028)	(0.025)	(0.031)	(0.034)	(0.030)	(0.040)	(0.030)	(0.034)	
Mission church	0.174	-0.011	0.022	0.054	-0.005	-0.005	-0.018	-0.006	0.057	0.009	-0.059	0.024	1,040
	(0.050)	(0.060)	(0.064)	(0.064)	(0.064)	(0.070)	(0.054)	(0.048)	(0.064)	(0.062)	(0.057)	(0.064)	
Revival church	0.489	0.033	0.098	-0.033	-0.024	0.107	0.003	-0.021	-0.075	-0.098	-0.016	0.008	1,040
	(0.056)	(0.077)	(0.076)	(0.076)	(0.079)	(0.085)	(0.064)	(0.074)	(0.069)	(0.073)	(0.063)	(0.062)	
Reported prior salongo participation	0.034	-0.134	-0.098	0.048	0.113	-0.165	0.019	0.217	0.161	-0.031	-0.247**	0.099	1,040
	(0.067)	(0.124)	(0.117)	(0.125)	(0.106)	(0.124)	(0.150)	(0.145)	(0.155)	(0.126)	(0.119)	(0.172)	
Salongo contribution index	0.145	-0.224	-0.319**	-0.032	-0.053	-0.112	0.174	0.088	0.233	0.252*	-0.225*	-0.226	1,040
	(0.123)	(0.147)	(0.136)	(0.164)	(0.164)	(0.171)	(0.131)	(0.130)	(0.142)	(0.143)	(0.135)	(0.141)	
Biometric support index	0.040	-0.075	0.033	0.111	0.028	0.112	0.120	0.121	-0.107	0.047	-0.268*	-0.040	1,040
	(0.120)	(0.146)	(0.139)	(0.136)	(0.152)	(0.152)	(0.142)	(0.132)	(0.122)	(0.152)	(0.146)	(0.144)	
Household size	8.380	-0.815	-1.130*	-0.228	-0.200	-1.212**	0.020	1.305**	0.394	0.835	1.028*	0.349	1,040
	(0.447)	(0.552)	(0.575)	(0.546)	(0.857)	(0.574)	(0.551)	(0.521)	(0.528)	(0.632)	(0.557)	(0.634)	
Adult share	0.426	0.041	0.027	0.002	0.039	0.025	-0.056	-0.033	0.028	0.018	0.003	0.032	1,039
	(0.017)	(0.035)	(0.032)	(0.026)	(0.029)	(0.029)	(0.037)	(0.037)	(0.034)	(0.035)	(0.032)	(0.028)	
Distance to centroid (m)	233.250	-1.463	1.625	-7.836	-4.202	5.058	-5.825	-18.727	-15.452	-1.795	5.550	7.423	1,033
	(31.416)	(15.191)	(14.124)	(13.777)	(15.463)	(15.604)	(14.104)	(14.445)	(12.589)	(12.635)	(12.018)	(13.108)	
Panel B. Additional baseline-survey characteristics													
Demographic and economic characteristics													
Log income	11.474	-0.162	0.243	-0.020	0.008	0.183	0.277	0.209	-0.209	-0.366*	0.411	0.370	1,024
	(0.207)	(0.262)	(0.181)	(0.258)	(0.244)	(0.186)	(0.283)	(0.272)	(0.209)	(0.194)	(0.250)	(0.248)	
Avenue tenure	17.272	-3.351*	1.168	-1.778	-0.400	-2.469	2.079	1.358	-5.445***	-3.081	1.596	-0.589	1,012
	(1.690)	(1.791)	(2.177)	(2.224)	(2.131)	(1.785)	(1.733)	(1.873)	(2.064)	(2.242)	(2.029)	(2.172)	

Continued on next page

Assignment Balance and Pretreatment Participation (continued)

	No leader				Leader − no leader, within environment								
Characteristic	$M0$ mean	$M1$ $-M0$	$H0$ $-M0$	$H1$ $-M0$	T $M0$	R $M0$	T $M1$	R $M1$	T $H0$	R $H0$	T $H1$	R $H1$	N
Traditional authority and local embeddedness													
Any traditional role	0.098	-0.042	-0.043	-0.053	-0.037	-0.050	-0.043	0.052	0.019	0.030	-0.008	-0.019	1,032
	(0.031)	(0.037)	(0.036)	(0.035)	(0.039)	(0.037)	(0.026)	(0.041)	(0.033)	(0.037)	(0.028)	(0.026)	
Traditional head role	0.065	-0.043	-0.043	-0.043	-0.016	-0.064**	-0.009	0.038	-0.009	0.027	-0.009	-0.021	1,032
	(0.029)	(0.030)	(0.030)	(0.031)	(0.034)	(0.027)	(0.019)	(0.028)	(0.019)	(0.031)	(0.019)	(0.016)	
Secrets network: same territory	0.286	-0.065	0.075	0.079	-0.069	-0.053	-0.014	-0.003	0.053	-0.028	0.020	-0.033	1,032
	(0.054)	(0.060)	(0.070)	(0.062)	(0.064)	(0.062)	(0.056)	(0.053)	(0.080)	(0.070)	(0.064)	(0.062)	
Material network: same territory	0.383	-0.101	0.148***	0.176***	-0.074	-0.014	0.077	0.036	0.097*	-0.034	-0.058	0.012	1,032
	(0.051)	(0.064)	(0.056)	(0.064)	(0.057)	(0.065)	(0.065)	(0.061)	(0.055)	(0.060)	(0.060)	(0.070)	
Planned salongo hours	17.293	0.224	-1.174	-0.203	-0.554	-0.812	1.282	-0.365	0.697	0.184	1.279	-1.201	1,022
	(1.704)	(1.706)	(1.391)	(1.588)	(1.873)	(1.688)	(2.358)	(1.987)	(1.623)	(1.403)	(1.632)	(1.651)	
Religious authority and church embeddedness													
Any religious role	0.391	-0.007	0.033	-0.062	-0.042	-0.022	-0.120*	-0.015	0.001	-0.078	0.059	0.081	1,032
	(0.057)	(0.072)	(0.066)	(0.065)	(0.073)	(0.077)	(0.069)	(0.075)	(0.062)	(0.062)	(0.064)	(0.071)	
Religious head role	0.098	-0.043	-0.043	-0.043	-0.025	-0.062*	-0.055**	-0.007	0.019	0.017	-0.018	0.005	1,032
	(0.027)	(0.034)	(0.034)	(0.034)	(0.040)	(0.032)	(0.024)	(0.030)	(0.034)	(0.033)	(0.030)	(0.037)	
Secrets network: same church	0.202	-0.021	-0.042	-0.010	-0.067	-0.031	-0.047	0.036	0.012	0.030	0.030	-0.035	1,032
	(0.042)	(0.049)	(0.049)	(0.051)	(0.047)	(0.057)	(0.048)	(0.054)	(0.056)	(0.044)	(0.055)	(0.049)	
Material network: same church	0.213	0.012	-0.019	0.006	-0.043	0.006	0.009	0.031	0.069	0.038	-0.011	0.033	1,032
	(0.037)	(0.050)	(0.044)	(0.044)	(0.048)	(0.050)	(0.056)	(0.053)	(0.052)	(0.050)	(0.046)	(0.045)	
Sunday attendance	3.467	-0.031	-0.152	-0.097	-0.079	0.178	-0.061	0.209	0.003	0.235	0.231	0.282**	1,032
	(0.087)	(0.144)	(0.148)	(0.136)	(0.127)	(0.124)	(0.162)	(0.145)	(0.162)	(0.147)	(0.151)	(0.135)	
Weekday attendance	1.967	-0.286	-0.413**	-0.198	-0.450*	-0.222	-0.183	0.040	-0.067	0.060	-0.210	-0.079	1,032
	(0.187)	(0.211)	(0.171)	(0.224)	(0.240)	(0.218)	(0.236)	(0.215)	(0.211)	(0.167)	(0.271)	(0.237)	
Log religious contribution	6.365	-0.283	0.023	0.000	-0.046	-0.056	-0.225	0.165	0.132	-0.190	-0.006	0.319	999
	(0.307)	(0.417)	(0.370)	(0.350)	(0.437)	(0.347)	(0.495)	(0.456)	(0.354)	(0.393)	(0.415)	(0.348)	
Knowledge of local authorities													
Knows authorities	0.312	0.019	-0.004	-0.044	-0.015	-0.028	-0.048	0.005	0.016	-0.017	0.035	0.000	1,032
	(0.034)	(0.040)	(0.035)	(0.037)	(0.040)	(0.039)	(0.041)	(0.038)	(0.036)	(0.035)	(0.038)	(0.038)	
Knows quartier chief	0.304	0.015	0.043	0.004	0.001	-0.033	0.005	-0.001	-0.016	-0.017	0.042	-0.022	1,032
	(0.057)	(0.051)	(0.047)	(0.047)	(0.054)	(0.058)	(0.060)	(0.054)	(0.053)	(0.050)	(0.056)	(0.052)	
Knows locality chief	0.359	0.026	0.011	-0.084	-0.041	0.040	-0.033	0.026	-0.023	-0.019	0.053	0.059	1,032
	(0.055)	(0.065)	(0.058)	(0.055)	(0.063)	(0.057)	(0.072)	(0.068)	(0.058)	(0.057)	(0.053)	(0.057)	
Knows chief phone	0.076	-0.021	0.011	-0.032	-0.014	-0.038	0.044	0.091**	-0.012	-0.037	0.019	0.067*	1,032
	(0.025)	(0.035)	(0.037)	(0.033)	(0.034)	(0.031)	(0.036)	(0.042)	(0.040)	(0.034)	(0.028)	(0.039)	
Chief help index	1.446	0.264	-0.029	0.018	0.064	0.103	-0.090	-0.327*	0.317*	0.029	-0.138	0.089	1,032
	(0.152)	(0.174)	(0.172)	(0.174)	(0.182)	(0.161)	(0.194)	(0.172)	(0.170)	(0.160)	(0.174)	(0.190)	

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Assignment Balance and Pretreatment Participation (continued)

Characteristic	No leader				Leader − no leader, within environment								
	$M0$ mean	$M1$ $-M0$	$H0$ $-M0$	$H1$ $-M0$	T $M0$	R $M0$	T $M1$	R $M1$	T $H0$	R $H0$	T $H1$	R $H1$	N
Standard psychological measures													
Big Five Inventory (main index)	0.068	-0.040	-0.109	-0.114	-0.115*	-0.051	-0.047	0.009	0.031	0.114	0.037	0.027	1,032
	(0.048)	(0.067)	(0.071)	(0.074)	(0.065)	(0.072)	(0.070)	(0.076)	(0.075)	(0.072)	(0.079)	(0.073)	
Extraversion	3.261	-0.063	-0.053	-0.047	-0.004	-0.047	0.005	-0.014	0.033	0.006	-0.052	-0.035	1,031
	(0.057)	(0.075)	(0.082)	(0.081)	(0.081)	(0.078)	(0.080)	(0.070)	(0.081)	(0.079)	(0.080)	(0.088)	
Agreeableness	3.957	-0.066	-0.114	-0.176*	-0.263***	-0.012	-0.073	0.000	-0.085	0.054	0.145	0.125	1,032
	(0.064)	(0.089)	(0.092)	(0.099)	(0.091)	(0.091)	(0.092)	(0.101)	(0.097)	(0.096)	(0.116)	(0.094)	
Conscientiousness	3.957	0.039	-0.076	-0.099	-0.079	-0.203*	-0.128	0.014	0.014	0.040	0.165*	0.014	1,030
	(0.069)	(0.100)	(0.091)	(0.092)	(0.088)	(0.104)	(0.105)	(0.097)	(0.084)	(0.099)	(0.095)	(0.096)	
Neuroticism	2.924	0.005	-0.027	-0.039	-0.033	-0.138	-0.176*	-0.220**	-0.068	-0.135	0.139	-0.004	1,032
	(0.062)	(0.091)	(0.107)	(0.100)	(0.087)	(0.096)	(0.093)	(0.098)	(0.115)	(0.114)	(0.102)	(0.110)	
Openness	3.266	-0.027	-0.152	-0.099	-0.084	-0.030	-0.145	-0.199*	0.077	0.170	0.013	-0.007	1,032
	(0.078)	(0.105)	(0.112)	(0.098)	(0.104)	(0.098)	(0.109)	(0.116)	(0.119)	(0.126)	(0.090)	(0.095)	
Emotional intelligence	4.353	-0.064	-0.071	0.028	-0.129*	-0.076	-0.017	0.001	-0.013	0.066	-0.116*	-0.044	1,032
	(0.057)	(0.066)	(0.066)	(0.062)	(0.071)	(0.080)	(0.073)	(0.062)	(0.070)	(0.067)	(0.060)	(0.066)	
Locus of control	4.065	-0.083	-0.163	0.191	0.008	0.149	0.238	-0.125	0.082	0.217	-0.246	0.045	1,032
	(0.144)	(0.171)	(0.171)	(0.175)	(0.174)	(0.173)	(0.186)	(0.182)	(0.179)	(0.204)	(0.182)	(0.185)	
Social desirability	0.574	-0.009	-0.028	-0.031	-0.015	-0.043*	0.000	0.020	0.016	0.031*	0.046**	0.020	1,032
	(0.019)	(0.019)	(0.019)	(0.019)	(0.020)	(0.022)	(0.021)	(0.021)	(0.020)	(0.017)	(0.021)	(0.019)	
Religious psychological measures													
Positive RCOPE	3.850	-0.025	-0.087	-0.080	-0.087	-0.032	0.000	0.012	0.049	-0.071	0.032	0.089	1,032
	(0.043)	(0.053)	(0.064)	(0.067)	(0.063)	(0.062)	(0.051)	(0.049)	(0.068)	(0.078)	(0.071)	(0.065)	
Negative RCOPE	3.391	0.203**	0.220***	0.140	0.121	0.203**	-0.090	-0.012	-0.093	-0.140	-0.083	0.030	1,032
	(0.070)	(0.083)	(0.084)	(0.089)	(0.101)	(0.088)	(0.081)	(0.088)	(0.099)	(0.088)	(0.100)	(0.095)	
Intrinsic religiosity	3.376	-0.046	-0.036	-0.046	-0.034	-0.002	0.005	0.002	-0.078*	-0.023	0.027	-0.053	1,032
	(0.022)	(0.037)	(0.036)	(0.038)	(0.043)	(0.033)	(0.038)	(0.039)	(0.042)	(0.045)	(0.048)	(0.045)	
Extrinsic religiosity: personal	4.446	0.026	-0.043	0.136*	-0.188**	-0.019	-0.040	-0.092	-0.036	-0.035	-0.101	-0.179**	1,030
	(0.060)	(0.081)	(0.079)	(0.074)	(0.091)	(0.092)	(0.090)	(0.080)	(0.083)	(0.094)	(0.084)	(0.086)	
Extrinsic religiosity: social	2.743	-0.139	0.022	-0.157	-0.193	-0.069	0.056	-0.081	-0.111	-0.206	0.166	0.075	1,032
	(0.118)	(0.125)	(0.126)	(0.136)	(0.127)	(0.134)	(0.136)	(0.137)	(0.121)	(0.126)	(0.148)	(0.150)	
Panel C. Post-assignment, pre-intervention participation													
Participation rate over six common pre-treatment days	0.404	-0.069*	-0.029	-0.009	-0.126***	-0.073**	0.002	0.040	-0.022	0.018	-0.085**	-0.068*	1,040
	(0.057)	(0.038)	(0.042)	(0.034)	(0.040)	(0.036)	(0.043)	(0.046)	(0.040)	(0.043)	(0.040)	(0.040)	

Notes: We compare residents' characteristics across the randomized groups before the first meeting. Panel A covers the characteristics used to form the groups, Panel B adds measures from the baseline survey, and Panel C reports administrative participation measured after assignment but before residents learned their treatment. The first column is the average for no-leader residents in mixed groups without disclosure ($M0$). The next three columns compare the other no-leader environments with that group. The final eight columns compare chief and pastor assignments with no leader in the same environment. M and H mean mixed and homogeneous groups; 0 and 1 mean group and individual disclosure. Comparisons are made within neighborhood, and standard errors in parentheses allow for residents assigned to the same group. The table is discussed in Section 3; Figure 2 summarizes the design. Stars use unadjusted two-sided p-values. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B-5 – ROSTER FORMATION, REPLACEMENTS, AND COMPOSITION FIDELITY

Leader	Composition	Disclosure	Assigned residents	Replacement share	Mean territories	Final chief-territory share	Composition fidelity
No leader	Mixed	Without disclosure	92	21.7	3.04	22.8	96.4
		With disclosure	92	18.5	3.07	22.8	100.0
	Homogeneous	Without disclosure	92	17.4	1.07	97.8	92.9
		With disclosure	92	22.8	1.07	97.8	92.6
Traditional	Mixed	Without disclosure	84	26.2	2.61	23.8	96.4
		With disclosure	84	27.4	2.82	28.6	100.0
	Homogeneous	Without disclosure	84	21.4	1.07	96.4	92.9
		With disclosure	84	25.0	1.11	95.2	89.3
Religious	Mixed	Without disclosure	84	29.8	2.79	26.2	96.4
		With disclosure	84	21.4	2.82	26.2	100.0
	Homogeneous	Without disclosure	84	15.5	1.07	97.6	92.9
		With disclosure	84	20.2	1.18	94.0	85.7
Total			1,040	22.2	1.98	60.7	94.6

Notes: Did the groups formed for the meetings follow the original assignments? Replacement share counts the percentage of residents whose final roster entry differs from their initial assignment. Mean territories counts the origins represented among ordinary residents, and Final chief-territory share gives the percentage from the chief's territory. We count composition as correct when all observed residents of a homogeneous group come from that territory, or when a mixed group contains at least two territories. Territory is observed in 335 of the 336 groups. All shares are percentages.
There are 92 residents in the no-leader cells rather than 84. In Waves 1–2, a different ordinary resident occupied the fourth seat in each no-leader team, and these residents remain in the analysis. From Wave 3, one person filled the seat in all four no-leader teams and was excluded, as were the repeated chief and pastor. Section 3 discusses the table; Figure 2 places these groups in the experiment.

Table B-6 – TREATMENT DELIVERY AT THE ACCOUNTABILITY MEETINGS

Leader	Composition	Disclosure	Assigned residents	Resident attendance	Realized group size	Full-group share	Leader attendance	M1 announcement verified	M2 record verified
No leader	Mixed	Without disclosure	92	95.7	3.82	82.1	–	96.4	100.0
		With disclosure	92	95.7	3.75	75.0	–	100.0	100.0
	Homogeneous	Without disclosure	92	97.8	3.82	85.7	–	100.0	100.0
		With disclosure	92	91.3	3.64	64.3	–	100.0	100.0
Traditional	Mixed	Without disclosure	84	94.0	3.82	85.7	100.0	100.0	100.0
		With disclosure	84	96.4	3.86	85.7	96.4	96.4	100.0
	Homogeneous	Without disclosure	84	92.9	3.79	78.6	100.0	100.0	100.0
		With disclosure	84	94.0	3.82	82.1	100.0	100.0	100.0
Religious	Mixed	Without disclosure	84	95.2	3.86	89.3	100.0	100.0	100.0
		With disclosure	84	96.4	3.89	89.3	100.0	100.0	100.0
	Homogeneous	Without disclosure	84	96.4	3.89	89.3	100.0	100.0	100.0
		With disclosure	84	94.0	3.82	82.1	100.0	100.0	100.0
Total			1,040	95.0	3.82	82.4	99.6	99.4	100.0

Notes: We report meeting attendance and whether the assigned information was delivered. Resident attendance refers to ordinary residents on the analysis roster. Realized group size counts everyone scheduled for the team, including the leader or no-leader participant; "full group" means that all four attended. Leader attendance verifies that the assigned chief or pastor was present. The last two columns verify that the assigned information was announced at Meeting 1 and that the team's performance record was shown at Meeting 2. Delivery requires verification at the meeting. Attendance and verification are percentages; group size is a count. No-leader cells contain 92 residents rather than 84 because Waves 1–2 used a different fourth ordinary resident in each no-leader team, and those residents remain in the analysis sample. From Wave 3 onward, the same person filled that role in all four teams and was excluded, like the repeated leaders. The table is discussed in Section 3; Figure 2 shows the experimental cells.

Table B-7 – ADMINISTRATIVE, SURVEY, AND AUDIO COVERAGE

Leader	Composition	Disclosure	Assigned residents	Baseline survey	Meeting 1 usable	Meeting 2 usable	Endline survey	Audio recorded	Administrative outcome
No leader	Mixed	Without disclosure	92	100.0	95.7	93.5	88.0	92.9	100.0
		With disclosure	92	98.9	95.7	90.2	95.7	96.4	100.0
	Homogeneous	Without disclosure	92	100.0	97.8	94.6	93.5	85.7	100.0
		With disclosure	92	98.9	91.3	92.4	97.8	96.4	100.0
Traditional	Mixed	Without disclosure	84	100.0	94.0	92.9	94.0	96.4	100.0
		With disclosure	84	98.8	96.4	90.5	92.9	96.4	100.0
	Homogeneous	Without disclosure	84	98.8	92.9	85.7	94.0	92.9	100.0
		With disclosure	84	98.8	94.0	92.9	97.6	89.3	100.0
Religious	Mixed	Without disclosure	84	100.0	95.2	89.3	85.7	89.3	100.0
		With disclosure	84	100.0	96.4	94.0	95.2	100.0	100.0
	Homogeneous	Without disclosure	84	100.0	96.4	90.5	96.4	100.0	100.0
		With disclosure	84	98.8	94.0	86.9	86.9	82.1	100.0
Total			1,040	99.4	95.0	91.2	93.2	93.2	100.0

Notes: For each data source, we report the share of residents or groups with usable observations. The survey columns count usable resident interviews; Meeting 1 and Meeting 2 exclude forms that recorded the resident as unavailable. Audio is the share of randomized groups with a verified recording. Administrative coverage means that the resident has the participation outcome used in the main results. Every covered resident contributes the same six-day window after Meeting 1 through the day of Meeting 2. All entries are percentages. No-leader cells contain 92 residents rather than 84 because Waves 1–2 used a different fourth ordinary resident in each no-leader team, and those residents remain in the analysis sample. From Wave 3 onward, the same person filled that role in all four teams and was excluded, like the repeated leaders. The table is discussed in Section 3; Figure 2 shows the experimental cells.

C Scope of Authority

C.1 Spillover Exposure and Radius Robustness

Residents assigned to different accountability groups live near one another. Could a resident respond to a nearby group's treatment as well as to their own? If so, our comparisons could partly reflect the treatments received by neighbors. We measure whether residents live near treated households from other groups, add this measure to the main specification, and check whether the results depend on how far we look for nearby households.

The analysis holds fixed the main sample of 1,040 participants in 336 assigned groups across 28 neighborhoods, the outcome, the pretreatment participation control, the design fixed effects, the 11 LASSO-selected baseline controls, and group-level clustering. The exposure coefficient measures an association with participation. Because proximity is not randomized, it does not identify an indirect treatment effect.

Exposure at the end of the wave. Let d_{ij} denote the distance between respondents i and j and G_i respondent i's assigned accountability group. For radius r, we measure exposure on the wave's final treatment date as

$$E_i(r) = \mathbf{1}\left\{\sum_{j \neq i} \mathbf{1}\{d_{ij} \leq r\}\mathbf{1}\{G_j \neq G_i\} > 0\right\}. \tag{4}$$

The sum runs over study households in respondent i's wave, all of whose groups have been treated by that date. Thus, a respondent is exposed when at least one household from another accountability group lies within the chosen radius. Nearby members of the respondent's own group are excluded because their treatment is part of the respondent's assigned intervention rather than a spillover.

Figure C-1 illustrates the 25-meter definition in neighborhoods near the 20th, 50th, and 80th percentiles of exposure at the end of the wave. Some households appear inside a gold buffer but remain classified as unexposed because the nearby treated household belongs to the same assigned group.

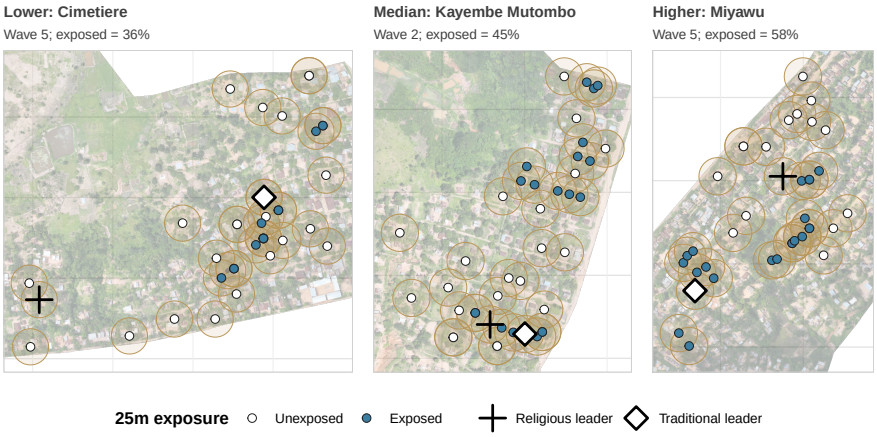


Figure C-1: TERMINAL SPILLOVER EXPOSURE IN THREE NEIGHBORHOODS

Notes: The three neighborhoods are those nearest the 20th, 50th, and 80th percentiles of the share exposed within 25 meters. All panels are measured on the final treatment date. Gold disks are 25-meter buffers around treated study households. Filled points are main-sample respondents with at least one treated out-of-group household within 25 meters; white points are unexposed. Crosses mark religious leaders and diamonds traditional leaders. The property registry TIFF appears as a translucent background. A respondent inside a buffer remains unexposed when the nearby household belongs to the respondent's own assigned group.

Radius choice and additive adjustment. At larger radii, more residents count as exposed, leaving fewer unexposed residents for comparison. At 25 meters, 483 of 1,040 participants are exposed, or 46.4 percent of the sample; at 30 meters the share is 55.4 percent. We choose 25 meters because it is the tested radius closest to an even split, without using participation outcomes. Exposure rises to 79.4 percent at 50 meters and 89.4 percent at 70 meters.

Figure C-2 adds $E_i(r)$ to the main specification at each radius while holding the sample and remaining specification fixed. It reports each of the eight leader effects separately by composition and disclosure. The effects change little: across 10–70 meters, each varies by at most 0.51 percentage points. Controlling for nearby households' treatment therefore does not account for the paper's main treatment effects.

Figure C-3 reports the exposure coefficient from the same regressions. At 25 meters, it is 2.08 percentage points (SE 1.99). The exposure measure is therefore not strongly associated with participation at the selected radius.

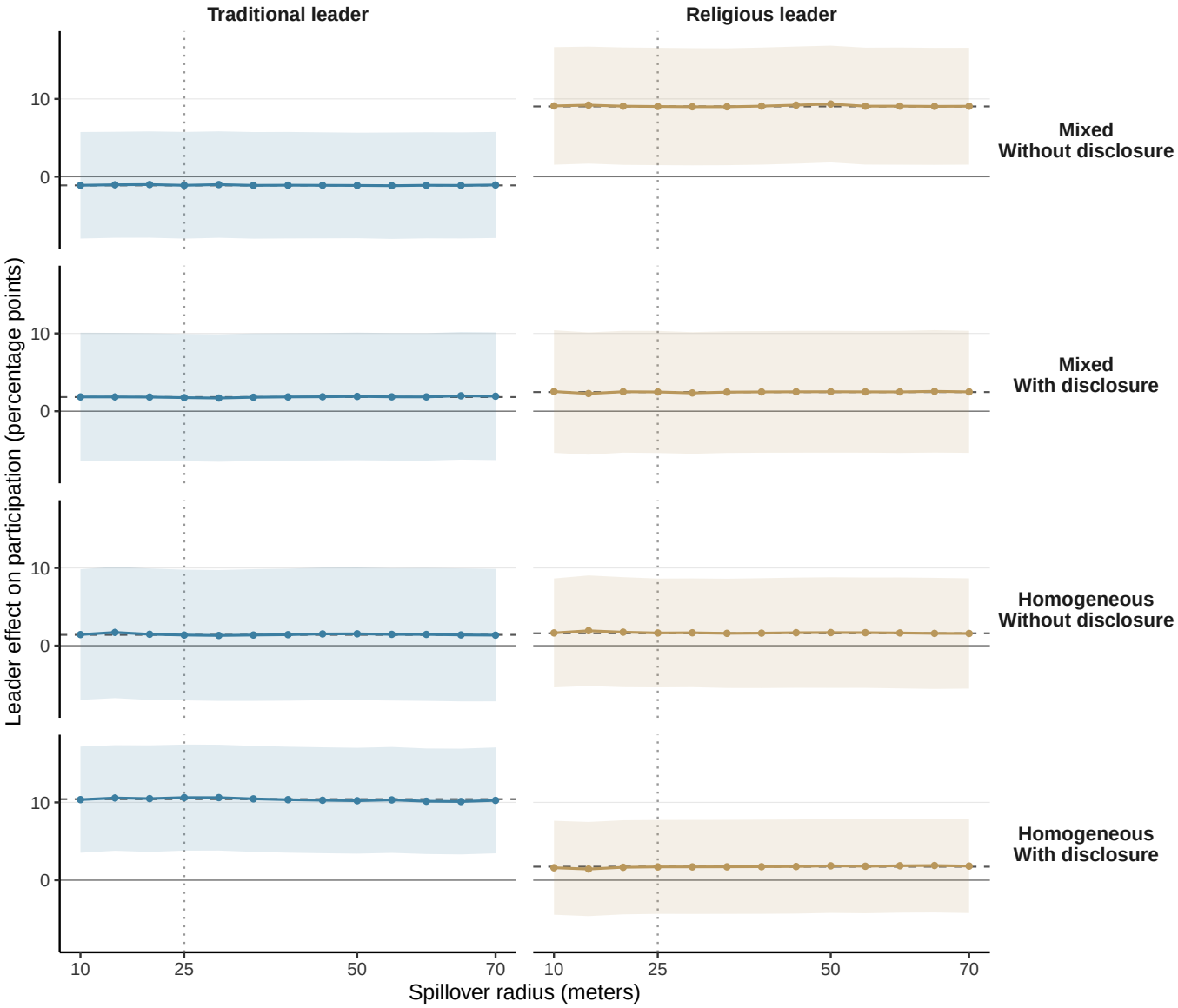


Figure C-2: LEADER EFFECTS ACROSS SPILLOVER RADII

Notes: The eight panels report traditional and religious leader effects relative to no leader within each composition-by-disclosure condition. Each regression adds the indicator for exposure within the given radius at the end of the wave to the preferred specification, retaining baseline participation, the same 11 LASSO-selected baseline controls, and neighborhood-by-composition-by-disclosure fixed effects. The sample is fixed at 1,040 residents in 336 assigned groups across 28 neighborhoods. Blue and gold lines denote traditional and religious leaders, respectively. Shaded regions are pointwise 95 percent confidence intervals clustered by assigned group. Horizontal dashed lines show the corresponding effects from the same specification without an exposure control; vertical dotted lines mark 25 meters. All panels use the same vertical scale.

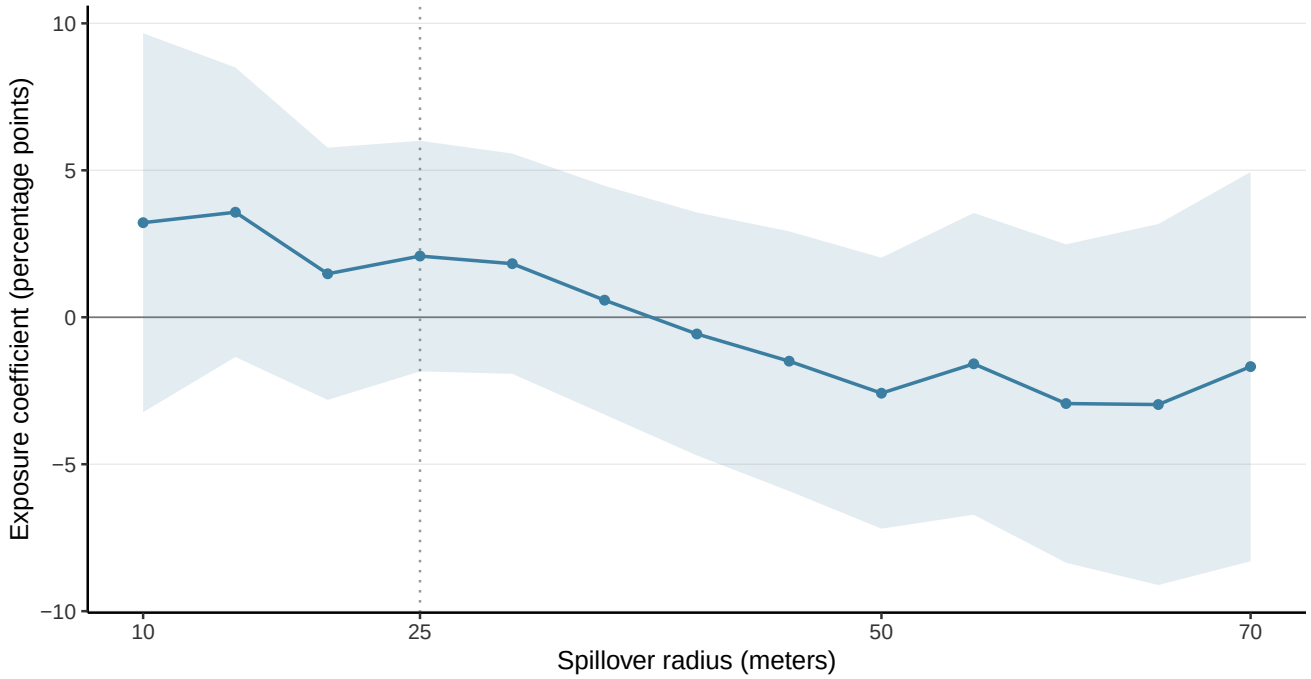


Figure C-3: SPILLOVER EXPOSURE COEFFICIENT ACROSS RADII

Notes: The line reports the coefficient on the indicator for exposure within the given radius at the end of the wave from the same regressions as Figure C-2. Each regression includes all eight leader-by-composition-by-disclosure effects, baseline participation, the same 11 LASSO-selected baseline controls, and neighborhood-by-composition-by-disclosure fixed effects. The sample is fixed at 1,040 residents in 336 assigned groups across 28 neighborhoods. Shading gives pointwise 95 percent confidence intervals clustered by assigned group. The vertical dotted line marks 25 meters. Exposure depends on where residents live. Its coefficient measures an association with participation; the design does not identify an indirect treatment effect.

C.2 Baseline Control Selection

The participation analyses include baseline participation and the design fixed effects. The preferred specification adds the 25-meter spillover measure and 11 baseline controls selected by LASSO. We use single-equation outcome LASSO to select baseline characteristics that predict participation (Belloni et al., 2014), keeping baseline participation, leader-by-composition effects, design fixed effects, and spillover exposure in the regression throughout. The selected controls are education, gender, age, NGO provision of poor relief and schools, and response-category indicators for chief performance in salongo (category 4), community cooperation (category 2), understanding others' emotions (category 3), government security performance (category 2 or higher), avenue-chief performance (category 2 or higher), and immediate ancestral rewards (category 2 or higher). We use this same set for alternative participation measures (attendance days, hours, effort, and asinh hours), observation windows, and participation robustness and heterogeneity specifications that include LASSO controls. We do not select controls again for these variants. Each alternative measure keeps its own pretreatment outcome; each alternative window keeps the common pre-meeting participation baseline. Missing values are filled at their main-sample means; no missingness indicators were selected.

C.3 Alternative Outcomes and Inference

Table C-1 – LABOR CONTRIBUTION OUTCOMES BY GROUP COMPOSITION

	Levels				Asinh robustness
	Participation rate	Attendance days	Total hours	Total effort	Asinh total hours
Panel A. Leader effects relative to no leader					
Traditional leader					
Mixed	0.004	0.041	-0.308	-0.084	-0.007
	(0.027)	(0.166)	(0.628)	(0.552)	(0.113)
Homogeneous	0.060**	0.315*	1.378**	1.174**	0.235**
	(0.028)	(0.163)	(0.639)	(0.540)	(0.103)
Religious leader					
Mixed	0.058**	0.369**	0.864	1.047*	0.164
	(0.028)	(0.169)	(0.669)	(0.558)	(0.109)
Homogeneous	0.017	0.071	0.047	0.188	0.079
	(0.024)	(0.142)	(0.580)	(0.498)	(0.087)
Panel B. Treatment-effect contrasts					
Mixed: T − R	-0.054*	-0.328*	-1.173*	-1.131*	-0.171
	(0.030)	(0.179)	(0.688)	(0.592)	(0.118)
	[Holm p = 0.228]	[Holm p = 0.228]	[Holm p = 0.228]	[Holm p = 0.228]	[Holm p = 0.148]
Homogeneous: T − R	0.043	0.244	1.331**	0.986*	0.156
	(0.027)	(0.159)	(0.613)	(0.537)	(0.101)
	[Holm p = 0.226]	[Holm p = 0.226]	[Holm p = 0.122]	[Holm p = 0.202]	[Holm p = 0.123]
Δ_C	0.098**	0.572**	2.504***	2.116***	0.327**
	(0.041)	(0.242)	(0.928)	(0.806)	(0.156)
	[Holm p = 0.033]	[Holm p = 0.033]	[Holm p = 0.029]	[Holm p = 0.029]	[Holm p = 0.037]
Panel C. No-leader reference means and diagnostics					
Mixed	0.583	3.446	13.210	11.334	2.669
	(0.018)	(0.108)	(0.420)	(0.358)	(0.072)
Homogeneous	0.568	3.364	12.703	10.872	2.594
	(0.017)	(0.103)	(0.425)	(0.352)	(0.062)
Wald: homogeneous = mixed	p = 0.536	p = 0.585	p = 0.396	p = 0.358	p = 0.431
Observations	1,040	1,040	1,040	1,040	1,040
Group-neighborhood clusters	336	336	336	336	336
Neighborhoods	28	28	28	28	28
Matched pretreatment outcome control	Yes	Yes	Yes	Yes	Yes
25m spillover control	Yes	Yes	Yes	Yes	Yes
Frozen participation LASSO controls	Yes	Yes	Yes	Yes	Yes
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes	Yes	Yes

Notes: We measure contributions over the six days after Meeting 1 through the day of Meeting 2, excluding the first-meeting day. Panel A compares each leader assignment with no leader. Panel B reports direct traditional-minus-religious contrasts and Δ_C. Standard errors in parentheses are clustered by group within neighborhood. Brackets report Holm-adjusted p-values for the same contrast within each outcome family. Panel C reports model-implied no-leader means and equality tests. Models include the matched pre-treatment outcome, the 25-meter spillover indicator, the same 11 baseline controls selected for primary participation, and neighborhood × disclosure × composition fixed effects. We use the controls selected for the main participation outcome across measures and observation windows. Participation is the daily rate; attendance is the number of labor days; hours and effort are window totals, with nonattendance coded zero. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The inverse-hyperbolic-sine transformation of hours retains residents who contributed no labor, so the sample remains the same. We apply the Holm correction separately to the four outcomes in levels and to the single asinh-hours outcome.

Table C-2 – CASH AND IN-KIND CONTRIBUTIONS BY GROUP COMPOSITION

	Levels				Asinh robustness	
	Any cash contribution	Cash total	Any in-kind contribution	In-kind value	Asinh cash total	Asinh in-kind value
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	0.030*	14.971	-0.005	-53.318	0.221*	-0.067
	(0.017)	(27.229)	(0.032)	(101.388)	(0.125)	(0.248)
Homogeneous	0.013	-1.271	0.007	-1131.884*	0.086	-0.059
	(0.014)	(14.186)	(0.040)	(677.600)	(0.097)	(0.316)
Religious leader						
Mixed	0.006	-7.053	0.053	-6.317	0.046	0.419
	(0.016)	(20.885)	(0.034)	(99.060)	(0.115)	(0.266)
Homogeneous	0.001	-13.396	0.001	-957.190	-0.005	-0.051
	(0.013)	(13.200)	(0.041)	(684.773)	(0.094)	(0.329)
Panel B. Treatment-effect contrasts						
Mixed: T − R	0.024	22.024	-0.058	-47.001	0.175	-0.486*
	(0.020)	(27.104)	(0.037)	(112.707)	(0.147)	(0.288)
	[Holm p = 0.702]	[Holm p = 0.834]	[Holm p = 0.481]	[Holm p = 0.834]	[Holm p = 0.235]	[Holm p = 0.185]
Homogeneous: T − R	0.012	12.125	0.007	-174.694	0.090	-0.009
	(0.013)	(11.360)	(0.043)	(486.568)	(0.091)	(0.339)
	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 0.641]	[Holm p = 0.980]
Δ_C	-0.012	-9.898	0.064	-127.693	-0.084	0.477
	(0.024)	(29.392)	(0.057)	(499.382)	(0.173)	(0.445)
	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 0.626]	[Holm p = 0.567]
Panel C. No-leader reference means and diagnostics						
Mixed	0.022	38.315	0.266	543.478	0.161	2.122
	(0.009)	(15.616)	(0.021)	(61.742)	(0.063)	(0.160)
Homogeneous	0.016	21.739	0.288	1619.022	0.122	2.327
	(0.010)	(11.417)	(0.026)	(611.892)	(0.069)	(0.211)
Wald: homogeneous = mixed	p = 0.679	p = 0.392	p = 0.513	p = 0.080	p = 0.682	p = 0.437
Observations	1,040	1,040	1,040	1,040	1,040	1,040
Group-neighborhood clusters	336	336	336	336	336	336
Neighborhoods	28	28	28	28	28	28
Matched pretreatment outcome control	Yes	Yes	Yes	Yes	Yes	Yes
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes	Yes	Yes	Yes

Notes: We measure contributions over the six days after Meeting 1 through the day of Meeting 2, excluding the first-meeting day. Panel A compares each leader assignment with no leader. Panel B reports direct traditional-minus-religious contrasts and Δ_C. Standard errors in parentheses are clustered by group within neighborhood. Brackets report Holm-adjusted p-values for the same contrast within each outcome family. Panel C reports model-implied no-leader means and equality tests. Models include the matched pre-treatment outcome and neighborhood × disclosure × composition fixed effects. For cash and in-kind contributions, we report whether a resident contributed during the window and the total value in Congolese francs (CDF). Combined material contributions are the sum of cash and the recorded value of in-kind contributions. Before estimation, we validate the source-data identity combined material value = cash + in-kind. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The inverse-hyperbolic-sine specifications retain residents who made no contribution during the window. We apply the Holm correction separately to the four outcomes in levels and to the two asinh outcomes.

Table C-3 – LEADER EFFECTS BY GROUP COMPOSITION

	Disclosure pooled		
	Baseline	+ Spillover	+ LASSO
Panel A. Leader effects relative to no leader			
Traditional leader			
Mixed	0.72	0.68	0.37
	(2.86)	(2.86)	(2.74)
Homogeneous	4.82*	4.88*	5.98**
	(2.91)	(2.90)	(2.78)
Religious leader			
Mixed	6.29**	6.28**	5.80**
	(2.94)	(2.95)	(2.80)
Homogeneous	0.83	0.83	1.66
	(2.52)	(2.52)	(2.37)
Panel B. Treatment-effect contrasts			
Mixed: T − R	-5.57*	-5.60*	-5.43*
	(2.98)	(3.00)	(2.96)
Homogeneous: T − R	3.98	4.05	4.32
	(2.94)	(2.92)	(2.72)
Δ_C: composition gradient	9.55**	9.65**	9.75**
	(4.20)	(4.20)	(4.06)
Panel C. No-leader reference means and diagnostics			
Mixed	58.33	58.33	58.33
	(1.99)	(1.99)	(1.79)
Homogeneous	56.79	56.79	56.79
	(1.76)	(1.76)	(1.73)
Wald: all no-leader cells equal	p = 0.562	p = 0.562	p = 0.536
Observations	1,040	1,040	1,040
Group-neighborhood clusters	336	336	336
Neighborhoods	28	28	28
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes
Pretreatment participation control	Yes	Yes	Yes
25m spillover control	No	Yes	Yes
LASSO-selected baseline controls	No	No	Yes

Notes: Panel A reports leader effects relative to no leader; Panel B reports direct leader contrasts; Panel C reports no-leader reference means. Leader effects are pooled across disclosure; Δ_C measures the change in the traditional-minus-religious gap from mixed to homogeneous composition. The outcome is average participation on six eligible days after Meeting 1 through the Meeting 2 day, in percentage points. Every column controls for pretreatment participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds the 25-meter spillover indicator; the third also adds 11 LASSO-selected baseline controls. Group-clustered standard errors appear in parentheses. Section 4.2 and Figure 3 discuss these estimates. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C-4 – RANDOMIZATION INFERENCE AND ALTERNATIVE CLUSTERING BY GROUP COMPOSITION

	Randomization inference			Alternative clustering		
	Estimate	Clustered SE	RI p	Group	Leader cluster	Neighborhood
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	0.72	(2.86)	0.817	0.72	0.72	0.72
				(2.86)	(3.15)	(3.74)
Homogeneous	4.82*	(2.91)	0.118	4.82*	4.82	4.82
				(2.91)	(2.99)	(3.37)
Religious leader						
Mixed	6.29**	(2.94)	0.046	6.29**	6.29**	6.29*
				(2.94)	(3.06)	(3.62)
Homogeneous	0.83	(2.52)	0.788	0.83	0.83	0.83
				(2.52)	(2.78)	(2.92)
Panel B. Treatment-effect contrasts						
Mixed: T − R	-5.57*	(2.98)	0.089	-5.57*	-5.57*	-5.57
				(2.98)	(3.11)	(3.50)
Homogeneous: T − R	3.98	(2.94)	0.227	3.98	3.98	3.98
				(2.94)	(3.36)	(4.21)
Δ_C: composition gradient	9.55**	(4.20)	0.039	9.55**	9.55**	9.55
				(4.20)	(4.63)	(5.67)
Panel C. No-leader reference means and diagnostics						
Mixed	58.33	(1.99)		58.33	58.33	58.33
				(1.99)	(2.19)	(2.21)
Homogeneous	56.79	(1.76)		56.79	56.79	56.79
				(1.76)	(1.61)	(1.63)
Wald: all no-leader cells equal	p = 0.562			p = 0.562	p = 0.546	p = 0.551
Observations	1,040			1,040	1,040	1,040
Clusters				336	84	28
Neighborhoods	28			28	28	28
Neighborhood × disclosure × composition FEs				Yes	Yes	Yes
Pretreatment participation control	Yes			Yes	Yes	Yes
Group-neighborhood clusters	336					
NQC fixed effects	Yes					
Formation-stage permutations	5,000					

Notes: Randomization-inference columns: We check whether the main results depend on the method of inference, keeping the specification and comparisons the same. The outcome is average post-first-meeting salongo participation, in percentage points. All models include the six-day pretreatment participation control and neighborhood × disclosure × composition fixed effects. Panel A compares each leader assignment with no leader in the same fixed-effect cell. Panel B reports the same leader contrasts as the main table; Δ_C is the difference between the Traditional and Religious composition gradients. Panel C reports model-implied no-leader reference means and clustered delta-method tests that the displayed no-leader means are equal. Standard errors are clustered one-way at the group–neighborhood level. To obtain two-sided randomization-inference (RI) p-values, we form new groups 5,000 times among the residents in the realized sample. We retain each wave's group sizes, random filling order, composition and fallback rules, gender × wealth targets, territorial diversity in mixed groups, and treatment coding. We do not resample residents or repeat the original selection of the best-balanced assignment. These RI tests therefore describe the conditional group-formation replay, rather than the full rerandomization design. The random seed is 20260731. Stars use clustered asymptotic inference: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Alternative-clustering columns:* We retain the main sample, outcomes, neighborhood × disclosure × composition fixed effects, pretreatment participation control, leader effects, contrasts, and no-leader comparisons. The columns change only the level at which standard errors are clustered. Group clusters are the 336 experimental group–neighborhood cells. Leader-group clusters are the 84 neighborhood-wave × leader-arm cells, each containing the four groups assigned to traditional, religious, or no leader. Neighborhood clustering uses 28 neighborhood-wave assignments. Panel B and Panel C inference uses the clustered delta method. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4 Alternative Explanations

C.4.1 Endogenous Leader Behavior

Table C-5 – LEADER SPEECH IN THE FIRST MEETING

	Speaking time (minutes) (1)	Attendance and performance (%) (2)	Divine reward and solidarity (%) (3)	Material support (%) (4)
I. Group composition (pooled across disclosure)				
Panel A. Adjusted levels				
Traditional leader				
Mixed	2.69	47.18	29.32	23.49
	(0.14)	(3.06)	(2.81)	(2.54)
Homogeneous	2.72	39.59	28.09	32.33
	(0.15)	(3.38)	(3.10)	(2.79)
Religious leader				
Mixed	3.46	22.27	51.01	26.72
	(0.15)	(2.79)	(2.91)	(2.10)
Homogeneous	3.45	27.28	49.27	23.45
	(0.15)	(2.84)	(2.96)	(2.15)
Panel B. Comparisons				
Traditional leader				
Homogeneous minus mixed	0.03	-7.60	-1.24	8.84
	(0.29)	(6.44)	(5.91)	(5.33)
Religious leader				
Homogeneous minus mixed	0.00	5.01	-1.74	-3.27
	(0.29)	(5.63)	(5.87)	(4.25)
II. Group composition by disclosure				
Panel C. Adjusted levels				
Traditional leader				
Mixed, without disclosure	2.65	49.96	29.51	20.53
	(0.17)	(5.53)	(4.15)	(3.63)
Mixed, with disclosure	2.74	44.35	29.14	26.52
	(0.19)	(3.31)	(3.44)	(3.58)
Homogeneous, without disclosure	2.53	38.02	32.82	29.16
	(0.23)	(4.77)	(4.75)	(5.17)
Homogeneous, with disclosure	2.93	41.27	23.16	35.57
	(0.15)	(6.89)	(5.66)	(5.53)
Religious leader				
Mixed, without disclosure	3.75	24.06	47.64	28.29
	(0.22)	(4.62)	(5.69)	(5.03)
Mixed, with disclosure	3.18	20.69	54.05	25.26
	(0.18)	(3.75)	(5.16)	(3.62)
Homogeneous, without disclosure	3.22	28.80	48.48	22.72
	(0.20)	(3.94)	(4.61)	(3.15)
Homogeneous, with disclosure	3.74	25.32	50.32	24.35
	(0.28)	(4.07)	(5.11)	(4.32)
Panel D. Comparisons				
Traditional leader				
Disclosure effect, mixed	0.09	-5.62	-0.37	5.99
	(0.23)	(6.72)	(5.12)	(5.10)
Disclosure effect, homogeneous	0.40	3.25	-9.66	6.41
	(0.24)	(9.66)	(8.37)	(9.12)
Religious leader				
Disclosure effect, mixed	-0.56**	-3.37	6.41	-3.03
	(0.27)	(6.21)	(9.15)	(7.61)
Disclosure effect, homogeneous	0.52	-3.48	1.84	1.64
	(0.37)	(5.59)	(7.69)	(6.09)
Leader–meetings, chiefs	103	103	103	103
Leader–meetings, pastors	103	103	103	103
Neighborhoods per leader type	28	28	28	28

Notes: Each observation is an assigned chief or pastor speaking at a recorded first meeting. We compare homogeneous and mixed groups for each leader type, pooling disclosure conditions, then add disclosure and its interaction with composition to estimate with-minus-without-disclosure differences within each composition. Each regression includes leader fixed effects; to calculate adjusted levels, we use the same observed leader weights within each type. The regressions include neither a pretreatment speech measure nor a participation control.
Speaking time is measured in minutes. To calculate topic shares, we assign each passage to one of the three topics described in Section 5.3 and weight it by its duration. Shares are percentages of modeled speaking time. Additional passages are classified using the same fitted model and vocabulary, leaving earlier assignments unchanged. Differences are in minutes for speaking time and percentage points for topic shares. Standard errors in parentheses are clustered by neighborhood. Stars on differences use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C-6 – ARRIVING TOGETHER WITH THE ASSIGNED LEADER

	Chain 30 sec. (1)	Chain 1 min. (main) (2)	Direct gap 1 min. (3)	Direct gap 2 min. (4)
I. Group composition (pooled across disclosure)				
Panel A. Adjusted levels				
Traditional leader				
Mixed	5.67	8.22	3.87	6.74
	(1.41)	(1.58)	(0.75)	(1.10)
Homogeneous	6.98	10.52	4.96	7.84
	(1.63)	(2.27)	(1.21)	(1.78)
Religious leader				
Mixed	4.74	5.80	2.91	5.64
	(1.04)	(1.09)	(0.62)	(0.97)
Homogeneous	3.97	5.24	2.75	4.09
	(1.35)	(1.57)	(0.65)	(1.09)
Panel B. Comparisons				
Traditional leader				
Homogeneous minus mixed	1.31	2.29	1.09	1.10
	(1.54)	(1.65)	(1.27)	(1.65)
Religious leader				
Homogeneous minus mixed	-0.77	-0.56	-0.17	-1.55
	(1.15)	(1.37)	(0.72)	(1.20)
II. Group composition by disclosure				
Panel C. Adjusted levels				
Traditional leader				
Mixed, without disclosure	7.16	9.41	4.86	8.43
	(2.04)	(2.25)	(1.27)	(1.61)
Mixed, with disclosure	4.18	7.03	2.87	5.05
	(1.23)	(1.55)	(0.81)	(1.26)
Homogeneous, without disclosure	5.63	9.35	3.85	6.99
	(1.60)	(2.29)	(1.23)	(1.90)
Homogeneous, with disclosure	8.33	11.68	6.06	8.69
	(2.02)	(2.63)	(1.51)	(2.18)
Religious leader				
Mixed, without disclosure	4.96	6.73	2.53	5.90
	(1.18)	(1.36)	(0.69)	(1.21)
Mixed, with disclosure	4.51	4.87	3.30	5.38
	(1.27)	(1.22)	(0.96)	(1.19)
Homogeneous, without disclosure	3.34	4.71	1.97	3.63
	(1.30)	(1.57)	(0.59)	(1.17)
Homogeneous, with disclosure	4.60	5.78	3.53	4.55
	(1.60)	(1.77)	(1.00)	(1.22)
Panel D. Comparisons				
Traditional leader				
Disclosure effect, mixed	-2.98	-2.38	-1.98	-3.37*
	(1.85)	(2.22)	(1.51)	(1.88)
Disclosure effect, homogeneous	2.69	2.33	2.21*	1.70
	(1.60)	(1.89)	(1.29)	(2.01)
Religious leader				
Disclosure effect, mixed	-0.45	-1.86	0.76	-0.52
	(1.30)	(1.39)	(1.12)	(1.40)
Disclosure effect, homogeneous	1.26	1.06	1.55	0.91
	(1.10)	(1.12)	(0.99)	(0.97)
Respondents, chiefs	336	336	336	336
Respondents, pastors	336	336	336	336
Neighborhoods per leader type	28	28	28	28

Notes: Each column measures the percentage of six post-meeting days satisfying the indicated co-arrival definition; nonattendance is zero and both people must contribute labor. Chains link successive check-ins separated by no more than the indicated interval, including all recorded attendees. The one-minute chain is the main definition. Direct gap limits the absolute time between the resident's and leader's check-ins, without transitive links. Each regression controls for its own measure over the six pretreatment days. We estimate separate regressions for chiefs and pastors. The pooled regressions compare homogeneous with mixed groups across disclosure conditions. We then add disclosure and its interaction with composition to compare groups with and without disclosure within each composition. We evaluate predicted levels at the common baseline mean. All levels are percentages and differences are percentage points. Neighborhood-clustered standard errors are in parentheses. For the main-text peer comparison, each day's number of co-arriving ordinary groupmates is divided by the fixed number of ordinary groupmates, including nonattenders. The one-minute measure gives equal weight to homogeneous and mixed groups and adjusts each partner measure for its own pretreatment value. A joint regression permits separate composition effects and baseline slopes by leader type and partner measure, with neighborhood-clustered inference. The pooled chief–ordinary-groupmate difference has $p = 0.309$. Stars on differences use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C-7 – INTERACTIONS WITH THE ASSIGNED LEADER

	Any interaction (1)	Discussed salongo (2)	Asked to participate (3)	Social consequences (4)	Divine consequences (5)
I. Group composition (pooled across disclosure)					
Panel A. Adjusted levels					
Traditional leader					
Mixed	50.30	39.61	13.78	2.11	3.33
	(3.93)	(5.01)	(3.38)	(1.13)	(1.90)
Homogeneous	46.00	35.33	12.65	2.00	2.67
	(4.86)	(4.51)	(2.83)	(1.12)	(1.23)
Religious leader					
Mixed	40.68	36.72	10.95	1.33	3.16
	(3.64)	(3.81)	(2.80)	(0.97)	(1.24)
Homogeneous	44.30	34.76	14.08	2.74	3.94
	(4.24)	(4.51)	(3.24)	(1.63)	(2.01)
Panel B. Comparisons					
Traditional leader					
Homogeneous minus mixed	-4.30	-4.28	-1.13	-0.11	-0.66
	(4.09)	(4.19)	(3.90)	(1.70)	(2.02)
Religious leader					
Homogeneous minus mixed	3.62	-1.96	3.13	1.41	0.78
	(4.28)	(5.07)	(2.74)	(1.37)	(2.29)
II. Group composition by disclosure					
Panel C. Adjusted levels					
Traditional leader					
Mixed, without disclosure	49.89	37.18	7.05	1.47	1.23
	(4.11)	(6.02)	(2.96)	(1.37)	(1.41)
Mixed, with disclosure	50.70	42.25	21.12	2.82	5.63
	(6.68)	(6.95)	(5.84)	(1.95)	(3.88)
Homogeneous, without disclosure	50.07	36.11	11.52	1.27	1.42
	(5.91)	(5.30)	(3.41)	(1.48)	(1.35)
Homogeneous, with disclosure	42.24	34.62	13.70	2.68	3.81
	(6.55)	(6.31)	(3.81)	(1.81)	(2.04)
Religious leader					
Mixed, without disclosure	45.39	42.74	14.69	2.66	5.36
	(5.82)	(6.03)	(4.39)	(1.87)	(2.54)
Mixed, with disclosure	36.19	30.98	7.40	0.07	1.07
	(5.43)	(5.34)	(2.79)	(0.15)	(1.16)
Homogeneous, without disclosure	43.10	32.17	14.33	4.12	3.67
	(5.58)	(5.05)	(3.62)	(2.32)	(2.19)
Homogeneous, with disclosure	45.51	37.38	13.81	1.33	4.22
	(6.58)	(6.51)	(4.33)	(1.32)	(2.29)
Panel D. Comparisons					
Traditional leader					
Disclosure effect, mixed	0.81	5.07	14.06**	1.35	4.41
	(7.67)	(8.24)	(6.15)	(2.46)	(4.30)
Disclosure effect, homogeneous	-7.83	-1.50	2.18	1.41	2.39
	(7.90)	(7.42)	(4.48)	(2.45)	(2.44)
Religious leader					
Disclosure effect, mixed	-9.21	-11.77	-7.29	-2.59	-4.28
	(8.61)	(8.49)	(4.71)	(1.76)	(3.00)
Disclosure effect, homogeneous	2.41	5.20	-0.52	-2.79	0.56
	(8.74)	(7.37)	(4.60)	(1.84)	(1.97)
Respondents, chiefs	299	299	299	299	299
Respondents, pastors	302	302	302	302	302
Neighborhoods per leader type	28	28	28	28	28

Notes: The sample consists of ordinary respondents' reports about their assigned leader. Outcomes record any interaction, discussion of salongo, and whether the leader asked the respondent to participate or mentioned social or divine consequences. Reports about unknown persons are excluded. Regressions control for respondent pretreatment participation. We estimate separate regressions for chiefs and pastors. The pooled regressions compare homogeneous with mixed groups across disclosure conditions. We then add disclosure and its interaction with composition to compare groups with and without disclosure within each composition. We evaluate predicted levels at the common baseline mean. All levels are percentages and differences are percentage points. Neighborhood-clustered standard errors are in parentheses. Stars on differences use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4.2 Majority Membership

Table C-8 – SHARED TERRITORIAL ORIGIN WITH THE NEIGHBORHOOD CHIEF

	Baseline participation control			+ Spillover and LASSO		
	Different territory	Same territory	Difference	Different territory	Same territory	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	-1.05	6.09	7.13	-1.28	5.26	6.54
	(3.35)	(7.31)	(8.56)	(3.35)	(6.65)	(8.00)
Homogeneous, pooled		4.85*			6.01**	
		(2.91)			(2.78)	
Religious leader						
Mixed	3.41	14.69**	11.27	3.51	12.47*	8.96
	(3.54)	(7.26)	(8.66)	(3.35)	(6.68)	(7.93)
Homogeneous, pooled		0.85			1.68	
		(2.53)			(2.38)	
Panel B. Implied no-leader counterfactuals						
Common no-leader comparison						
Mixed	58.92	56.35	-2.57	58.92	56.35	-2.57
	(2.23)	(5.17)	(5.92)	(1.98)	(4.96)	(5.63)
Homogeneous, pooled		56.79			56.79	
		(1.76)			(1.73)	
Observations		1,040			1,040	
Group clusters		336			336	
Groups containing respondents in each category (first / second)						
Chief, mixed		56 / 34			56 / 34	
Chief homogeneous groups (pooled)		56			56	
Pastor, mixed		55 / 36			55 / 36	
Pastor homogeneous groups (pooled)		56			56	

Notes: The difference column reports the effect in the second category minus the effect in the first. Both categories compare the respondent's territorial origin with the neighborhood chief's origin. The territorial-origin comparison is defined in every assignment arm. Heterogeneity is estimated only in mixed groups; homogeneous effects are pooled. Panel B averages fitted predictions over actual no-leader respondents within each composition and, in mixed groups, territorial-origin category, retaining their observed covariates and fixed effects. Within each composition, we pool disclosure conditions when estimating leader effects, their interactions with the characteristic, and any separate category levels. Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds spillover and 11 LASSO-selected baseline controls; Appendix Section C.5 details their selection and missing-value treatment. Standard errors in parentheses are clustered by group and treat category membership as fixed. Stars use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Section 4.3 discusses the estimates. In the specification with spillover and LASSO controls, the pastor effect among the chief's coethnics in mixed groups minus the pooled pastor effect in homogeneous groups is 10.79 percentage points (SE 7.09; $p = 0.129$), using the full coefficient covariance matrix.

C.4.3 Pastors' Territorial Origins

Table C-9 – LEADER EFFECTS UNDER THE RELIGIOUS-LEADER TERRITORY RESTRICTION

	Leader by composition		
	Baseline	+ Spillover	+ LASSO
Panel A. Leader effects relative to no leader			
Traditional leader			
Mixed	0.65	0.60	0.24
	(2.71)	(2.70)	(2.62)
Homogeneous	4.79*	4.89*	6.04**
	(2.88)	(2.87)	(2.76)
Religious leader			
Mixed	10.05***	10.12***	8.46**
	(3.52)	(3.51)	(3.45)
Homogeneous	-1.43	-1.50	-0.84
	(3.33)	(3.31)	(3.03)
Panel B. Treatment-effect contrasts			
Mixed: T − R	-9.40***	-9.52***	-8.22**
	(3.43)	(3.44)	(3.45)
Homogeneous: T − R	6.22*	6.38*	6.88**
	(3.74)	(3.71)	(3.38)
Δ_C: composition gradient	15.62***	15.91***	15.10***
	(5.08)	(5.07)	(4.84)
Panel C. No-leader reference means and diagnostics			
Mixed	58.33	58.33	58.33
	(1.92)	(1.90)	(1.77)
Homogeneous	56.79	56.79	56.79
	(1.76)	(1.76)	(1.73)
Wald: homogeneous = mixed	p = 0.555	p = 0.553	p = 0.534
Observations	908	908	908
Group-neighborhood clusters	292	292	292
Neighborhoods	28	28	28
Design fixed effects	Yes	Yes	Yes
Pretreatment participation control	Yes	Yes	Yes
25m spillover control	No	Yes	Yes
LASSO-selected baseline controls	No	No	Yes

Notes: Could pastors' limited influence in homogeneous groups reflect a lack of ethnic ties? We repeat the pooled scope specification after excluding 132 residents assigned a pastor in the 11 neighborhoods where the pastor and chief originate from different territories. Residents assigned a chief or no leader all remain. Panel A gives leader effects pooling disclosure, Panel B compares chief and pastor effects, and Panel C gives predicted no-leader means. The first column adjusts for participation before treatment and neighborhood-by-composition-by-disclosure fixed effects. Moving to the second column adds nearby spillover exposure; moving to the third adds the 11 baseline controls selected by LASSO, retaining the preceding adjustments. Parentheses give standard errors clustered by group. See Section 4.2 for the territorial comparison. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4.4 Leadership by Example

Table C-10 – LEADERSHIP BY EXAMPLE: LEADER PRETREATMENT PARTICIPATION

	Group-clustered SE			Neighborhood-clustered SE		
	Participation at/below leader-type median	Participation above leader-type median	Difference	Participation at/below leader-type median	Participation above leader-type median	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	0.91	0.51	-0.40	0.91	0.51	-0.40
	(3.59)	(4.00)	(5.05)	(4.26)	(5.50)	(6.50)
Homogeneous	7.96**	2.12	-5.85	7.96	2.12	-5.85
	(3.93)	(3.84)	(5.21)	(5.79)	(3.78)	(6.99)
Religious leader						
Mixed	4.91	8.09**	3.18	4.91	8.09*	3.18
	(3.84)	(4.00)	(5.21)	(4.75)	(4.68)	(6.16)
Homogeneous	3.65	-2.87	-6.52	3.65	-2.87	-6.52
	(3.40)	(3.35)	(4.53)	(4.19)	(4.45)	(6.23)
Panel B. Implied no-leader counterfactuals						
No leader, mixed		58.33			58.33	
		(1.99)			(2.21)	
No leader, homogeneous		56.79			56.79	
		(1.79)			(1.70)	
Observations			1,040			
Group clusters			336			

Notes: The difference column compares the effect above the median with the effect at or below it. Panel B shows predicted participation without a leader, averaged over the no-leader residents in each composition using their observed covariates and fixed effects. Because leader participation is undefined in no-leader groups, each block uses one pooled no-leader mean for both categories. Entries compare leader effects above and below the type-specific median of leaders' participation during the common week before any meeting. The cutpoints are 0.80 for chiefs and 0.83 for pastors; ties enter the at-or-below-median category. Sunday is excluded and each leader rate uses four to six observed eligible days. The two blocks report the same composition-pooled regression with standard errors clustered by 336 groups or by 28 neighborhoods. The latter allows dependence across groups assigned to the same leader. All 1,040 participants are retained. The regression includes baseline participation, neighborhood-by-composition-by-disclosure fixed effects, and leader-by-composition interactions with the leader participation category. Because the characteristic is undefined without a leader, it enters only through interactions with leader assignment. Section 4.2 discusses the interpretation. Stars use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4.5 Pretreatment Participation and Baseline Adjustment

Table C-11 – PRETREATMENT DYNAMICS: POOLED ACROSS DISCLOSURE

	All six pretreatment days			Excluding day 1		
Panel A. Linear trends						
	Estimate	Standard error	p-value	Estimate	Standard error	p-value
Traditional leader						
Mixed	-0.50	(0.92)	0.587	-0.60	(1.06)	0.576
Homogeneous	-1.48	(0.86)	0.087	-0.87	(1.15)	0.453
Religious leader						
Mixed	1.10	(0.84)	0.190	1.01	(1.03)	0.325
Homogeneous	-2.21	(0.90)	0.014	-1.76	(1.20)	0.145
Panel B. Early versus late pretreatment						
	Estimate	Standard error	p-value	Estimate	Standard error	p-value
Traditional leader						
Mixed	-0.33	(2.93)	0.911	0.06	(2.97)	0.985
Homogeneous	-4.23	(2.99)	0.158	-2.40	(3.41)	0.483
Religious leader						
Mixed	2.05	(2.69)	0.446	1.05	(2.79)	0.707
Homogeneous	-4.82	(3.19)	0.131	-2.50	(3.57)	0.485
Panel C. Saturated day-specific paths						
	F	Restrictions	p-value	F	Restrictions	p-value
Traditional leader						
Mixed	0.97	5	0.435	1.19	4	0.316
Homogeneous	1.66	5	0.145	1.15	4	0.335
Religious leader						
Mixed	1.17	5	0.323	1.26	4	0.286
Homogeneous	2.75	5	0.019	1.97	4	0.099
Respondents		1,040			1,040	
Group–neighborhood clusters		336			336	
Neighborhoods		28			28	

Notes: Panels A and B report percentage-point estimates, clustered standard errors, and p-values for each of the four leader-by-composition paths, pooling disclosure conditions. Panel A estimates the change in the leader-minus-no-leader gap per campaign day. Panel B compares days 1–3 with days 4–6 in the full window and days 2–3 with days 4–6 after excluding day 1. Panel C jointly tests the day-specific coefficients for each path, taking campaign day 6 as the reference. We report the F statistic, number of restrictions, and p-value for this test. All regressions include respondent fixed effects and neighborhood × information × composition × campaign-day fixed effects. Standard errors are clustered one-way by assigned group within neighborhood.

Table C-12 – BASELINE IMBALANCE AND TREATMENT-SPECIFIC BASELINE ADJUSTMENT

	Leader by composition				Environment specific		
	Baseline adjustment	+ Spillover	+ LASSO		Baseline adjustment	+ Spillover	+ LASSO
Panel A. Leader effects relative to no leader							
Traditional leader							
Mixed	0.73	0.68	0.23	Without disclosure	-0.20	-0.23	-1.34
	(2.82)	(2.83)	(2.68)		(3.47)	(3.47)	(3.40)
				With disclosure	1.56	1.49	1.55
					(4.40)	(4.39)	(4.03)
Homogeneous	4.13	4.20	5.40**	Without disclosure	-0.50	-0.53	1.12
	(2.83)	(2.83)	(2.73)		(4.50)	(4.49)	(4.31)
				With disclosure	8.87***	9.06***	9.79***
					(3.26)	(3.27)	(3.25)
Religious leader							
Mixed	6.01**	6.00**	5.38*	Without disclosure	8.74**	8.72**	8.43**
	(2.93)	(2.94)	(2.81)		(3.98)	(3.99)	(3.85)
				With disclosure	3.29	3.28	2.20
					(4.29)	(4.32)	(4.00)
Homogeneous	1.10	1.10	1.78	Without disclosure	1.39	1.39	1.38
	(2.57)	(2.57)	(2.42)		(3.93)	(3.92)	(3.67)
				With disclosure	0.84	0.83	2.23
					(3.26)	(3.26)	(3.07)
Panel B. Treatment-effect contrasts							
Mixed: T − R	-5.29*	-5.32*	-5.15*	Mixed, with: T − R	-1.73	-1.79	-0.66
	(2.94)	(2.96)	(2.89)		(4.36)	(4.38)	(4.14)
				Mixed, without: T − R	-8.95**	-8.95**	-9.77**
					(3.88)	(3.90)	(4.02)
Homogeneous: T − R	3.03	3.11	3.62	Homogeneous, with: T − R	8.02**	8.23**	7.57**
	(2.85)	(2.85)	(2.68)		(3.73)	(3.72)	(3.46)
				Homogeneous, without: T − R	-1.89	-1.92	-0.26
					(4.18)	(4.16)	(3.98)
Δ_C: composition gradient	8.32**	8.43**	8.77**	Δ_E: H, with − M, without	16.97***	17.18***	17.34***
	(4.10)	(4.11)	(3.98)		(5.39)	(5.42)	(5.31)
Panel C. No-leader reference means and diagnostics							
Mixed	58.33	58.33	58.33	Without disclosure	58.88	58.88	58.88
	(1.98)	(1.97)	(1.78)		(2.45)	(2.44)	(2.23)
				With disclosure	57.79	57.79	57.79
					(3.11)	(3.11)	(2.80)
Homogeneous	56.79	56.79	56.79	Without disclosure	57.79	57.79	57.79
	(1.73)	(1.73)	(1.72)		(2.91)	(2.91)	(2.77)
				With disclosure	55.80	55.80	55.80
					(1.84)	(1.84)	(2.01)
Wald: all no-leader cells equal	p = 0.557	p = 0.557	p = 0.534		p = 0.770	p = 0.770	p = 0.773
Panel D. Joint tests							
All composition coefficients zero	0.135	0.133	0.084	All environment coefficients zero	0.062	0.059	0.038
Observations	1,040	1,040	1,040		1,040	1,040	1,040
Group–neighborhood clusters	336	336	336		336	336	336
Neighborhoods	28	28	28		28	28	28
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes		Yes	Yes	Yes
Assignment-cell baseline slopes	Yes	Yes	Yes		Yes	Yes	Yes
25m spillover control	No	Yes	Yes		No	Yes	Yes
LASSO-selected baseline controls	No	No	Yes		No	No	Yes

Notes: Each observation is a respondent's mean salongo participation after the first meeting, in percentage points, within the assigned treatment cell. We allow the relationship between pretreatment and subsequent participation to differ across all 12 assignment cells. Every specification includes neighborhood × information × composition fixed effects, and pretreatment participation is centered at the full-sample mean. Panels A–D report, respectively, leader effects, leader contrasts, model-implied no-leader means and equality tests, and joint tests of the displayed leader effects. $\Delta_C = (\tau^T_H - \tau^T_M) - (\tau^R_H - \tau^R_M)$ and $\Delta_E = (\tau^T_{HI} - \tau^T_{MN}) - (\tau^R_{HI} - \tau^R_{MN})$. Spillover columns add 25m exposure; LASSO columns additionally include 11 LASSO-selected baseline controls. We use the delta method for inference, with standard errors clustered by group within neighborhood. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.5 Resident and Group Characteristics

Tables C-13–C-18 estimate Equation (2), pooling both leader effects and heterogeneity across disclosure within each composition. How a characteristic enters the regression depends on whom we can measure it for. When it is observed in every assignment arm, we allow its association with participation to differ by composition. When it rates the assigned leader, we interact it only with leader assignment, since there is no corresponding rating in no-leader groups. We split effects by shared territory with the chief only in mixed groups, because all homogeneous-group residents share his territory. Ethnic identification uses the baseline ranking of six identities, scoring own-territory identification from 1 (least important) to 6 (most important). Group ethnic identification averages these scores among the three ordinary-member seats. We use the available scores: three in 330 groups and two in six. We exclude the leader and include each ordinary member's own score where available, retaining all 1,040 respondents. High identification is strictly above the group median of 10/3, with ties in the lower category. We add no separate controls or interactions for individual ethnic identification. The comparison describes heterogeneity across groups; it does not isolate a causal peer effect.

Table C-14 further divides each group-identification category in mixed groups by whether residents share the neighborhood chief's territory. The regression interacts both characteristics with each leader type and includes the corresponding no-leader category terms; homogeneous-group effects retain the group-identification split. No-leader levels average fitted predictions over actual no-leader respondents within each joint identity and coethnicity category.

Authority and opinion assessments are matched to the assigned leader using the meeting membership records. Missing values for 33 residents are filled with the observed mean among ordinary members of the same group. Three residents in one group lack an observed group mean and are excluded. This gives 1,037 observations in 335 groups for both assessments. Standard errors condition on the filled categories. For these two measures, the no-leader reference pools respondents within each composition; it does not represent low- and high-assessment no-leader subgroups. Differences between rating categories may also reflect pre-existing differences in residents' willingness to participate, rather than differences in their response to the leader. These reference levels average fitted predictions over actual no-leader respondents within composition, retaining their observed covariates and fixed effects. For group ethnic identification, reference levels average fitted predictions over actual no-leader respondents within each observed composition and identity category, retaining their own covariates and fixed effects. For shared territory with the chief, reference levels average fitted predictions over actual no-leader respondents within composition and, in mixed groups, territorial-origin category, retaining their observed covariates and fixed effects. For the other characteristics observed in every arm, we predict no-leader levels for the same set of chief- and pastor-arm residents within each composition and category.

Religious orientation follows the intrinsic/extrinsic distinction of Allport and Ross (1967) and the revised measurement of Gorsuch and McPherson (1989). In the nine-item survey adaptation, intrinsic religiosity averages items 1, 3 reversed, 4, 5, and 9 reversed. Extrinsic religiosity averages items 2, 6, 7, and 8, combining the personal-comfort item with the three social-contact items with equal weight per item. Responses run from 1 to 5; reversal subtracts the response from 6. Each scale requires at least three valid responses. Both analyses retain 1,034 residents in 336 groups, excluding six without usable baseline items. Five residents contribute four intrinsic items, and five contribute three extrinsic items. We choose the cutpoint that most nearly divides residents into equally sized categories, using only the distribution of the religiosity scale. Scores strictly above it enter the higher category. The intrinsic cutpoint is 3.2 (498 lower and 536 higher); the extrinsic cutpoint is 2.75 (493 lower and 541 higher). Each scale enters a separate regression, with the same specifications and reference-prediction convention as the other resident characteristics.

The first specification includes baseline participation and the design fixed effects. The second adds the corresponding 25-meter spillover measure and the 11 baseline controls described in Appendix Section C.2. Tests in this exploratory heterogeneity family are unadjusted for multiple comparisons.

Table C-13 – GROUP ETHNIC IDENTIFICATION AND LEADER EFFECTIVENESS

	Baseline participation control			+ Spillover and LASSO		
	Low identity	High identity	Difference	Low identity	High identity	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	-2.14	5.59	7.72	-2.47	5.22	7.69
	(3.58)	(5.82)	(7.35)	(3.51)	(5.32)	(6.81)
Homogeneous	3.65	6.67	3.02	3.42	9.68**	6.26
	(3.91)	(4.39)	(6.13)	(3.85)	(3.96)	(5.73)
Religious leader						
Mixed	0.07	14.85***	14.78**	0.13	13.18***	13.05**
	(3.90)	(4.60)	(6.15)	(3.78)	(4.20)	(5.78)
Homogeneous	7.58**	-7.33	-14.91**	6.77**	-4.35	-11.13*
	(3.41)	(4.94)	(6.79)	(3.27)	(4.56)	(6.36)
Panel B. Predicted no-leader participation by identity category						
No-leader respondents in each category						
Mixed	61.88	52.42	-9.47**	61.88	52.42	-9.47***
	(2.29)	(3.51)	(4.19)	(2.21)	(2.86)	(3.62)
Homogeneous	57.48	55.84	-1.63	57.48	55.84	-1.63
	(2.05)	(2.87)	(3.52)	(2.14)	(2.70)	(3.44)
Observations	1,040			1,040		
Group clusters	336			336		
Groups containing respondents in each category (first / second)						
Chief, mixed	36 / 20			36 / 20		
Chief, homogeneous	33 / 23			33 / 23		
Pastor, mixed	29 / 27			29 / 27		
Pastor, homogeneous	29 / 27			29 / 27		

Notes: The difference column compares the leader effect in groups with higher ethnic identification with the effect in groups with lower identification. Baseline respondents ranked six identities; own-territory identification is scored from 1 (least important) to 6 (most important). We average available scores among the three ordinary-member seats, excluding the leader and including the respondent's own score where available: three scores in 330 groups and two in six. It is assigned to all 1,040 outcome respondents in 336 groups. High identification is strictly above the group median of 10/3; ties are low. There are 603 low- and 437 high-category respondents. Panel B shows predicted participation without a leader for each composition and identity category. We average over the no-leader residents in that category, retaining their observed covariates and fixed effects; intervals reflect uncertainty in the fitted model. Within each composition, we pool disclosure conditions when estimating leader effects, their interactions with the characteristic, and the category levels. Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds spillover and 11 LASSO-selected baseline controls; Appendix Section C.5 details their selection and missing-value treatment. Neither block adds separate individual-identity controls or interactions. Standard errors in parentheses are clustered by group and treat category membership as fixed. Stars use unadjusted exploratory tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Section 4.3 and Figure 5A discuss the estimates.

Table C-14 – GROUP ETHNIC IDENTIFICATION AND COETHNICITY IN MIXED GROUPS

	Baseline participation control			+ Spillover and LASSO		
	Different territory	Same territory	Difference	Different territory	Same territory	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Lower group identification	-2.67	-1.20	1.47	-2.80	-2.11	0.69
	(4.06)	(9.82)	(11.20)	(4.03)	(9.35)	(10.74)
Higher group identification	1.60	15.62	14.02	1.15	15.08	13.93
	(6.64)	(11.86)	(13.84)	(6.52)	(10.54)	(12.99)
Religious leader						
Lower group identification	-6.52	21.12**	27.64***	-5.23	17.00**	22.23**
	(4.47)	(8.70)	(10.25)	(4.24)	(8.57)	(9.90)
Higher group identification	16.69***	9.69	-7.00	14.76***	9.01	-5.75
	(5.56)	(11.93)	(14.21)	(5.10)	(10.68)	(12.76)
Panel B. Predicted no-leader participation by identity and coethnicity						
No-leader respondents in each category						
Lower group identification	61.30	64.00	2.70	61.30	64.00	2.70
	(2.77)	(6.23)	(7.30)	(2.52)	(6.42)	(7.23)
Higher group identification	54.81	45.10	-9.71	54.81	45.10	-9.71
	(3.71)	(9.27)	(10.37)	(3.13)	(8.26)	(9.39)
Observations	1,040			1,040		
Group clusters	336			336		
Mixed groups containing respondents in each category (different / same territory)						
Chief, lower identification	36 / 19			36 / 19		
Chief, higher identification	20 / 15			20 / 15		
Pastor, lower identification	29 / 18			29 / 18		
Pastor, higher identification	26 / 18			26 / 18		
No leader, lower identification	35 / 19			35 / 19		
No leader, higher identification	21 / 16			21 / 16		

Notes: Do shared ethnic origins matter more in groups that identify strongly with ethnicity? We compare residents of mixed groups who share the neighborhood chief's territorial origin with residents who come from other origins, separately by the group's ethnic identification. The difference subtracts effects for other-origin residents from effects for the chief's coethnics. We use the chief's origin to define coethnicity even when a group is assigned a pastor or no leader. Group identification is the average of ordinary members' available baseline own-territory scores, ranging from 1–6 with higher scores giving the identity more importance. Groups strictly above the fixed group median, 10/3, are higher; groups at the median or below are lower.

We retain all 1,040 residents in 336 groups in the regression, although the table displays mixed groups. In those groups, leader assignment, group identification, and coethnicity are fully interacted, with the corresponding category terms for no-leader residents. We also estimate homogeneous-group effects separately by group identification, leaving them out of this table. All effects pool disclosure. Panel B averages predictions over actual no-leader residents in each observed identity-by-coethnicity category, using their own covariates and fixed effects; its standard errors reflect model uncertainty. Higher-identification coethnics number 19 residents in 15 chief groups and 17 residents in 16 no-leader groups.

Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects. Spillover and the same 11 LASSO-selected baseline controls enter the second block, following the selection and missing-value treatment in Appendix Section C.5. Parentheses give group-clustered standard errors. Stars use unadjusted exploratory tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Section 4.3 discusses this breakdown of Figure 4A.

Table C-15 – PERCEIVED SECULAR AUTHORITY OF THE ASSIGNED LEADER

	Baseline participation control			+ Spillover and LASSO		
	Score ≤ 4	Score > 4	Difference	Score ≤ 4	Score > 4	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	-2.33	8.25*	10.58*	-2.10	6.76*	8.86*
	(3.44)	(4.75)	(6.02)	(3.31)	(3.95)	(4.97)
Homogeneous	2.90	7.85**	4.95	4.25	8.62**	4.36
	(3.65)	(3.92)	(4.98)	(3.35)	(4.01)	(4.82)
Religious leader						
Mixed	4.68	9.14**	4.45	5.40*	6.61*	1.20
	(3.38)	(3.79)	(4.24)	(3.26)	(3.64)	(4.11)
Homogeneous	1.31	-0.08	-1.39	1.69	1.52	-0.17
	(3.29)	(3.54)	(4.69)	(3.20)	(3.57)	(4.89)
Panel B. Implied no-leader counterfactuals						
Common no-leader comparison						
Mixed, pooled		58.33			58.33	
		(1.91)			(1.72)	
Homogeneous, pooled		56.79			56.79	
		(1.75)			(1.72)	
Observations		1,037			1,037	
Group clusters		335			335	
Groups containing respondents in each category (first / second)						
Chief, mixed		47 / 34			47 / 34	
Chief, homogeneous		53 / 43			53 / 43	
Pastor, mixed		52 / 38			52 / 38	
Pastor, homogeneous		51 / 39			51 / 39	

Notes: The difference column reports the effect in the second category minus the effect in the first. The columns distinguish pre-discussion secular-authority scores at or below 4 from scores above 4. We use meeting rosters to match each assessment to the assigned leader. Missing scores for 33 residents are filled with their group's observed mean; three residents without a group mean are excluded. No-leader references are pooled by composition because the assessment is undefined without a leader. Panel B shows predicted participation without a leader, averaged over the no-leader residents in each composition using their observed covariates and fixed effects. Within each composition, we pool disclosure conditions when estimating leader effects, their interactions with the characteristic, and any separate category levels. Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds spillover and 11 LASSO-selected baseline controls; Appendix Section C.5 details their selection and missing-value treatment. Standard errors in parentheses are clustered by group and treat category membership as fixed. Stars use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Section 4.3 and Figure 4B discuss the estimates.

Table C-16 – OPINION OF THE ASSIGNED LEADER

	Baseline participation control			+ Spillover and LASSO		
	Index at/below median	Index above median	Difference	Index at/below median	Index above median	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	-4.05	8.36**	12.41**	-3.51	6.79*	10.30**
	(3.56)	(4.00)	(5.23)	(3.38)	(3.56)	(4.51)
Homogeneous	0.95	8.12**	7.16	2.68	8.71**	6.03
	(4.18)	(3.44)	(5.05)	(4.04)	(3.37)	(4.98)
Religious leader						
Mixed	4.90	7.70*	2.80	4.07	7.56**	3.49
	(3.51)	(3.94)	(4.73)	(3.39)	(3.73)	(4.54)
Homogeneous	2.92	-1.23	-4.14	1.64	1.56	-0.08
	(3.92)	(3.18)	(5.04)	(3.67)	(3.12)	(4.89)
Panel B. Implied no-leader counterfactuals						
Common no-leader comparison						
Mixed, pooled		58.33			58.33	
		(1.93)			(1.74)	
Homogeneous, pooled		56.79			56.79	
		(1.76)			(1.72)	
Observations		1,037			1,037	
Group clusters		335			335	
Groups containing respondents in each category (first / second)						
Chief, mixed		45 / 39			45 / 39	
Chief, homogeneous		46 / 48			46 / 48	
Pastor, mixed		48 / 46			48 / 46	
Pastor, homogeneous		44 / 47			44 / 47	

Notes: The difference column reports the effect in the second category minus the effect in the first. The columns distinguish opinion-index scores at or below the median from scores above it. The four-item pre-discussion opinion index measures trustworthiness, hard work, morality, and inspiration. The median opinion-index score is 0.20195; ties enter the first category. We use meeting rosters to match each assessment to the assigned leader. Missing indices for 33 residents are filled with their group's observed mean; three residents without a group mean are excluded. No-leader references are pooled by composition because the assessment is undefined without a leader. Panel B shows predicted participation without a leader, averaged over the no-leader residents in each composition using their observed covariates and fixed effects. Within each composition, we pool disclosure conditions when estimating leader effects, their interactions with the characteristic, and any separate category levels. Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds spillover and 11 LASSO-selected baseline controls; Appendix Section C.5 details their selection and missing-value treatment. Standard errors in parentheses are clustered by group and treat category membership as fixed. Stars use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Section 4.3 discusses the estimates.

Table C-17 – INTRINSIC RELIGIOSITY AND LEADER EFFECTIVENESS

	Baseline participation control			+ Spillover and LASSO		
	Score ≤ 3.2	Score > 3.2	Difference	Score ≤ 3.2	Score > 3.2	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	-1.00	0.76	1.76	-2.27	1.92	4.19
	(4.83)	(4.27)	(7.04)	(4.59)	(4.14)	(6.68)
Homogeneous	3.70	3.79	0.09	5.21	4.92	-0.30
	(4.47)	(4.52)	(6.83)	(4.34)	(4.12)	(6.32)
Religious leader						
Mixed	-0.58	11.75**	12.33*	0.71	9.89**	9.18
	(4.88)	(4.61)	(7.44)	(5.01)	(4.27)	(7.36)
Homogeneous	-4.01	5.34	9.35	-2.18	5.28	7.46
	(4.43)	(3.61)	(6.29)	(4.20)	(3.46)	(6.00)
Panel B. Implied no-leader counterfactuals						
Common no-leader comparison						
Mixed	59.21	55.27	-3.94	58.06	56.62	-1.44
	(3.50)	(3.02)	(5.09)	(3.52)	(2.70)	(5.00)
Homogeneous	57.19	52.17	-5.02	55.47	52.27	-3.20
	(3.16)	(2.75)	(4.70)	(3.08)	(2.48)	(4.32)
Observations		1,034			1,034	
Group clusters		336			336	
Groups containing respondents in each category (first / second)						
Chief, mixed		48 / 49			48 / 49	
Chief, homogeneous		50 / 44			50 / 44	
Pastor, mixed		49 / 50			49 / 50	
Pastor, homogeneous		48 / 49			48 / 49	

Notes: The difference column reports the effect in the second category minus the effect in the first. Intrinsic religiosity averages baseline items 1, 3 (reversed), 4, 5, and 9 (reversed), requiring at least three valid responses. Responses range from 1 to 5. The nine-item survey adaptation follows the intrinsic/extrinsic distinction of Allport and Ross (1967) and revised measurement of Gorsuch and McPherson (1989). The discrete cutpoint 3.2 minimizes the difference in category sizes, using only the scale distribution: 498 residents are at or below it and 536 above. Six residents lack usable baseline items; 1,034 residents in 336 groups remain. Panel B reports common no-leader predictions within each category. Within each composition, we pool disclosure conditions when estimating leader effects, their interactions with the characteristic, and the category levels. Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds spillover and 11 LASSO-selected baseline controls; Appendix Section C.5 details their selection and missing-value treatment. Standard errors in parentheses are clustered by group and treat category membership as fixed. Stars use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Section 4.3 and Figure 5B discuss the estimates.

Table C-18 – EXTRINSIC RELIGIOSITY AND LEADER EFFECTIVENESS

	Baseline participation control			+ Spillover and LASSO		
	Score $\leq$ 2.75	Score $>$ 2.75	Difference	Score $\leq$ 2.75	Score $>$ 2.75	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	-6.41	6.21	12.62*	-4.30	3.46	7.76
	(5.00)	(4.22)	(7.19)	(4.52)	(3.84)	(6.28)
Homogeneous	3.76	4.16	0.40	5.96	4.77	-1.19
	(4.00)	(4.64)	(6.48)	(3.91)	(4.38)	(6.20)
Religious leader						
Mixed	7.01	5.14	-1.87	7.73*	3.63	-4.10
	(4.51)	(3.98)	(6.17)	(4.35)	(3.86)	(6.05)
Homogeneous	4.07	-2.18	-6.25	4.49	-0.95	-5.44
	(3.17)	(4.44)	(5.82)	(3.20)	(4.25)	(5.80)
Panel B. Implied no-leader counterfactuals						
Common no-leader comparison						
Mixed	60.30	54.16	-6.14	60.95	54.24	-6.71
	(3.50)	(2.65)	(4.65)	(3.29)	(2.33)	(4.35)
Homogeneous	52.69	56.13	3.44	52.41	54.86	2.46
	(2.44)	(3.03)	(4.17)	(2.48)	(2.94)	(4.17)
Observations	1,034			1,034		
Group clusters	336			336		
Groups containing respondents in each category (first / second)						
Chief, mixed	46 / 45			46 / 45		
Chief, homogeneous	44 / 46			44 / 46		
Pastor, mixed	42 / 47			42 / 47		
Pastor, homogeneous	46 / 44			46 / 44		

Notes: The difference column reports the effect in the second category minus the effect in the first. Extrinsic religiosity averages baseline items 2, 6, 7, and 8, pooling personal comfort and social contact with equal weight per item and requiring at least three valid responses. Responses range from 1 to 5. The nine-item survey adaptation follows the intrinsic/extrinsic distinction of Allport and Ross (1967) and revised measurement of Gorsuch and McPherson (1989). The discrete cutpoint 2.75 minimizes the difference in category sizes, using only the scale distribution: 493 residents are at or below it and 541 above. Six residents lack usable baseline items; 1,034 residents in 336 groups remain. Panel B reports common no-leader predictions within each category. Within each composition, we pool disclosure conditions when estimating leader effects, their interactions with the characteristic, and the category levels. Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds spillover and 11 LASSO-selected baseline controls; Appendix Section C.5 details their selection and missing-value treatment. Standard errors in parentheses are clustered by group and treat category membership as fixed. Stars use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Section 4.3 discusses the estimates.

Table C-19 – DENOMINATIONAL TIES AND LEADER EFFECTIVENESS

	Denomination relative to the neighborhood pastor		
	Different denomination	Same denomination	Difference
Panel A. Leader effects relative to no leader			
Traditional leader			
Mixed	1.39	-6.70	-8.09
	(3.31)	(9.46)	(10.66)
Homogeneous	3.14	11.99	8.85
	(3.34)	(9.15)	(10.20)
Religious leader			
Mixed	7.61**	-3.69	-11.30
	(3.11)	(10.14)	(10.85)
Homogeneous	-0.96	12.26	13.23
	(2.90)	(8.33)	(9.29)
Panel B. Implied no-leader counterfactuals			
No leader, mixed	56.93	55.66	-1.26
	(2.22)	(5.87)	(6.55)
No leader, homogeneous	54.89	50.26	-4.64
	(1.95)	(5.51)	(5.95)
Observations		1,032	
Group clusters		336	

Notes: The difference column reports the effect in the second category minus the effect in the first. Panel B reports model-implied no-leader participation in percentage points. Predictions set leader terms to zero and baseline participation to its estimation-sample mean, averaging the fixed effects and category levels over the combined chief and pastor participants within each composition and category. Both categories compare the respondent's denomination with the neighborhood pastor's denomination. The comparison is defined for residents in every assignment arm, including no leader. Effects and interactions are pooled across disclosure within each composition; the model also includes separate composition-specific category levels, baseline participation, and neighborhood-by-composition-by-disclosure fixed effects. The sample includes 1,032 residents in 336 groups. Forty-two pastor-arm groups contain at least one resident who shares the pastor's denomination. Standard errors in parentheses are clustered by accountability group. Section 4.3 discusses these estimates. Stars use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.6 Which Leaders Can Exercise Their Authority?

We ask whether the assigned leader is equipped to exercise the relevant form of authority. Tables C-20–C-22 estimate Equation (2), pooling both leader effects and their interactions with leader characteristics across disclosure within each composition. The first specification controls for baseline participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds spillover exposure and 11 LASSO-selected baseline controls. We include leader attributes through their interactions with leader assignment and omit a separate level term, since they are undefined for residents without an assigned leader. Standard errors are clustered by randomized group; the tables also report neighborhood-clustered standard errors and tests of the higher-minus-lower contrasts. These comparisons draw on only 28 leaders of each type, and we do not adjust the exploratory tests for multiple comparisons.

Chiefs benefit from local roots; pastors from religious conviction. If chiefs' authority comes from being embedded in local ethnic networks, chiefs who have lived in the neighborhood longer should be better able to sustain participation. A longer time living in the neighborhood gives them more knowledge of residents, stronger local ties, and greater standing in the community. These are exactly the types of relationships through which chiefs would be able to enforce participation in local collective action. Pastors, on the other hand, should depend less on local tenure because their authority should not rely as strongly on neighborhood networks or local standing. Instead, their effectiveness should depend more on how strongly they themselves believe that God's oversight rewards cooperation. Thus, we expect pastor effects to be stronger among pastors with stronger beliefs in divine enforcement.

Table C-20 shows how chief effects vary with avenue tenure, comparing chiefs who have lived on the avenue for 26 years or fewer with chiefs with longer tenure. In homogeneous groups, chiefs with longer tenure raise participation by 10.3 percentage points (SE 4.5), compared with 0.1 points for chiefs with shorter tenure (SE 3.2). The 10.2-point difference is suggestive (SE 5.2; $p = 0.051$); it remains similar with additional controls, although neighborhood-clustered inference is less precise ($p = 0.143$ without additional controls and $p = 0.152$ with them; Table C-20). The chief-tenure difference in mixed groups is close to zero. Pastor effects do not increase with their own avenue tenure: in homogeneous groups, their effect falls from 7.1 to -5.4 points, a -12.6-point difference (SE 4.4; $p = 0.005$). Local rootedness therefore has different associations with the two leaders' effectiveness.

Table C-21 turns to pastors' conviction in divine moral enforcement. We combine questions from Lang et al. (2019) on whether God rewards good people and punishes bad people into a single measure, then divide leaders into lower and higher categories within each type. In mixed groups, pastors with stronger convictions raise participation by 12.2 percentage points (SE 3.7), compared with 0.6 points for pastors with weaker convictions (SE 4.1). The 11.6-point difference is statistically significant (SE 5.1; $p = 0.024$). The adjusted contrast remains positive with neighborhood-clustered inference ($p = 0.025$). In homogeneous groups, the corresponding 7.4-point difference is less precise (SE 4.6; $p = 0.107$). Chief effects do not significantly vary with chiefs' conviction (Table C-21). These comparisons support the interpretation that pastors' reach depends on their own conviction in the religious obligation they invoke.

Leadership roles across institutions. Some leaders who participated in the campaign were primarily recognized for one role but also held a position in the other institution. Thus, some chiefs also reported serving as pastors, and some pastors reported serving as chiefs. Table C-22 compares leaders who head both institutions with those who do not. The cross-institution sample is small: five chiefs also head a religious institution, and seven pastors also head a traditional institution. In mixed groups, the chief effect is 7.4 points among those who head both institutions (SE 5.4), compared with -0.7 points among the other chiefs (SE 3.2). The 8.1-point difference is imprecise (SE 6.0; $p = 0.178$). Pastors who also head a traditional institution do not have larger effects in homogeneous groups; their higher-minus-lower difference is -4.8 points (SE 4.4; $p = 0.276$). The directions are consistent with religious roles extending chiefs' reach across territorial boundaries, but these pooled estimates do not establish an asymmetry between the two types of leader.

Table C-20 – LEADER EFFECTS BY YEARS OF RESIDENCE ON THE AVENUE

	Baseline participation control			+ Spillover and LASSO		
	Chief: $\leq$ 26 years Pastor: $\leq$ 27 years	Chief: $>$ 26 years Pastor: $>$ 27 years	Difference	Chief: $\leq$ 26 years Pastor: $\leq$ 27 years	Chief: $>$ 26 years Pastor: $>$ 27 years	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	0.53	0.97	0.43	0.17	0.59	0.42
	(3.87)	(3.71)	(5.02)	(3.83)	(3.42)	(4.81)
	[5.06]	[4.71]	[6.35]	[4.59]	[3.59]	[5.29]
Homogeneous	0.11	10.32**	10.21*	1.95	10.64**	8.69*
	(3.20)	(4.47)	(5.21)	(3.12)	(4.19)	(4.91)
	[3.40]	[5.82]	[6.76]	[2.80]	[5.44]	[5.89]
Religious leader						
Mixed	5.21	7.38*	2.18	5.35	6.29*	0.94
	(4.01)	(3.84)	(5.21)	(3.91)	(3.62)	(5.05)
	[4.96]	[4.53]	[6.13]	[4.80]	[3.83]	[5.68]
Homogeneous	7.14**	-5.42	-12.56***	7.50**	-4.13	-11.63***
	(3.32)	(3.30)	(4.44)	(2.94)	(3.25)	(4.09)
	[3.92]	[4.34]	[5.95]	[3.25]	[4.05]	[5.20]
Panel B. Implied no-leader counterfactuals						
Common no-leader comparison						
Mixed	58.33	58.33	–	58.33	58.33	–
	(1.99)	(1.99)		(1.79)	(1.79)	
Homogeneous	56.79	56.79	–	56.79	56.79	–
	(1.77)	(1.77)		(1.75)	(1.75)	
Observations	1,040			1,040		
Group clusters	336			336		
Neighborhood clusters	28			28		
Pretreatment participation control	Yes			Yes		
Spillover and LASSO controls	No			Yes		
Traditional, mixed: neighborhood p	0.946			0.937		
Traditional, homogeneous: neighborhood p	0.143			0.152		
Religious, mixed: neighborhood p	0.726			0.869		
Religious, homogeneous: neighborhood p	0.044			0.034		

Notes: The difference column reports the effect in the second category minus the effect in the first. Residence is measured in years living on the avenue. The first category includes chiefs with at most 26 years and pastors with at most 27 years. There are 15 chiefs in the first category and 13 in the second, and 14 pastors in each category. Within each composition, we pool disclosure conditions for both leader effects and their interactions with the characteristic. Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects; the second adds spillover exposure and 11 LASSO-selected baseline controls. The characteristic enters only through its interactions with leader assignment. Parentheses report group-clustered standard errors; square brackets report neighborhood-clustered standard errors for the same estimates. Tests at the foot of the table compare the two categories using neighborhood clustering. Panel B shows predicted participation without a leader, averaged over the no-leader residents in each composition using their observed covariates and fixed effects. The same no-leader mean appears in both categories because residents without an assigned leader cannot be divided by that leader's characteristic. Effects are percentage points. Section C.6 discusses the results. Stars use unadjusted group-clustered tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C-21 – LEADER EFFECTS BY BELIEF IN DIVINE REWARDS AND PUNISHMENTS

	Baseline participation control			+ Spillover and LASSO		
	Chief: score ≤ 3.5 Pastor: score ≤ 4.5	Chief: score > 3.5 Pastor: score > 4.5	Difference	Chief: score ≤ 3.5 Pastor: score ≤ 4.5	Chief: score > 3.5 Pastor: score > 4.5	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	3.00	-1.38	-4.38	3.12	-2.26	-5.38
	(3.60)	(3.86)	(4.93)	(3.66)	(3.54)	(4.82)
	[4.11]	[5.10]	[5.71]	[3.66]	[4.05]	[4.73]
Homogeneous	6.64**	3.10	-3.53	7.21**	4.76	-2.45
	(3.30)	(4.44)	(5.27)	(3.37)	(4.06)	(4.98)
	[3.54]	[5.91]	[7.06]	[3.95]	[5.00]	[6.40]
Religious leader						
Mixed	0.59	12.17***	11.58**	-0.15	11.92***	12.06**
	(4.09)	(3.66)	(5.12)	(3.91)	(3.51)	(4.97)
	[5.06]	[4.18]	[5.73]	[4.30]	[3.95]	[5.10]
Homogeneous	-2.79	4.57	7.37	-0.41	3.72	4.13
	(3.27)	(3.47)	(4.56)	(3.04)	(3.31)	(4.25)
	[3.54]	[4.67]	[6.13]	[3.13]	[4.31]	[5.47]
Panel B. Implied no-leader counterfactuals						
Common no-leader comparison						
Mixed	58.33	58.33	–	58.33	58.33	–
	(2.01)	(2.01)		(1.78)	(1.78)	
Homogeneous	56.79	56.79	–	56.79	56.79	–
	(1.74)	(1.74)		(1.72)	(1.72)	
Observations	1,040			1,040		
Group clusters	336			336		
Neighborhood clusters	28			28		
Pretreatment participation control	Yes			Yes		
Spillover and LASSO controls	No			Yes		
Traditional, mixed: neighborhood p	0.450			0.265		
Traditional, homogeneous: neighborhood p	0.621			0.705		
Religious, mixed: neighborhood p	0.053			0.025		
Religious, homogeneous: neighborhood p	0.240			0.457		

Notes: We compare leader effects according to the leader's baseline beliefs that God rewards good behavior and punishes improper behavior, combined in an index. The cutpoint is 3.5 for chiefs and 4.5 for pastors. There are 14 leaders of each type on each side of the cutpoint, with scores equal to it entering the first category.

The characteristic enters only through its interaction with leader assignment. We pool disclosure within each composition for both the leader effects and these interactions. The difference reported is the second category's effect less the first, in percentage points. Participation before treatment and neighborhood-by-composition-by-disclosure fixed effects enter both blocks. We add spillover exposure and the same 11 LASSO-selected baseline controls in the second block. Residents assigned no leader have no leader characteristic on which to divide them. For that reason, the reference mean in Panel B is common to both categories. We calculate it within each composition from the no-leader residents, predicting participation with each resident's observed covariates and fixed effects and then averaging these predictions. For each estimate we report group-clustered standard errors in parentheses and neighborhood-clustered standard errors in square brackets. The category tests at the bottom use neighborhood clustering. See Section C.6 for the results. Stars use unadjusted group-clustered tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C-22 – LEADER EFFECTS BY JOINT CHIEF AND PASTOR ROLES

	Baseline participation control			+ Spillover and LASSO		
	Does not hold both roles	Chief and pastor roles	Difference	Does not hold both roles	Chief and pastor roles	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed	-0.74	7.37	8.11	-1.02	6.63	7.65
	(3.17)	(5.41)	(6.01)	(3.09)	(5.46)	(6.17)
	[4.18]	[5.96]	[6.63]	[3.71]	[3.92]	[4.82]
Homogeneous	6.23*	-1.58	-7.81	7.55**	-1.15	-8.69
	(3.32)	(3.99)	(4.84)	(3.12)	(4.67)	(5.34)
	[3.88]	[5.25]	[6.24]	[3.46]	[6.40]	[6.90]
Religious leader						
Mixed	5.43*	8.82	3.39	4.68	9.06*	4.38
	(3.23)	(5.64)	(6.18)	(3.12)	(5.39)	(6.02)
	[4.51]	[2.84]	[4.73]	[3.97]	[4.05]	[5.28]
Homogeneous	2.05	-2.76	-4.81	3.32	-3.25	-6.58
	(2.94)	(3.73)	(4.41)	(2.74)	(3.46)	(4.03)
	[3.59]	[3.83]	[5.06]	[3.26]	[3.48]	[4.63]
Panel B. Implied no-leader counterfactuals						
Common no-leader comparison						
Mixed	58.33	58.33	–	58.33	58.33	–
	(2.01)	(2.01)		(1.81)	(1.81)	
Homogeneous	56.79	56.79	–	56.79	56.79	–
	(1.78)	(1.78)		(1.76)	(1.76)	
Observations		1,040			1,040	
Group clusters		336			336	
Neighborhood clusters		28			28	
Pretreatment participation control		Yes			Yes	
Spillover and LASSO controls		No			Yes	
Traditional, mixed: neighborhood p		0.232			0.124	
Traditional, homogeneous: neighborhood p		0.222			0.218	
Religious, mixed: neighborhood p		0.480			0.413	
Religious, homogeneous: neighborhood p		0.350			0.167	

Notes: The difference column reports the effect in the second category minus the effect in the first. The second category comprises leaders recorded as both an avenue/customary chief and a pastor, cure, or reverend. There are 5 such chiefs and 7 such pastors. The first category contains the remaining 23 chiefs and 21 pastors. Within each composition, we pool disclosure conditions for both leader effects and their interactions with the characteristic. Both blocks include baseline participation and neighborhood-by-composition-by-disclosure fixed effects; the second adds spillover exposure and 11 LASSO-selected baseline controls. The characteristic enters only through its interactions with leader assignment. Parentheses report group-clustered standard errors; square brackets report neighborhood-clustered standard errors for the same estimates. Tests at the foot of the table compare the two categories using neighborhood clustering. Panel B shows predicted participation without a leader, averaged over the no-leader residents in each composition using their observed covariates and fixed effects. The same no-leader mean appears in both categories because residents without an assigned leader cannot be divided by that leader's characteristic. Effects are percentage points. Section C.6 discusses the results. Stars use unadjusted group-clustered tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Source of Enforcement

D.1 Main Effects, Alternative Outcomes, and Inference

Table D-1 – LEADER EFFECTS BY COMPOSITION AND DISCLOSURE

	Composition and disclosure		
	Baseline	+ Spillover	+ LASSO
Panel A. Leader effects relative to no leader			
Traditional leader			
Mixed, without disclosure	-0.29	-0.31	-1.11
	(3.48)	(3.48)	(3.48)
Mixed, with disclosure	1.75	1.67	1.75
	(4.53)	(4.52)	(4.16)
Homogeneous, without disclosure	-0.31	-0.34	1.37
	(4.46)	(4.45)	(4.27)
Homogeneous, with disclosure	9.95***	10.13***	10.62***
	(3.63)	(3.63)	(3.47)
Religious leader			
Mixed, without disclosure	9.18**	9.16**	9.02**
	(3.97)	(3.98)	(3.83)
Mixed, with disclosure	3.41	3.40	2.48
	(4.30)	(4.32)	(3.99)
Homogeneous, without disclosure	1.66	1.67	1.64
	(3.81)	(3.80)	(3.56)
Homogeneous, with disclosure	0.01	-0.01	1.71
	(3.27)	(3.26)	(3.07)
Panel B. Treatment-effect contrasts			
Mixed, with: T − R	-1.66	-1.73	-0.73
	(4.40)	(4.41)	(4.15)
Mixed, without: T − R	-9.47**	-9.47**	-10.13**
	(3.98)	(4.01)	(4.17)
Homogeneous, with: T − R	9.94**	10.13**	8.91**
	(4.03)	(4.00)	(3.67)
Homogeneous, without: T − R	-1.97	-2.01	-0.27
	(4.13)	(4.12)	(3.90)
Δ_E: H, with − M, without	19.41***	19.60***	19.04***
	(5.67)	(5.69)	(5.57)
Panel C. No-leader reference means and diagnostics			
Mixed, without disclosure	58.88	58.88	58.88
	(2.46)	(2.44)	(2.24)
Mixed, with disclosure	57.79	57.79	57.79
	(3.14)	(3.14)	(2.81)
Homogeneous, without disclosure	57.79	57.79	57.79
	(2.91)	(2.91)	(2.76)
Homogeneous, with disclosure	55.80	55.80	55.80
	(1.96)	(1.98)	(2.05)
Wald: all no-leader cells equal	p = 0.789	p = 0.790	p = 0.781
Observations	1,040	1,040	1,040
Group-neighborhood clusters	336	336	336
Neighborhoods	28	28	28
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes
Pretreatment participation control	Yes	Yes	Yes
25m spillover control	No	Yes	Yes
LASSO-selected baseline controls	No	No	Yes

Notes: Panel A reports leader effects relative to no leader; Panel B reports direct leader contrasts; Panel C reports no-leader reference means. Leader effects vary by composition and disclosure; Δ_E compares the traditional-minus-religious gap in homogeneous groups with disclosure with the gap in mixed groups without disclosure. The outcome is average participation on six eligible days after Meeting 1 through the Meeting 2 day, in percentage points. Every column controls for pretreatment participation and neighborhood-by-composition-by-disclosure fixed effects. The second adds the 25-meter spillover indicator; the third also adds 11 LASSO-selected baseline controls. Group-clustered standard errors appear in parentheses. In the preferred specification, disclosure changes the chief effect in homogeneous groups by 9.24 percentage points (SE 5.52; $p = 0.095$) and the pastor effect in mixed groups by −6.55 points (SE 5.49; $p = 0.233$); both contrasts use the full coefficient covariance matrix. Section 5 and Figure 6 discuss these estimates. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D-2 – LABOR CONTRIBUTION OUTCOMES BY COMPOSITION AND DISCLOSURE

	Levels				Asinh robustness
	Participation rate	Attendance days	Total hours	Total effort	Asinh total hours
Panel A. Leader effects relative to no leader					
Traditional leader					
Mixed, without disclosure	-0.011	-0.027	-0.669	-0.256	-0.033
	(0.035)	(0.217)	(0.855)	(0.737)	(0.163)
Mixed, with disclosure	0.017	0.103	0.027	0.074	0.015
	(0.042)	(0.246)	(0.906)	(0.816)	(0.155)
Homogeneous, without disclosure	0.014	0.049	0.663	0.386	0.031
	(0.043)	(0.248)	(0.906)	(0.812)	(0.151)
Homogeneous, with disclosure	0.106***	0.583***	2.098**	1.968***	0.440***
	(0.035)	(0.207)	(0.887)	(0.703)	(0.134)
Religious leader					
Mixed, without disclosure	0.090**	0.580**	1.701*	1.522**	0.309**
	(0.038)	(0.235)	(0.937)	(0.756)	(0.151)
Mixed, with disclosure	0.025	0.151	0.001	0.555	0.015
	(0.040)	(0.238)	(0.931)	(0.809)	(0.154)
Homogeneous, without disclosure	0.016	0.035	-0.032	-0.137	0.029
	(0.036)	(0.210)	(0.806)	(0.706)	(0.129)
Homogeneous, with disclosure	0.017	0.108	0.130	0.519	0.131
	(0.031)	(0.186)	(0.816)	(0.690)	(0.113)
Panel B. Treatment-effect contrasts					
Mixed, without: T − R	-0.101**	-0.607**	-2.369**	-1.778**	-0.342**
	(0.042)	(0.256)	(0.965)	(0.843)	(0.174)
	[Holm p = 0.058]	[Holm p = 0.058]	[Holm p = 0.058]	[Holm p = 0.058]	[Holm p = 0.050]
Mixed, with: T − R	-0.007	-0.048	0.025	-0.481	0.000
	(0.042)	(0.249)	(0.961)	(0.834)	(0.160)
	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]
Homogeneous, without: T − R	-0.003	0.015	0.695	0.523	0.003
	(0.039)	(0.229)	(0.821)	(0.767)	(0.145)
	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 0.986]
Homogeneous, with: T − R	0.089**	0.474**	1.967**	1.449*	0.309**
	(0.037)	(0.217)	(0.903)	(0.749)	(0.136)
	[Holm p = 0.063]	[Holm p = 0.088]	[Holm p = 0.088]	[Holm p = 0.088]	[Holm p = 0.024]
Δ_E	0.190***	1.081***	4.337***	3.226***	0.651***
	(0.056)	(0.336)	(1.324)	(1.129)	(0.221)
	[Holm p = 0.003]	[Holm p = 0.004]	[Holm p = 0.004]	[Holm p = 0.005]	[Holm p = 0.003]
Panel C. No-leader reference means and diagnostics					
Mixed, without disclosure	0.589	3.478	13.438	11.571	2.683
	(0.022)	(0.140)	(0.578)	(0.460)	(0.099)
Mixed, with disclosure	0.578	3.413	12.983	11.098	2.654
	(0.028)	(0.165)	(0.615)	(0.553)	(0.105)
Homogeneous, without disclosure	0.578	3.457	12.877	11.272	2.622
	(0.028)	(0.161)	(0.626)	(0.527)	(0.095)
Homogeneous, with disclosure	0.558	3.272	12.529	10.473	2.565
	(0.021)	(0.126)	(0.573)	(0.461)	(0.078)
Wald: four cells equal	p = 0.781	p = 0.690	p = 0.736	p = 0.391	p = 0.798
Observations	1,040	1,040	1,040	1,040	1,040
Group-neighborhood clusters	336	336	336	336	336
Neighborhoods	28	28	28	28	28
Matched pretreatment outcome control	Yes	Yes	Yes	Yes	Yes
25m spillover control	Yes	Yes	Yes	Yes	Yes
Frozen participation LASSO controls	Yes	Yes	Yes	Yes	Yes
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes	Yes	Yes

Notes: We measure contributions over the six days after Meeting 1 through the day of Meeting 2, excluding the first-meeting day. Panel A compares each leader assignment with no leader. Panel B reports direct traditional-minus-religious contrasts and Δ_E. Standard errors in parentheses are clustered by group within neighborhood. Brackets report Holm-adjusted p-values for the same contrast within each outcome family. Panel C reports model-implied no-leader means and equality tests. Models include the matched pre-treatment outcome, the 25-meter spillover indicator, the same 11 baseline controls selected for primary participation, and neighborhood × disclosure × composition fixed effects. We use the controls selected for the main participation outcome across measures and observation windows. Participation is the daily rate; attendance is the number of labor days; hours and effort are window totals, with nonattendance coded zero. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The inverse-hyperbolic-sine transformation of hours retains residents who contributed no labor, so the sample remains the same. We apply the Holm correction separately to the four outcomes in levels and to the single asinh-hours outcome.

Table D-3 – CASH AND IN-KIND CONTRIBUTIONS BY COMPOSITION AND DISCLOSURE

	Levels				Asinh robustness	
	Any cash contribution	Cash total	Any in-kind contribution	In-kind value	Asinh cash total	Asinh in-kind value
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed, without disclosure	0.014	31.786	-0.074*	-258.738**	0.137	-0.653**
	(0.022)	(44.980)	(0.041)	(127.396)	(0.173)	(0.318)
Mixed, with disclosure	0.046*	-1.842	0.065	149.692	0.305*	0.518
	(0.025)	(30.774)	(0.047)	(154.389)	(0.182)	(0.366)
Homogeneous, without disclosure	0.012	6.736	-0.001	-404.699*	0.087	-0.180
	(0.018)	(8.137)	(0.058)	(216.039)	(0.119)	(0.459)
Homogeneous, with disclosure	0.013	-9.239	0.015	-1861.438	0.084	0.061
	(0.020)	(27.015)	(0.054)	(1332.055)	(0.150)	(0.428)
Religious leader						
Mixed, without disclosure	0.002	5.595	-0.011	-96.757	0.034	-0.112
	(0.026)	(29.524)	(0.039)	(138.257)	(0.178)	(0.300)
Mixed, with disclosure	0.010	-19.699	0.117**	81.611	0.059	0.950**
	(0.020)	(29.577)	(0.054)	(140.508)	(0.148)	(0.427)
Homogeneous, without disclosure	0.012	3.165	0.067	-0.268	0.076	0.486
	(0.020)	(8.133)	(0.059)	(237.941)	(0.133)	(0.472)
Homogeneous, with disclosure	-0.010	-29.877	-0.065	-1919.618	-0.086	-0.588
	(0.018)	(24.873)	(0.055)	(1336.608)	(0.131)	(0.441)
Panel B. Treatment-effect contrasts						
Mixed, without: T − R	0.012	26.190	-0.064	-161.981	0.103	-0.540
	(0.031)	(48.356)	(0.046)	(122.633)	(0.232)	(0.347)
	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 0.666]	[Holm p = 0.666]	[Holm p = 0.658]	[Holm p = 0.241]
Mixed, with: T − R	0.036	17.857	-0.052	68.082	0.247	-0.432
	(0.025)	(24.618)	(0.057)	(187.360)	(0.182)	(0.455)
	[Holm p = 0.647]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 0.350]	[Holm p = 0.350]
Homogeneous, without: T − R	-0.000	3.571	-0.067	-404.431*	0.011	-0.666
	(0.017)	(6.854)	(0.062)	(208.376)	(0.111)	(0.492)
	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 0.830]	[Holm p = 0.212]	[Holm p = 0.922]	[Holm p = 0.353]
Homogeneous, with: T − R	0.023	20.638	0.081	58.181	0.170	0.649
	(0.020)	(21.559)	(0.058)	(940.972)	(0.144)	(0.449)
	[Holm p = 0.735]	[Holm p = 0.735]	[Holm p = 0.647]	[Holm p = 0.951]	[Holm p = 0.299]	[Holm p = 0.299]
Δ_E	0.011	-5.552	0.144*	220.161	0.067	1.189**
	(0.037)	(52.956)	(0.074)	(948.659)	(0.273)	(0.568)
	[Holm p = 1.000]	[Holm p = 1.000]	[Holm p = 0.203]	[Holm p = 1.000]	[Holm p = 0.805]	[Holm p = 0.074]
Panel C. No-leader reference means and diagnostics						
Mixed, without disclosure	0.033	38.587	0.293	572.826	0.225	2.334
	(0.010)	(17.678)	(0.025)	(100.143)	(0.068)	(0.194)
Mixed, with disclosure	0.011	38.043	0.239	514.130	0.096	1.909
	(0.014)	(25.631)	(0.031)	(67.275)	(0.108)	(0.235)
Homogeneous, without disclosure	0.011	5.435	0.261	747.826	0.075	2.123
	(0.014)	(6.318)	(0.037)	(172.380)	(0.096)	(0.296)
Homogeneous, with disclosure	0.022	38.043	0.315	2490.217	0.170	2.531
	(0.013)	(21.843)	(0.037)	(1201.373)	(0.099)	(0.299)
Wald: four cells equal	p = 0.526	p = 0.122	p = 0.354	p = 0.236	p = 0.558	p = 0.349
Observations	1,040	1,040	1,040	1,040	1,040	1,040
Group-neighborhood clusters	336	336	336	336	336	336
Neighborhoods	28	28	28	28	28	28
Matched pretreatment outcome control	Yes	Yes	Yes	Yes	Yes	Yes
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes	Yes	Yes	Yes

Notes: We measure contributions over the six days after Meeting 1 through the day of Meeting 2, excluding the first-meeting day. Panel A compares each leader assignment with no leader. Panel B reports direct traditional-minus-religious contrasts and Δ_E. Standard errors in parentheses are clustered by group within neighborhood. Brackets report Holm-adjusted p-values for the same contrast within each outcome family. Panel C reports model-implied no-leader means and equality tests. Models include the matched pre-treatment outcome and neighborhood × disclosure × composition fixed effects. For cash and in-kind contributions, we report whether a resident contributed during the window and the total value in Congolese francs (CDF). Combined material contributions are the sum of cash and the recorded value of in-kind contributions. Before estimation, we validate the source-data identity combined material value = cash + in-kind. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The inverse-hyperbolic-sine specifications retain residents who made no contribution during the window. We apply the Holm correction separately to the four outcomes in levels and to the two asinh outcomes.

Table D-4 – RANDOMIZATION INFERENCE AND ALTERNATIVE CLUSTERING BY COMPOSITION AND DISCLOSURE

	Randomization inference			Alternative clustering		
	Estimate	Clustered SE	RI p	Group	Leader cluster	Neighborhood
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed, without disclosure	-0.29	(3.48)	0.951	-0.29	-0.29	-0.29
				(3.48)	(3.46)	(3.65)
Mixed, with disclosure	1.75	(4.53)	0.696	1.75	1.75	1.75
				(4.53)	(4.55)	(5.48)
Homogeneous, without disclosure	-0.31	(4.46)	0.943	-0.31	-0.31	-0.31
				(4.46)	(4.48)	(5.60)
Homogeneous, with disclosure	9.95***	(3.63)	0.028	9.95***	9.95***	9.95**
				(3.63)	(3.65)	(4.22)
Religious leader						
Mixed, without disclosure	9.18**	(3.97)	0.038	9.18**	9.18**	9.18*
				(3.97)	(3.97)	(4.82)
Mixed, with disclosure	3.41	(4.30)	0.447	3.41	3.41	3.41
				(4.30)	(4.31)	(5.09)
Homogeneous, without disclosure	1.66	(3.81)	0.703	1.66	1.66	1.66
				(3.81)	(3.82)	(4.15)
Homogeneous, with disclosure	0.01	(3.27)	0.998	0.01	0.01	0.01
				(3.27)	(3.31)	(3.48)
Panel B. Treatment-effect contrasts						
Mixed, with: T − R	-1.66	(4.40)	0.724	-1.66	-1.66	-1.66
				(4.40)	(4.42)	(4.95)
Mixed, without: T − R	-9.47**	(3.98)	0.043	-9.47**	-9.47**	-9.47*
				(3.98)	(4.00)	(4.76)
Homogeneous, with: T − R	9.94**	(4.03)	0.033	9.94**	9.94**	9.94*
				(4.03)	(4.04)	(5.06)
Homogeneous, without: T − R	-1.97	(4.13)	0.654	-1.97	-1.97	-1.97
				(4.13)	(4.14)	(4.67)
Δ_E: H, with − M, without	19.41***	(5.67)	0.003	19.41***	19.41***	19.41**
				(5.67)	(6.00)	(7.56)
Panel C. No-leader reference means and diagnostics						
Mixed, without disclosure	58.88	(2.46)		58.88	58.88	58.88
				(2.46)	(2.47)	(2.50)
Mixed, with disclosure	57.79	(3.14)		57.79	57.79	57.79
				(3.14)	(3.15)	(3.19)
Homogeneous, without disclosure	57.79	(2.91)		57.79	57.79	57.79
				(2.91)	(2.92)	(2.96)
Homogeneous, with disclosure	55.80	(1.96)		55.80	55.80	55.80
				(1.96)	(1.97)	(1.99)
Wald: all no-leader cells equal	p = 0.789			p = 0.789	p = 0.733	p = 0.740
Observations	1,040			1,040	1,040	1,040
Clusters				336	84	28
Neighborhoods	28			28	28	28
Neighborhood × disclosure × composition FEs				Yes	Yes	Yes
Pretreatment participation control	Yes			Yes	Yes	Yes
Group-neighborhood clusters	336					
NQC fixed effects	Yes					
Formation-stage permutations	5,000					

Notes: Randomization-inference columns: We check whether the main results depend on the method of inference, keeping the specification and comparisons the same. The outcome is average post-first-meeting salongo participation, in percentage points. All models include the six-day pretreatment participation control and neighborhood × disclosure × composition fixed effects. Panel A compares each leader assignment with no leader in the same fixed-effect cell. Panel B reports the same leader contrasts as the main table; Δ_E is the Traditional-minus-Religious contrast in homogeneous groups with disclosure minus the same contrast in mixed groups without disclosure. Panel C reports model-implied no-leader reference means and clustered delta-method tests that the displayed no-leader means are equal. Standard errors are clustered one-way at the group–neighborhood level. To obtain two-sided randomization-inference (RI) p-values, we form new groups 5,000 times among the residents in the realized sample. We retain each wave's group sizes, random filling order, composition and fallback rules, gender × wealth targets, territorial diversity in mixed groups, and treatment coding. We do not resample residents or repeat the original selection of the best-balanced assignment. The random seed is 20260731. Stars use clustered asymptotic inference: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Alternative-clustering columns:* We retain the main sample, outcomes, neighborhood × disclosure × composition fixed effects, pretreatment participation control, leader effects, contrasts, and no-leader comparisons. The columns change only the level at which standard errors are clustered. Group clusters are the 336 experimental group–neighborhood cells. Leader-group clusters are the 84 neighborhood-wave × leader-arm cells, each containing the four groups assigned to traditional, religious, or no leader. Neighborhood clustering uses 28 neighborhood-wave assignments. Panel B and Panel C inference uses the clustered delta method. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.2 Additional Outcome Windows

Our main outcome measures participation over six administrative days, starting strictly after the first meeting and including the second-meeting day. Does the comparison between leaders depend on this window? Figure D-1 locates the main and alternative periods on the implementation timeline, and Table D-5 gives estimates for the main window, the intervening middle period, the final week of each wave, and all eligible post-treatment periods. In making these comparisons, we retain the same residents, fixed effects, common pretreatment participation control, 25-meter spillover indicator, 11 baseline controls selected for primary participation, and clustering level. The controls are selected once, rather than again for each window.

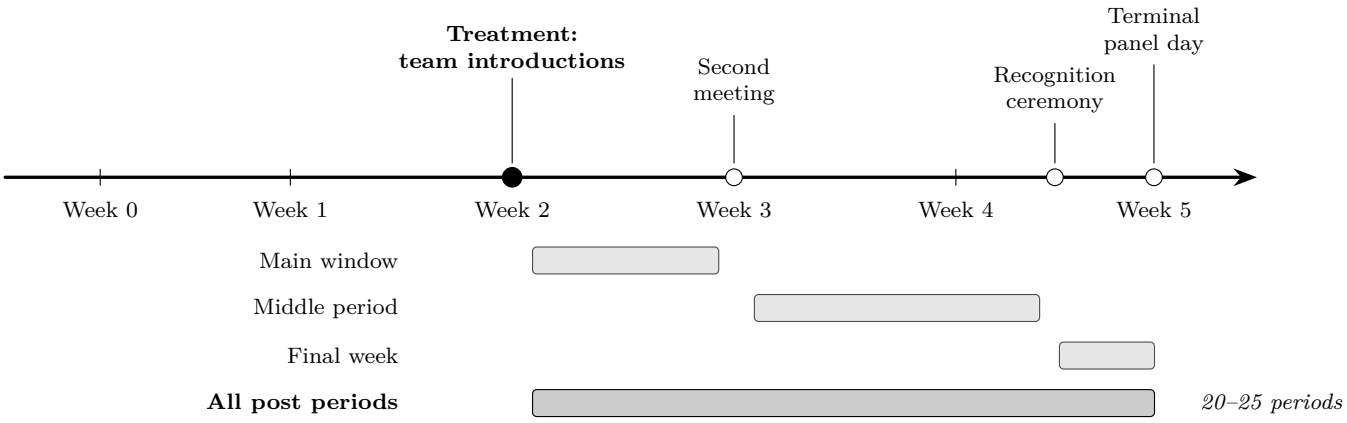


Figure D-1: Outcome windows used in the timing robustness analysis.

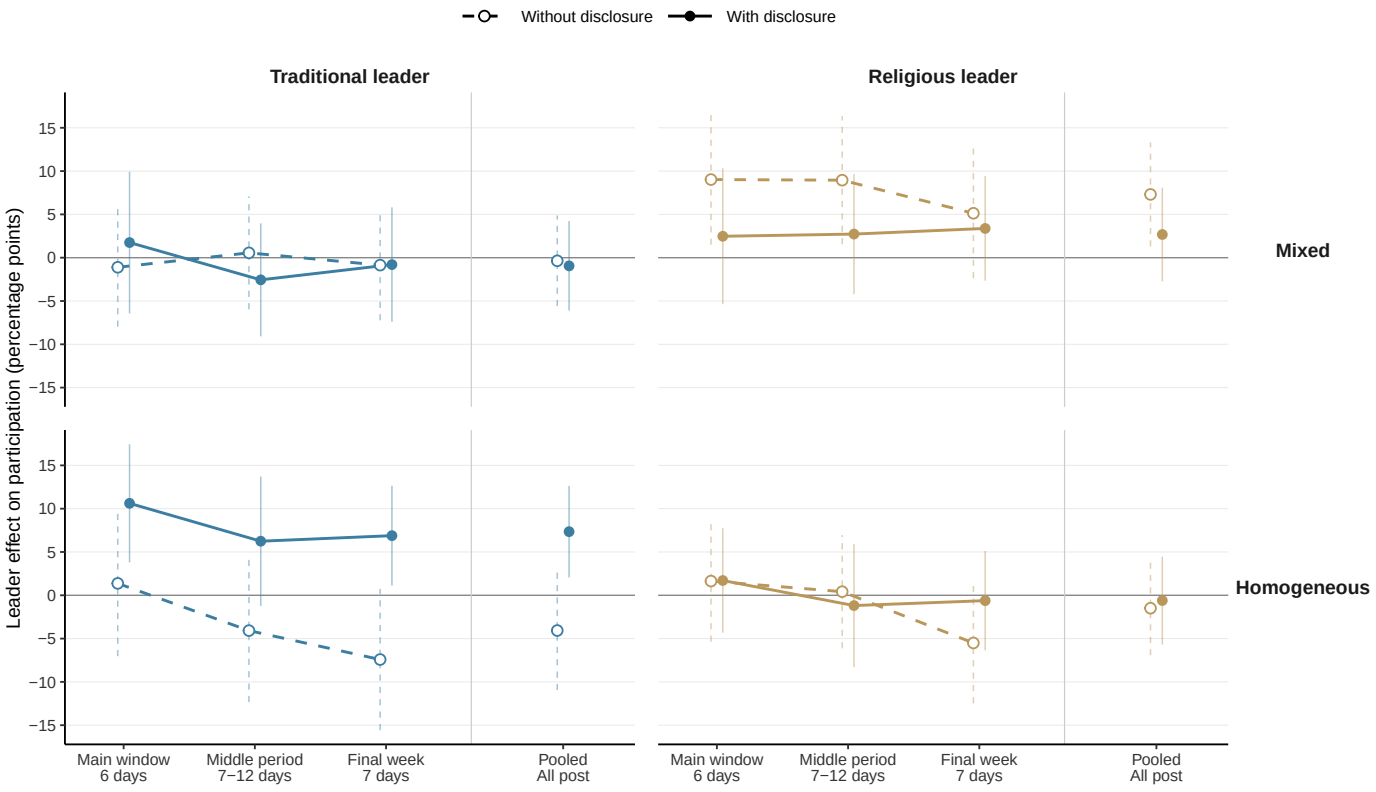


Figure D-2: LEADER EFFECTS ACROSS POST-TREATMENT OUTCOME WINDOWS

Notes: Points report each leader's effect on participation relative to no leader within the indicated composition and disclosure condition. Lines connect the main window, middle period, and final week; the separate points to the right pool all eligible post-treatment days. Vertical bars are 95 percent confidence intervals clustered at the group–neighborhood level. Estimates reproduce Appendix Table D-5, using 1,040 respondents and 336 clusters, neighborhood × disclosure × composition fixed effects, and pretreatment participation. All specifications also include the 25-meter spillover measure and the same 11 LASSO-selected baseline controls as the main specification. Appendix Figure D-1 describes the outcome windows.

Table D-5 – ROBUSTNESS TO ALTERNATIVE POST-OUTCOME WINDOWS

	Leader by composition					Environment specific			
	Main window	Middle period	Final week	All post		Main window	Middle period	Final week	All post
Panel A. Leader effects relative to no leader									
Traditional leader									
Mixed	0.37	-0.99	-0.81	-0.63	Without disclosure	-1.11	0.56	-0.86	-0.36
	(2.74)	(2.38)	(2.33)	(1.89)		(3.48)	(3.32)	(3.24)	(2.66)
					With disclosure	1.75	-2.56	-0.80	-0.94
						(4.16)	(3.32)	(3.36)	(2.63)
Homogeneous	5.98**	1.06	-0.29	1.62	Without disclosure	1.37	-4.08	-7.42*	-4.07
	(2.78)	(2.89)	(2.65)	(2.28)		(4.27)	(4.20)	(4.14)	(3.50)
					With disclosure	10.62***	6.24	6.88**	7.35***
						(3.47)	(3.80)	(2.93)	(2.68)
Religious leader									
Mixed	5.80**	5.86**	4.28*	5.02**	Without disclosure	9.02**	8.95**	5.13	7.31**
	(2.80)	(2.59)	(2.44)	(2.06)		(3.83)	(3.76)	(3.82)	(3.05)
					With disclosure	2.48	2.73	3.38	2.68
						(3.99)	(3.53)	(3.07)	(2.75)
Homogeneous	1.66	-0.40	-3.08	-1.06	Without disclosure	1.64	0.42	-5.51	-1.49
	(2.37)	(2.48)	(2.35)	(1.93)		(3.56)	(3.33)	(3.55)	(2.76)
					With disclosure	1.71	-1.19	-0.62	-0.60
						(3.07)	(3.61)	(2.92)	(2.58)
Panel B. Treatment-effect contrasts									
Mixed: T − R	-5.43*	-6.85**	-5.09**	-5.65***	Mixed, with: T − R	-0.73	-5.29	-4.17	-3.62
	(2.96)	(2.71)	(2.47)	(2.13)		(4.15)	(3.93)	(3.27)	(2.92)
					Mixed, without: T − R	-10.13**	-8.39**	-5.99	-7.67**
						(4.17)	(3.77)	(3.68)	(3.09)
Homogeneous: T − R	4.32	1.46	2.79	2.68	Homogeneous, with: T − R	8.91**	7.43**	7.50**	7.95***
	(2.72)	(2.66)	(2.43)	(2.09)		(3.67)	(3.47)	(3.04)	(2.66)
					Homogeneous, without: T − R	-0.27	-4.50	-1.91	-2.58
						(3.90)	(3.88)	(3.61)	(3.06)
Δ_C: composition gradient	9.75**	8.31**	7.88**	8.33***	Δ_E: H, with − M, without	19.04***	15.82***	13.49***	15.62***
	(4.06)	(3.82)	(3.48)	(3.01)		(5.57)	(5.16)	(4.80)	(4.09)
Panel C. No-leader reference means and diagnostics									
Mixed	58.33	42.81	26.63	39.65	Without disclosure	58.88	42.38	27.64	39.90
	(1.79)	(1.54)	(1.62)	(1.26)		(2.24)	(2.32)	(2.34)	(1.84)
					With disclosure	57.79	43.24	25.62	39.40
						(2.81)	(1.99)	(2.24)	(1.70)
Homogeneous	56.79	43.86	27.95	40.30	Without disclosure	57.79	45.53	32.45	42.75
	(1.73)	(1.89)	(1.80)	(1.50)		(2.76)	(2.55)	(2.87)	(2.26)
					With disclosure	55.80	42.18	23.45	37.85
						(2.05)	(2.78)	(1.99)	(1.89)
Wald: all no-leader cells equal	p = 0.536	p = 0.668	p = 0.585	p = 0.738		p = 0.781	p = 0.781	p = 0.071	p = 0.421
Observations	1,040	1,040	1,040	1,040		1,040	1,040	1,040	1,040
Group-neighborhood clusters	336	336	336	336		336	336	336	336
Neighborhoods	28	28	28	28		28	28	28	28
Post periods: mean	6.0	9.5	7.0	22.5		6.0	9.5	7.0	22.5
Post periods: range	6–6	7–12	7–7	20–25		6–6	7–12	7–7	20–25
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes
Pretreatment participation control	Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes
25m spillover control	Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes
11 LASSO-selected baseline controls	Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes

Notes: All columns control for pretreatment participation, 25-meter spillover exposure, and the same 11 LASSO-selected baseline controls as the main specification and include neighborhood × disclosure × composition fixed effects. The main window averages participation strictly after the assigned first meeting through and including the assigned second-meeting day. The middle period runs from the assigned second-meeting cutoff through the day before the final week; the final week follows the recognition ceremony and covers $T-6$ through T, where T is the last day in the panel. The all-post window uses every non-treatment day after the assigned first meeting. Panel A compares each leader assignment to no leader within the relevant fixed-effect cell. Panel B reports traditional-minus-religious contrasts and the composition and environment gradients. Panel C reports model-implied no-leader reference means and equality tests. Panel B and Panel C inference uses the clustered delta method. Standard errors are clustered one-way at the group-neighborhood level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.3 Alternative Explanations Under Disclosure

Panels C–D of Tables C-5, C-6, and C-7 extend the speech, co-arrival, and follow-up comparisons to disclosure. We also ask whether residents report more contact with leaders or ordinary teammates than residents in no-leader groups.

Table C-7 compares contact with the assigned leader across compositions and disclosure conditions. At the start of the second meeting, we asked residents about the preceding week: whether they interacted with the leader, discussed *salongo*, were asked to participate, or heard mentions of social or divine consequences. The table uses one report per respondent about the assigned leader, excluding respondent-leaders, ordinary-peer reports, and no-leader groups. There are 299 respondents in chief groups and 302 in pastor groups with nonmissing pretreatment participation. We estimate the regressions separately by leader type. Panels A–B report adjusted levels and composition comparisons from a regression with a homogeneous-group indicator and mean-centered respondent pretreatment participation, pooling disclosure conditions. Panels C–D add disclosure and its interaction with composition and report adjusted levels and disclosure comparisons. Adjusted levels use the same pooled leader-report baseline mean. Standard errors cluster by neighborhood, allowing correlation across groups assigned to the same leader. We estimate composition differences from the pooled regression; they need not equal a 50/50 average of the disclosure-specific levels.

For the analysis of outcomes directed toward specific group members, let j index the referenced person and let $J_{jg} = 1$ indicate that this person is the assigned leader. The general extension of equation (1) is

$$Y_{ij} = \lambda_{n(i),q(i),c(i)} + \rho Y_{ij0} + \sum_{e \in \mathcal{E}} \Big[\left(\tau_e^T + \kappa_e^T J_{jg}\right) T_{g(i)} + \left(\tau_e^R + \kappa_e^R J_{jg}\right) R_{g(i)} \Big] \mathbf{1}\{e(i) = e\} + X_i'\beta + \varepsilon_{ij}. \quad (5)$$

The outcome Y_{ij} concerns a particular person; Y_{ij0} is its pretreatment value when we include one. For an ordinary participant as the target, the leader-assignment effect is τ_e^L. For the assigned leader as the target, it is $\tau_e^L + \kappa_e^L$. The contact regressions in Table **??** use this treatment structure separately in each environment, with neighborhood fixed effects. They control for the respondent's pretreatment participation in place of the dyad-specific baseline outcome and additional controls. We cluster standard errors by randomized group within neighborhood. These reports establish whether contact occurred, but cannot tell us who initiated it or how intensive the follow-up was.

Leader contact relative to no-leader groups. Appendix Table **??** presents results for several types of recorded interactions. Reported contact with leaders is not greater in the environments where they induce more participation. In homogeneous groups with disclosure, reported contact with the assigned chief is only 5.0 percentage points higher (SE 6.5) than the rates of contact between ordinary members of no-leader groups. In mixed groups with disclosure, where the chief's participation effect is small, reported contact with the assigned chief is 20.8 points higher (SE 6.6) than contact between ordinary members of no-leader groups. Pastors show a similar mismatch. In mixed groups without disclosure, where the participation effect is largest, respondents report nearly the same rate of contact with the assigned pastor as between ordinary members of no-leader groups, a difference of 6.5 percentage points (SE 5.4). Requests, references to consequences, and interactions among ordinary participants show no offsetting pattern. If anything, respondents report more contact with chiefs where their effects on participation are weaker. The recorded contacts therefore provide little support for differences in follow-up effort as an explanation of where leaders mobilize participation.

Table D-6 – PASTOR EFFECTS BY CHIEF COETHNICITY AND DISCLOSURE

	Mixed groups		
	Baseline	+ Spillover	+ LASSO
Panel A. Pastor effects relative to no leader (percentage points)			
Not coethnic with chief			
Without disclosure	5.67	5.69	6.08
	(4.93)	(4.93)	(4.74)
With disclosure	1.15	0.95	0.79
	(4.98)	(5.00)	(4.60)
With minus without disclosure	-4.52	-4.74	-5.29
	(7.00)	(7.03)	(6.57)
Chief's coethnics			
Without disclosure	20.12*	20.00*	18.24**
	(10.38)	(10.44)	(8.86)
With disclosure	9.05	9.63	6.67
	(10.29)	(10.30)	(10.25)
With minus without disclosure	-11.07	-10.37	-11.56
	(14.59)	(14.68)	(13.52)
Difference in disclosure contrasts (coethnic minus non-coethnic)	-6.55	-5.63	-6.27
	(17.36)	(17.47)	(15.99)
Panel B. Predicted no-leader participation (percent)			
Not coethnic with chief			
Without disclosure	61.74	61.74	61.74
	(3.01)	(2.98)	(2.77)
With disclosure	56.10	56.10	56.10
	(3.26)	(3.27)	(2.83)
Chief's coethnics			
Without disclosure	49.21	49.21	49.21
	(7.04)	(7.12)	(6.35)
With disclosure	63.49	63.49	63.49
	(7.56)	(7.53)	(7.73)
Baseline participation control	Yes	Yes	Yes
Neighborhood × composition × disclosure FE	Yes	Yes	Yes
Spillover control	No	Yes	Yes
LASSO controls	No	No	Yes
Observations	1,040	1,040	1,040
Group clusters	336	336	336

Notes: The table compares pastor effects in mixed groups among residents who share the neighborhood chief's territorial origin and those who do not, separately with and without disclosure. Coethnicity refers to the chief in every assignment arm. Disclosure contrasts subtract without from with disclosure; the difference in contrasts subtracts non-coethnics from coethnics. Panel B averages fitted predictions over actual no-leader respondents in each observed coethnicity-by-disclosure category, retaining their covariates and fixed effects; these levels equal the observed subgroup means, and standard errors reflect model uncertainty. The joint regression retains all 1,040 residents and includes chief and pastor effects fully interacted with disclosure and coethnicity within mixed groups, corresponding no-leader category terms, and homogeneous-group leader effects by disclosure. Only mixed-group pastor effects are displayed. Among non-coethnics, the pastor and no-leader samples contain, respectively, 27/28 groups without disclosure and 28/28 with disclosure; for coethnics it is 18/18 and 18/17 groups, respectively. The regression also retains mixed groups assigned a chief: non-coethnics occur in 28 groups in each disclosure arm, and coethnics in 15 without and 19 with disclosure. All columns include baseline participation and the stated fixed effects; the second adds spillover, and the third adds the same 11 LASSO-selected baseline controls. Group-clustered standard errors are in parentheses. Stars denote unadjusted exploratory tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.3.1 Pastors' Territorial Origins

Table D-7 – LEADER EFFECTS UNDER THE RELIGIOUS-LEADER TERRITORY RESTRICTION: DISCLOSURE SPECIFICATION

	Environment specific		
	Baseline	+ Spillover	+ LASSO
Panel A. Leader effects relative to no leader			
Traditional leader			
Mixed, without disclosure	-0.36	-0.38	-1.25
	(3.27)	(3.26)	(3.32)
Mixed, with disclosure	1.67	1.57	1.67
	(4.33)	(4.31)	(3.95)
Homogeneous, without disclosure	-0.24	-0.28	1.48
	(4.42)	(4.40)	(4.24)
Homogeneous, with disclosure	9.83***	10.08***	10.62***
	(3.61)	(3.60)	(3.44)
Religious leader			
Mixed, without disclosure	9.56*	9.52*	8.76*
	(4.93)	(4.90)	(4.94)
Mixed, with disclosure	10.53**	10.73**	8.17*
	(5.04)	(5.04)	(4.78)
Homogeneous, without disclosure	-0.50	-0.45	-1.20
	(4.75)	(4.73)	(4.45)
Homogeneous, with disclosure	-2.35	-2.55	-0.46
	(4.59)	(4.54)	(4.08)
Panel B. Treatment-effect contrasts			
Mixed, without disclosure: T − R	-9.93**	-9.90**	-10.01*
	(4.89)	(4.89)	(5.26)
Mixed, with disclosure: T − R	-8.86*	-9.17*	-6.51
	(4.85)	(4.85)	(4.49)
Homogeneous, without disclosure: T − R	0.26	0.17	2.68
	(5.03)	(5.01)	(4.63)
Homogeneous, with disclosure: T − R	12.18**	12.63**	11.08**
	(5.41)	(5.35)	(4.82)
Δ_E: H, with − M, without	22.11***	22.54***	21.09***
	(7.31)	(7.28)	(7.16)
Panel C. No-leader reference means and diagnostics			
Mixed, without disclosure	58.88	58.88	58.88
	(2.29)	(2.26)	(2.13)
Mixed, with disclosure	57.79	57.79	57.79
	(3.08)	(3.08)	(2.84)
Homogeneous, without disclosure	57.79	57.79	57.79
	(2.92)	(2.91)	(2.83)
Homogeneous, with disclosure	55.80	55.80	55.80
	(1.93)	(1.94)	(1.95)
Wald: four no-leader cells equal	p = 0.770	p = 0.767	p = 0.755
Observations	908	908	908
Group-neighborhood clusters	292	292	292
Neighborhoods	28	28	28
Design fixed effects	Yes	Yes	Yes
Pretreatment participation control	Yes	Yes	Yes
25m spillover control	No	Yes	Yes
LASSO-selected baseline controls	No	No	Yes

Notes: Could ethnic mismatch explain pastors' effects under either disclosure condition? We repeat the disclosure specification after excluding 132 residents assigned a pastor in the 11 neighborhoods where the pastor and chief originate from different territories. Residents assigned a chief or no leader all remain. Panel A gives leader effects separately for each composition and disclosure condition, Panel B compares chief and pastor effects, and Panel C gives predicted no-leader means. The first column adjusts for participation before treatment and neighborhood-by-composition-by-disclosure fixed effects. Moving to the second column adds nearby spillover exposure; moving to the third adds the 11 baseline controls selected by LASSO, retaining the preceding adjustments. Parentheses give standard errors clustered by group. See Section 5 for the territorial comparison. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.3.2 Leadership by Example

Table D-8 – LEADERSHIP BY EXAMPLE: DISCLOSURE SPECIFICATION

	Group-clustered SE			Neighborhood-clustered SE		
	Participation at/below leader-type median	Participation above leader-type median	Difference	Participation at/below leader-type median	Participation above leader-type median	Difference
Panel A. Leader effects relative to no leader						
Traditional leader						
Mixed, without disclosure	0.41	-0.96	-1.37	0.41	-0.96	-1.37
	(5.15)	(4.43)	(6.53)	(5.62)	(4.85)	(7.45)
Mixed, with disclosure	1.41	2.00	0.59	1.41	2.00	0.59
	(4.97)	(6.67)	(7.68)	(5.80)	(8.00)	(8.92)
Homogeneous, without disclosure	2.63	-2.80	-5.43	2.63	-2.80	-5.43
	(5.14)	(6.29)	(7.39)	(6.23)	(7.93)	(8.98)
Homogeneous, with disclosure	13.32**	7.05	-6.27	13.32*	7.05	-6.27
	(5.74)	(4.31)	(7.07)	(6.70)	(5.39)	(8.84)
Religious leader						
Mixed, without disclosure	8.07	10.60*	2.53	8.07	10.60	2.53
	(5.00)	(5.88)	(7.37)	(5.87)	(7.60)	(9.30)
Mixed, with disclosure	1.74	5.60	3.86	1.74	5.60	3.86
	(5.80)	(5.37)	(7.28)	(6.80)	(5.96)	(8.02)
Homogeneous, without disclosure	5.10	-2.84	-7.94	5.10	-2.84	-7.94
	(5.07)	(4.88)	(6.33)	(5.39)	(5.46)	(6.50)
Homogeneous, with disclosure	2.19	-2.86	-5.05	2.19	-2.86	-5.05
	(4.39)	(4.69)	(6.37)	(5.02)	(5.08)	(7.47)
Panel B. Implied no-leader counterfactuals						
No leader, mixed, without disclosure		55.52			55.52	
		(2.47)			(2.46)	
No leader, mixed, with disclosure		57.52			57.52	
		(3.13)			(3.20)	
No leader, homogeneous, without disclosure		55.43			55.43	
		(3.12)			(3.11)	
No leader, homogeneous, with disclosure		52.73			52.73	
		(1.88)			(1.94)	
Observations			1,040			
Group clusters			336			
Neighborhood clusters			28			

Notes: The difference column compares the effect above the median with the effect at or below it. Panel B reports model-implied no-leader participation in percentage points, averaging over the same chief- and pastor-arm residents within each composition-by-disclosure environment and setting baseline participation to its estimation-sample mean. The characteristic is undefined without a leader, so each block reports one common reference rather than separate participation categories among no-leader groups. Entries compare leader effects above and below the type-specific median of leaders' participation during the common week before any meeting. The cutpoints are 0.80 for chiefs and 0.83 for pastors; ties enter the at-or-below-median category. Sunday is excluded and each leader rate uses four to six observed eligible days. The two blocks report the same disclosure-specific regression with standard errors clustered by 336 groups or by 28 neighborhoods. The latter allows dependence across groups assigned to the same leader. All 1,040 participants are retained. The regression includes baseline participation, neighborhood-by-composition-by-disclosure fixed effects, and leader-by-composition-by-disclosure interactions with the leader participation category. Because the characteristic is undefined without a leader, it enters only through interactions with leader assignment. Section 5 discusses the interpretation. Stars use unadjusted tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.3.3 Pretreatment Participation

Table D-9 – PRETREATMENT DYNAMICS

	All six pretreatment days			Excluding day 1		
Panel A. Linear trends						
	Estimate	Standard error	p-value	Estimate	Standard error	p-value
Traditional leader						
Mixed, without disclosure	-1.83	(1.32)	0.165	-1.29	(1.60)	0.421
Mixed, with disclosure	0.83	(1.26)	0.510	0.10	(1.40)	0.945
Homogeneous, without disclosure	-0.84	(1.35)	0.533	-0.00	(1.91)	0.999
Homogeneous, with disclosure	-2.12	(1.06)	0.047	-1.73	(1.27)	0.175
Religious leader						
Mixed, without disclosure	1.20	(1.21)	0.321	1.33	(1.58)	0.400
Mixed, with disclosure	1.00	(1.15)	0.385	0.69	(1.30)	0.596
Homogeneous, without disclosure	-2.75	(1.41)	0.052	-2.03	(1.81)	0.263
Homogeneous, with disclosure	-1.68	(1.10)	0.126	-1.49	(1.59)	0.349
Panel B. Early versus late pretreatment						
	Estimate	Standard error	p-value	Estimate	Standard error	p-value
Traditional leader						
Mixed, without disclosure	-3.13	(4.19)	0.456	-0.72	(4.31)	0.868
Mixed, with disclosure	2.47	(4.06)	0.543	0.83	(4.09)	0.839
Homogeneous, without disclosure	-2.20	(4.79)	0.647	-0.29	(5.52)	0.958
Homogeneous, with disclosure	-6.25	(3.53)	0.078	-4.51	(3.99)	0.260
Religious leader						
Mixed, without disclosure	2.83	(3.79)	0.457	2.26	(4.11)	0.583
Mixed, with disclosure	1.28	(3.80)	0.736	-0.16	(3.77)	0.967
Homogeneous, without disclosure	-6.96	(4.71)	0.141	-4.06	(5.00)	0.418
Homogeneous, with disclosure	-2.68	(4.25)	0.529	-0.94	(5.08)	0.854
Panel C. Saturated day-specific paths						
	F	Restrictions	p-value	F	Restrictions	p-value
Traditional leader						
Mixed, without disclosure	2.97	5	0.012	3.39	4	0.010
Mixed, with disclosure	0.51	5	0.768	0.50	4	0.737
Homogeneous, without disclosure	0.47	5	0.801	0.20	4	0.936
Homogeneous, with disclosure	1.83	5	0.106	1.76	4	0.137
Religious leader						
Mixed, without disclosure	1.56	5	0.170	1.94	4	0.104
Mixed, with disclosure	0.42	5	0.832	0.34	4	0.850
Homogeneous, without disclosure	2.72	5	0.020	2.17	4	0.072
Homogeneous, with disclosure	1.29	5	0.270	1.35	4	0.253
Respondents		1,040			1,040	
Group–neighborhood clusters		336			336	
Neighborhoods		28			28	

Notes: Panels A and B report percentage-point estimates, clustered standard errors, and p-values for each of the eight leader-by-information-by-composition paths. Panel A estimates the change in the leader-minus-no-leader gap per campaign day. Panel B compares days 1–3 with days 4–6 in the full window and days 2–3 with days 4–6 after excluding day 1. Panel C jointly tests the day-specific coefficients for each path, taking campaign day 6 as the reference. We report the F statistic, number of restrictions, and p-value for this test. All regressions include respondent fixed effects and neighborhood × information × composition × campaign-day fixed effects. Standard errors are clustered one-way by assigned group within neighborhood.

D.4 Meeting Discourse

We clean the first-meeting transcripts and divide speakers' turns into short passages. A single-topic mixture model for short texts groups passages into three topics using word and phrase frequencies (Quinn et al., 2010), without treatment assignments, speaker roles, or participation outcomes. Each passage is assigned to its most likely topic. We interpret the topics using characteristic words and representative passages: attendance and performance, material support, and divine reward and solidarity (Table D-10). We classify additional transcribed meetings using the same fitted model and vocabulary, keeping earlier assignments unchanged. The analysis contains 2,691 classified passages from 313 meetings.

We allocate each recorded turn's duration across its passages in proportion to their word counts, then sum minutes by speaker and topic. Figure D-5 uses 969 ordinary participants in recorded meetings, retaining 62 silent participants with zero minutes. Unrecorded meetings are excluded. Regressions compare each leader assignment with no leader within neighborhood-by-composition-by-disclosure cells, with standard errors clustered by assigned group. No pretreatment speech measure is available. These comparisons measure responses in the discussion; they do not identify the causal effect of particular messages on later participation.

Robustness across text models. We compare the single-topic model and its full-corpus refit (Quinn et al., 2010), standard and sparse-prior LDA (Blei et al., 2003), a correlated topic model (Blei and Lafferty, 2007), spherical k-means on multilingual E5 embeddings (Dhillon and Modha, 2001; Wang et al., 2024), and BERTopic with three topics (Grootendorst, 2022). Figures D-3 and D-4 compare leaders' pooled topic shares and their speech within each composition-by-disclosure environment. Figure D-6 compares ordinary participants' speaking levels, including the no-leader arm.

All models classify the same 2,691 passages and use the same analysis samples: 206 speaking leader-meetings and 969 ordinary participants. In models that assign each passage to one topic, we give that topic the passage's full duration. In mixed-membership models, we divide the duration using posterior topic weights. Leader figures show equal-weight means across leader-meetings. Participant levels use the same fixed-effect regressions as Figure D-5, averaging each assignment's predictions over the actual no-leader participants in the corresponding environment. Using the same residents for all three predictions means that differences in predicted levels recover the original treatment effects. Confidence intervals use the full covariance matrix and cluster by neighborhood for leaders and by assigned group for participants. They are pointwise and conditional on the fitted text measures. Chiefs' emphasis on attendance and pastors' emphasis on religious appeals persist across models. Pooling disclosure, pastor assignments increase participants' divine speech in both compositions, alongside more total speech in homogeneous groups and less attendance-and-performance talk in mixed groups. These broad directions persist across models; chiefs do not consistently shift participants toward attendance talk.

Table D-10 – EXAMPLES OF THE THREE SPEECH TOPICS

Topic and typical words	Illustrative speech
Attendance and performance *hours, week, absence, percentage, punctuality*	"Last week we completed only a few hours. We should reach five hours per person each day and come regularly." "I call on those who did not work last week to join us so that we can maintain the places where we live." "Everyone was present this week. Next week, let us work with the same commitment to improve our neighborhood." **Chief:** "Doing this work is like being on a football team: those who want to win compete for the cup. [. . .] We must improve our performance to win the contest. [. . .] We must encourage those who did not come and go and see them to find out why they were absent. We must help one another so that our group does better." **Resident:** "[. . .] I ask that encouragement become a habit in our group. If I am absent, I should be reminded to come and take part in the work [. . .]."
Requests for material support *tools, sacks, hoes, wheelbarrows, avenues*	"The municipality gave us hoes, machetes, spades, and sacks to improve our environment." "Volunteers brought sacks, and the work is progressing. We still need wheelbarrows, hoes, picks, and more sacks." "We now have the tools, and the people with strength to work are here. We can keep improving the neighborhood." **Resident:** "We have started salongo well, and now we are approaching the second week. We ask for help so that we can maintain our streets: we need tools and empty sacks. We will keep doing this voluntarily to maintain our neighborhood and our streets. We ask you to support us so that we can do this work."
Divine reward and solidarity *God, bless, reward, united, together*	"Salongo is group work, not one person's work. With God's support, let us have patience, peace, and gentleness." "Volunteers should not expect wages, but look to God, who rewards people according to their work." "A person reaps what they sow; God rewards each person according to their deeds." **Pastor:** "This work is voluntary and meant to help everyone. [. . .] It is not a person on earth who can pay us; it is God who will do so. I ask only for each person's willingness to work." **Resident:** "I liked what you said. [. . .] God will pay us, and let us respect the working hours, from nine until three." **Pastor, closing:** "[. . .] May God bless you, and may he also bless our group so that we can move forward."

Notes: The excerpts illustrate the appeals that distinguish the three topics. The left column lists words that frequently characterize each topic. The right column gives translated examples of chief and pastor speech that the model placed in that topic, followed by additional illustrative excerpts with speaker labels. These include residents' requests and responses to leaders. Excerpts are shortened for readability; ellipses mark internal omissions. The labelled exchanges illustrate the appeals and responses, rather than identifying causal effects of particular speeches. The topic model formed the categories from word patterns without using leader type or treatment assignment. The table is discussed in Section 5.3. Figure 7 shows the leaders' mean topic shares; Appendix Table C-5 adds speaking time and separates the experimental environments; Appendix Figure D-5 shows how leader assignment changes ordinary participants' speech.

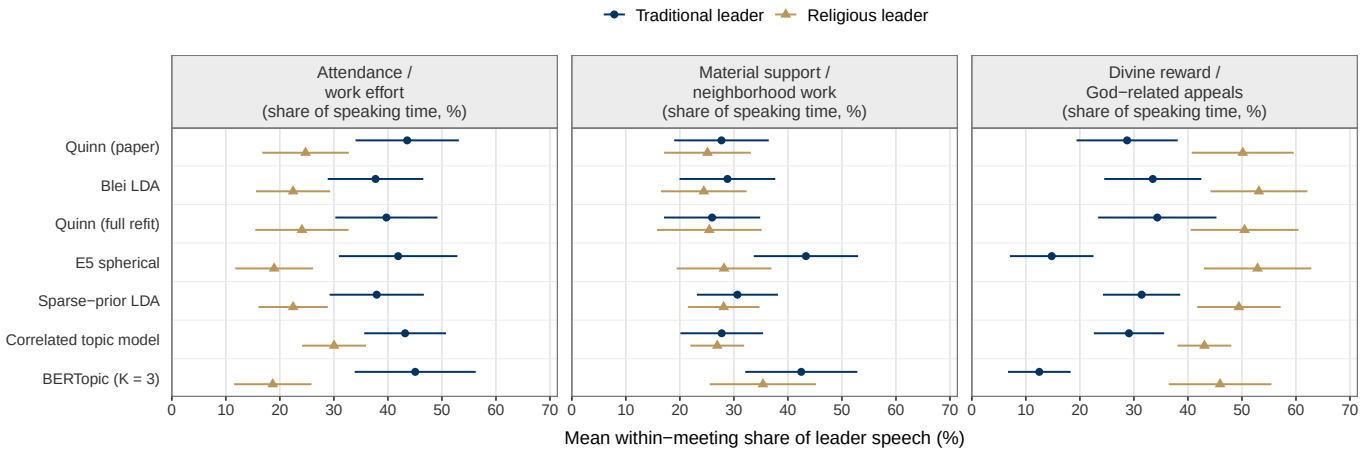


Figure D-3: LEADER TOPIC SHARES ACROSS TEXT MODELS

Notes: Points show mean within-meeting shares of leader speaking time, pooling composition and disclosure. Blue circles denote traditional leaders and gold triangles religious leaders. Each model uses the same 103 speaking leader-meetings of each type. Lines are pointwise 95 percent confidence intervals clustered by neighborhood (28). Topic headings group related themes whose precise meanings can vary across models. Appendix Section D.4 describes the models and aggregation.

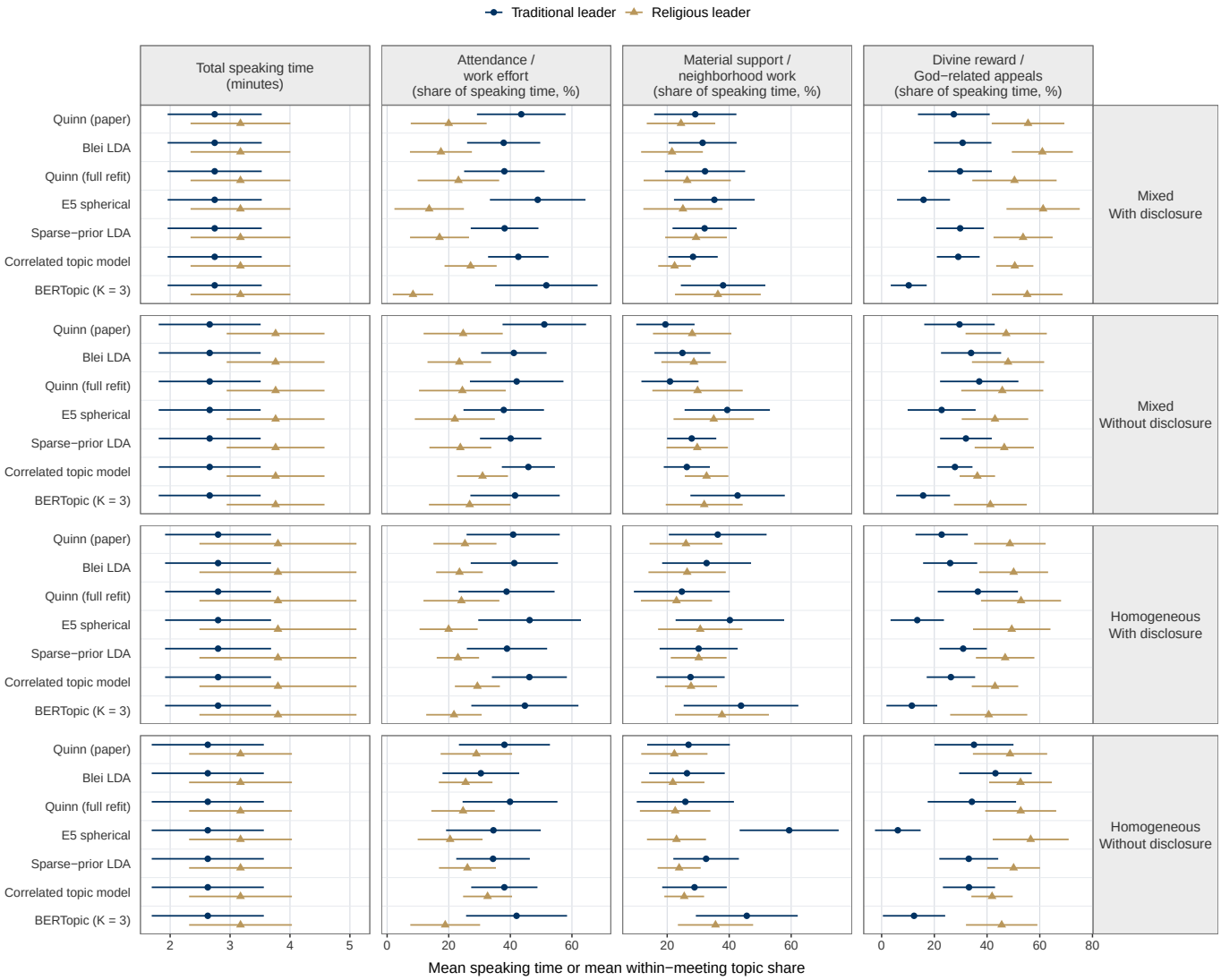


Figure D-4: LEADER SPEECH BY ENVIRONMENT ACROSS TEXT MODELS

Notes: Points show mean total speaking time and mean within-meeting topic shares, separately by group composition and disclosure. Blue circles denote traditional leaders and gold triangles religious leaders. All models use the same 206 speaking leader-meetings; total speaking time is identical across models. Lines are pointwise 95 percent confidence intervals clustered by neighborhood (28), conditional on the fitted text measures. Topic headings group related themes. Appendix Section D.4 describes the methods.

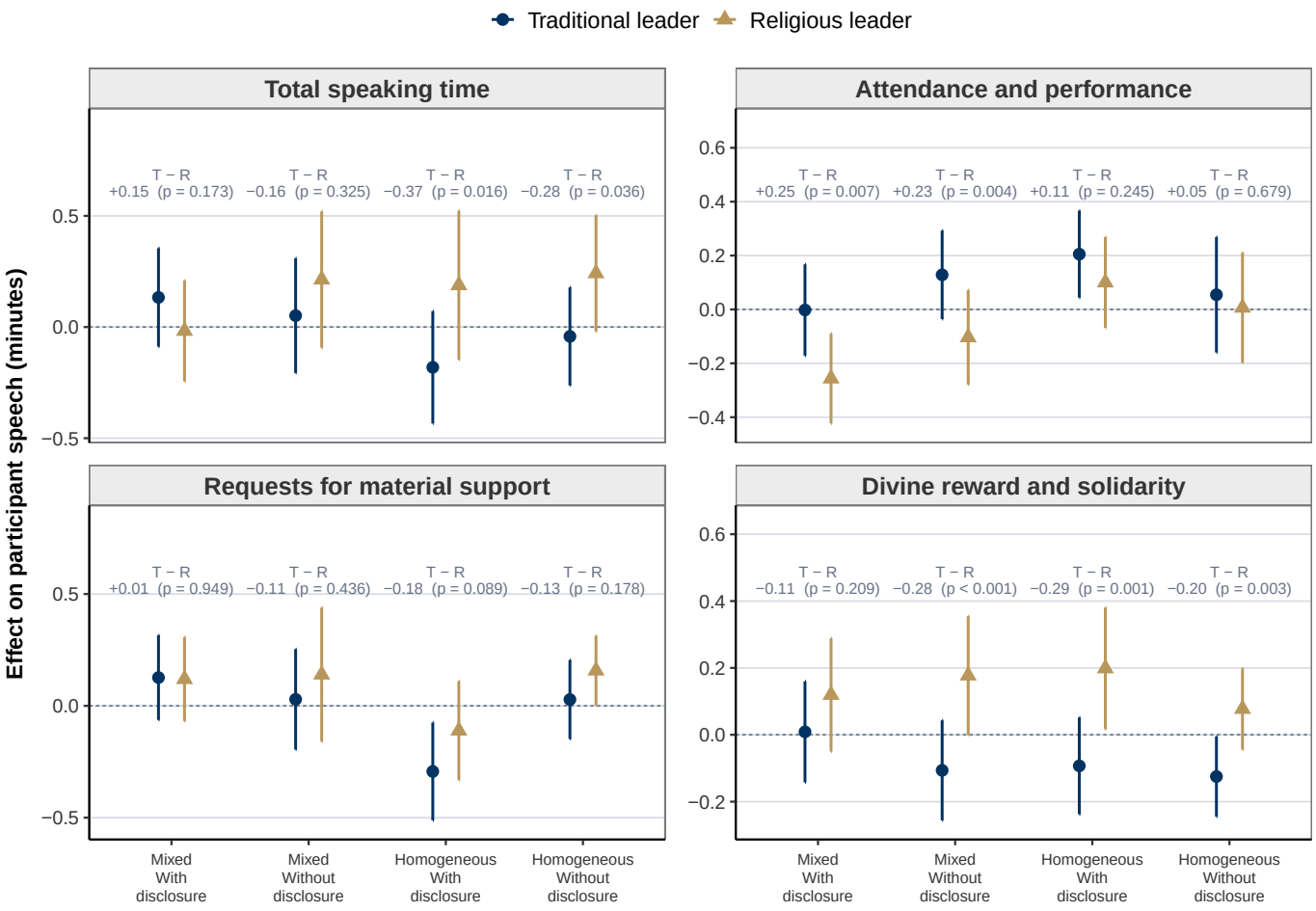


Figure D-5: LEADER ASSIGNMENT AND ORDINARY PARTICIPANTS' SPEECH

Notes: Each point shows how assigning a traditional leader or religious leader changes ordinary participants' speaking time relative to a no-leader group in the same environment. The four panels report total speaking time and time spent on each topic. Positive values mean that participants speak longer under the assigned leader. The T–R labels compare traditional leader and religious leader assignments directly. The topic pattern follows the leaders' own speech: traditional leader groups place relatively more emphasis on attendance and performance, while religious leader groups place relatively more emphasis on divine reward and solidarity. The sample contains 969 ordinary participants in 313 recorded meetings across 28 neighborhoods; 62 silent participants count as zero speaking time. Regressions include neighborhood-by-composition-by-disclosure fixed effects. Lines are 95 percent confidence intervals clustered by assigned group; T–R tests are unadjusted. Appendix Section D.4 describes the classification and duration measures. The figure is discussed in Section 5.3; Main Figure 7 and Table D-10 describe the leaders' speech.

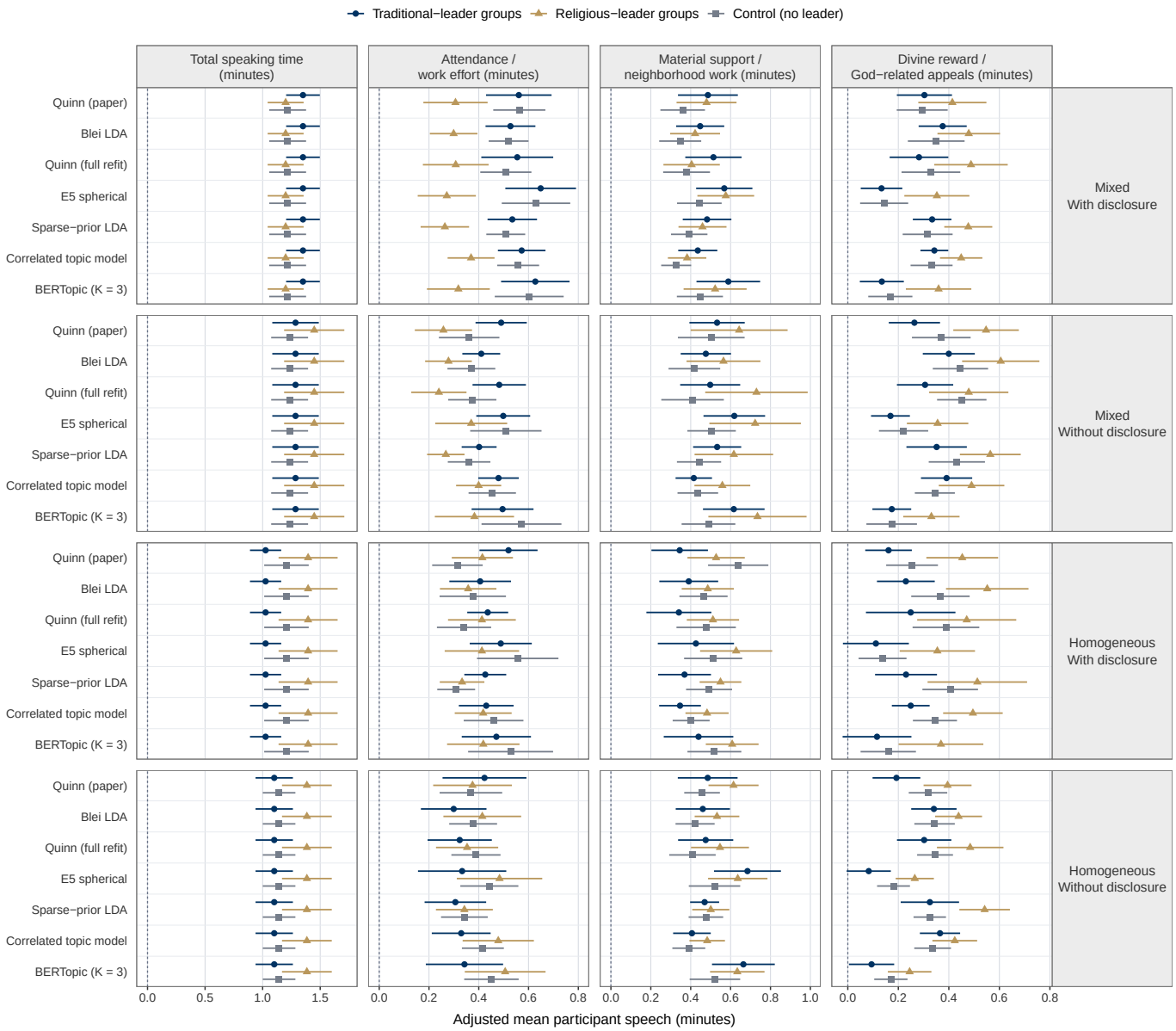


Figure D-6: PARTICIPANT SPEECH LEVELS ACROSS TEXT MODELS

Notes: Points show adjusted ordinary-participant speaking levels in traditional-leader groups (blue circles), religious-leader groups (gold triangles), and no-leader groups (gray squares). The sample contains 969 ordinary participants in 313 groups, including 62 silent participants. We include neighborhood-by-composition-by-disclosure fixed effects. Predictions for all three arms are averaged over the actual no-leader participants in each displayed environment. Lines are pointwise 95 percent confidence intervals using the full group-clustered covariance matrix, conditional on the fitted text measures. Total speaking time is identical across models. Appendix Section D.4 describes the methods.

D.5 Responses to Revealed Participation Histories

Table D-11 – LEADER EFFECTS BY DAYS OF PRETREATMENT PARTICIPATION

	0–1 days	2–6 days	Difference
Panel A. Leader effects relative to no leader			
Traditional leader			
Mixed, without disclosure	-7.04	4.38	11.42
	(5.93)	(5.26)	(8.91)
Mixed, with disclosure	7.84	-5.28	-13.12
	(7.01)	(5.05)	(9.46)
Homogeneous, without disclosure	-5.82	7.87	13.69
	(7.35)	(5.36)	(9.37)
Homogeneous, with disclosure	19.03***	1.72	-17.31**
	(6.06)	(4.15)	(7.93)
Religious leader			
Mixed, without disclosure	6.31	9.43	3.12
	(5.89)	(5.85)	(8.93)
Mixed, with disclosure	4.75	-2.27	-7.02
	(7.29)	(4.55)	(9.07)
Homogeneous, without disclosure	-2.45	6.14	8.59
	(6.83)	(4.99)	(9.43)
Homogeneous, with disclosure	0.81	4.20	3.39
	(5.19)	(4.31)	(7.07)
	0–1 days	2–6 days	Difference
Panel B. Implied no-leader counterfactuals			
Traditional leader			
Mixed, without disclosure	38.75	74.36	35.61***
	(3.80)	(3.74)	(6.05)
Mixed, with disclosure	37.76	80.62	42.87***
	(4.79)	(3.30)	(6.23)
Homogeneous, without disclosure	39.47	70.68	31.21***
	(5.39)	(3.65)	(6.98)
Homogeneous, with disclosure	25.88	76.85	50.97***
	(3.67)	(2.68)	(4.66)
Religious leader			
Mixed, without disclosure	38.75	74.36	35.61***
	(3.80)	(3.74)	(6.05)
Mixed, with disclosure	37.76	80.62	42.87***
	(4.79)	(3.30)	(6.23)
Homogeneous, without disclosure	39.47	70.68	31.21***
	(5.39)	(3.65)	(6.98)
Homogeneous, with disclosure	25.88	76.85	50.97***
	(3.67)	(2.68)	(4.66)
Observations		1,040	
Group clusters		336	

Notes: The difference column reports the effect in the second category minus the effect in the first. These are the estimates plotted in Figure 8 and discussed in Section 5.4. The categories are participation on 0–1 versus 2–6 eligible days before any meeting. The regression uses all 1,040 residents and 336 groups, baseline participation, 25-meter spillover exposure, 11 LASSO-selected baseline controls, neighborhood-by-composition-by-disclosure fixed effects, and separate pretreatment-category levels and leader interactions in every environment. To predict participation without a leader, we set the centered baseline, spillover, and LASSO controls to zero. We average the fixed-effect and category components over the treated residents used to estimate each displayed effect. Standard errors are clustered by group. Table **??** reports fixed-effect sensitivity. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. In mixed groups, disclosure changes the implied no-leader comparison by -0.99 points (SE 6.11) among residents who participated on 0–1 days and 6.26 points (SE 4.99) among residents who participated on 2–6 days; the corresponding changes in the pastor effect are -1.56 (SE 9.21) and -11.71 (SE 7.42).

D.6 Campaign Contributions and Leadership Alignment

Recorded contributions. We measure labor from recorded arrival and departure times, summing valid intervals without double-counting overlaps for all recorded attendees in the 28 treatment neighborhoods, Waves 1–7 (August 2025–April 2026), including nonexperimental residents. The 70,250 hours equal 8,781 eight-hour days; excluding spells rated as no effort leaves 69,003 hours. Cash contributions total 154,250 CDF. We value in-kind contributions, including materials and refreshments, at their recorded unit prices, giving 3,324,417 CDF. We do not count cash purchases again as in-kind contributions.

Worksites. GPS coverage spans 20 neighborhoods in Waves 1–5. Within neighborhoods, we pool different-day observations whose maximum pairwise distance is five meters; separately recorded same-day sites remain distinct. This yields 596 worksite histories (Figure D-7). We sum recorded returns, splitting session labor equally across separately recorded sites.

GPS records are unavailable for Waves 6–7. To estimate the count across all seven waves, we assign one worksite to each neighborhood on each non-Sunday wave workday. The four neighborhoods in Wave 6 have 30 such days and those in Wave 7 have 29, adding $4 \times 30 + 4 \times 29 = 236$ to the 596 GPS histories, for an estimated total of 832 worksites. This convention includes four neighborhood-days without attendance records. This total combines mapped histories and imputed neighborhood-workday counts; it does not independently verify distinct projects or their completion.

For the workload benchmark, we use the GPS histories and measure positive-effort person-hours from all recorded attendees at these sessions, including nonexperimental residents. Valid check-in/check-out intervals are nonnegative, no longer than 12 hours, and dated in 2025–2026; overlapping intervals are counted once. Spells rated as no effort (category 1) contribute zero. We first exclude 23 histories last observed on their neighborhood's final GPS-recorded day. Of the 573 remaining histories, five have unusable or unlinked labor records and two have zero positive labor. The remaining 566 histories contain 54,735.3375 positive-effort person-hours, giving

$$\frac{54{,}735.3375}{566} = 96.705543\ldots \approx 96.7 \quad \text{person-hours per history.}$$

This selection requires a later GPS observation in the neighborhood, rather than simply a later attendance-recorded workday. We use the observed mean to express additional hours in worksite workload equivalents. It does not measure the labor required to finish a project, and the records do not verify whether each project was completed.

Valuation. We apply Kananga's historical STEP II wage of \$3/day (Agence Congolaise de Presse, 2021), assuming eight hours as in the earlier STEP evaluation (Brandily et al., 2025). At \$0.375/hour, labor is worth \$26,344. Converting cash and goods at the fiscal workbook's 2,250 CDF/USD convention gives total contributions of \$27,890. The comparator is 205,682,923 CDF in reported provincial property-tax receipts for 2025 (Direction Générale des Recettes du Kasaï Central, 2026), or \$91,415. The comparison spans different periods and uses nominal values without inflation adjustment.

Alignment projection. We estimate six-day total hours using matched baseline hours, neighborhood-by-composition-by-disclosure fixed effects, spillover exposure, and the same 11 LASSO controls selected for participation. Retaining all 1,040 residents (520 per composition), we predict hours under chief/public disclosure for homogeneous groups and pastor/no disclosure for mixed groups. We retain each resident's covariates and use the fixed effect corresponding to the assigned disclosure condition. For each composition–disclosure cell, we multiply predictions by the ratio of full-period to six-day no-leader hours, averaging both over the same residents. This assumes effects operate throughout each 33–34-day wave, including before the first meeting; unobserved blackout days are not imputed.

The aligned total is 53,240 projected hours versus 45,989 with no leaders under the same disclosure rules: a gain of 7,251 hours. Using the unrounded mean above gives 75.0 worksite workload equivalents. The group-clustered conditional 95% interval is 2,424–12,078 additional hours; it does not account for uncertainty about policy selection, persistence of the effects, or conversion into worksite workloads. The projection also leaves aside leader supply, additional costs, and changes in spillovers.

Distance-cutoff sensitivity. The workload conversion depends on how recorded locations are grouped into sites. Table D-12 repeats the construction at cutoffs of 1, 5, 10, 20, and 50 meters, holding the policy estimates and projected additional hours fixed. At 10 and 20 meters, mean workloads are 116.3 and 141.7 person-hours, giving 62.3 and 51.2 workload equivalents, respectively, compared with 75.0 under the five-meter definition. These comparisons describe sensitivity to the worksite construction; they do not establish how many projects would be completed.

Table D-12 – Worksite Workload Benchmarks by Distance Cutoff

Distance cutoff (m)	Selected histories	Allocated person-hours	Mean person-hours	Workload equivalents
1	666	55,564.47	83.43	86.9
5	566	54,735.34	96.71	75.0
10	460	53,506.31	116.32	62.3
20	353	50,006.21	141.66	51.2
50	243	44,159.90	181.73	39.9

Notes: We repeat the spatial grouping within the same 20 neighborhoods in Waves 1–5, changing only the maximum distance between observations assigned to a worksite history. Complete-linkage grouping, separation of same-day sites, equal allocation of session labor across recorded sites, and the positive-labor and final-GPS-day eligibility rules are unchanged. We reapply eligibility at each cutoff, so the retained histories and allocated person-hours can change. Workload equivalents divide the same projected gain of 7,251.124 person-hours by each unrounded mean. These are sensitivities of the descriptive conversion; they do not count verified completed projects or change the policy treatment-effect estimates.

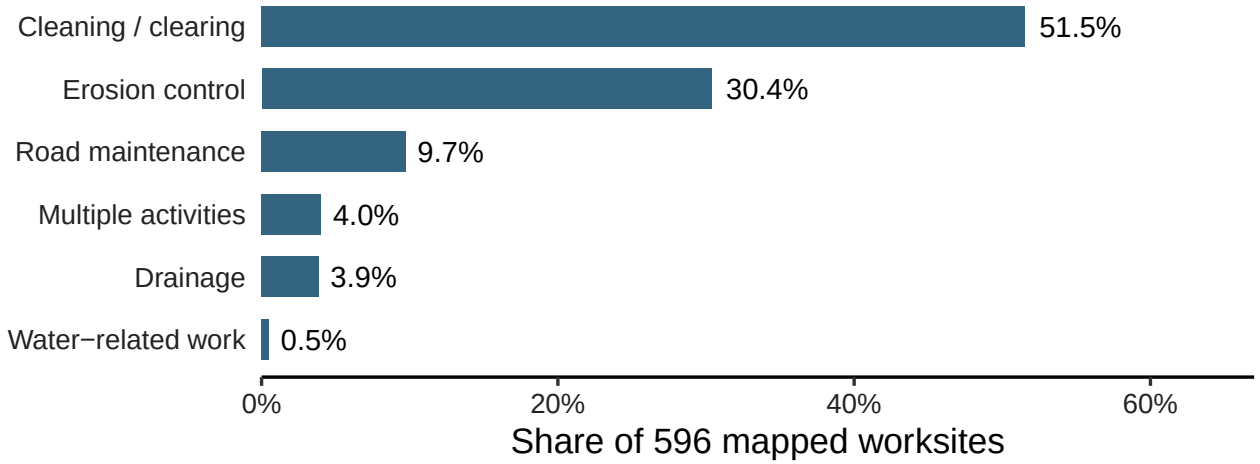


Figure D-7 – ACTIVITIES AT OBSERVED CAMPAIGN WORKSITES

Notes: Shares of the 596 location-based worksite histories in the 20 neighborhoods with GPS records (Waves 1–5). Revisits are pooled using a maximum five-meter distance, as described above. Multiple activities denotes histories with more than one recorded activity type. The counts describe observed worksites, not independently verified completed projects.

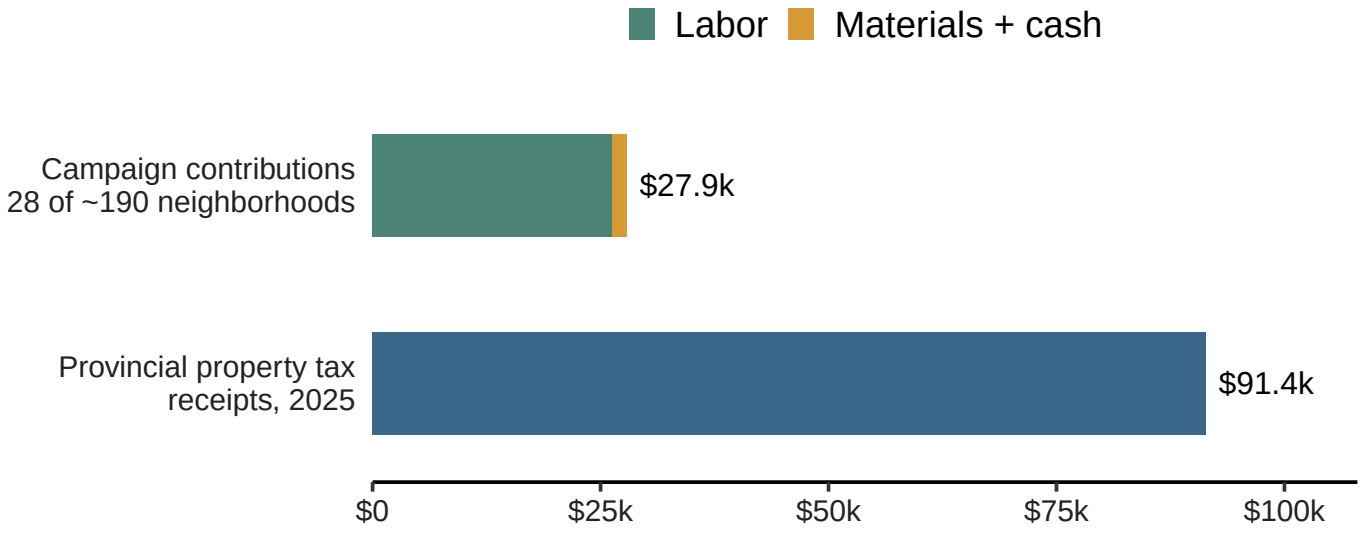


Figure D-8 – CAMPAIGN CONTRIBUTIONS AND PROPERTY-TAX RECEIPTS

Notes: Recorded campaign contributions combine labor valued at $0.375/hour with cash and recorded in-kind contributions. The $27,890 total equals 30.5% of reported 2025 Kasaï-Central provincial property-tax receipts, converted at 2,250 CDF/USD. Campaign contributions cover 28 neighborhoods during August 2025–April 2026. See Appendix D.6 for the valuation and coverage details.

E Beyond Participation

E.1 Cooperation Beliefs, Identity, and Governance

Table E-1 – LEADER EFFECTS ON COOPERATION NORMS AND DIVINE MONITORING BELIEFS

Panel A. Evolution in the two environments where leaders increased participation

	Traditional leader: homogeneous, with disclosure				Religious leader: mixed, without disclosure			
Outcome	Meeting 1	Meeting 2	Endline	Pooled	Meeting 1	Meeting 2	Endline	Pooled
Active participants are well regarded	0.01	-0.11	-0.01	-0.03	-0.06	-0.10	-0.01	-0.06
	(0.08)	(0.10)	(0.10)	(0.07)	(0.09)	(0.10)	(0.12)	(0.08)
Contributing is a moral obligation	-0.09	-0.11	0.01	-0.06	-0.09	-0.00	0.04	-0.02
	(0.07)	(0.08)	(0.10)	(0.07)	(0.08)	(0.09)	(0.09)	(0.06)
God rewards community support	-0.03	-0.01	0.03	0.00	-0.09	-0.05	0.02	-0.04
	(0.06)	(0.08)	(0.07)	(0.06)	(0.06)	(0.07)	(0.08)	(0.05)
God punishes neglect of community obligations	-0.14	-0.09	-0.15	-0.13	-0.21	-0.05	-0.06	-0.11
	(0.15)	(0.14)	(0.10)	(0.09)	(0.15)	(0.13)	(0.13)	(0.10)
Family p-value	0.889	0.591	0.573	0.677	0.431	1.000	1.000	1.000

Panel B. Pooled effects in every leader environment

	Well regarded	Moral obligation	God rewards	God punishes	Family p-value
Traditional leader					
Mixed, without disclosure	0.04	-0.05	-0.04	-0.13	0.610
	(0.08)	(0.07)	(0.05)	(0.09)	
Mixed, with disclosure	0.14*	0.12**	0.08	0.21**	0.149
	(0.08)	(0.06)	(0.05)	(0.10)	
Homogeneous, without disclosure	-0.04	0.05	-0.05	0.02	1.000
	(0.07)	(0.08)	(0.05)	(0.08)	
Homogeneous, with disclosure	-0.03	-0.06	0.00	-0.13	0.677
	(0.07)	(0.07)	(0.06)	(0.09)	
Religious leader					
Mixed, without disclosure	-0.06	-0.02	-0.04	-0.11	1.000
	(0.08)	(0.06)	(0.05)	(0.10)	
Mixed, with disclosure	-0.06	0.04	-0.03	0.13	1.000
	(0.09)	(0.07)	(0.05)	(0.12)	
Homogeneous, without disclosure	-0.07	0.10	0.00	0.07	0.565
	(0.06)	(0.07)	(0.05)	(0.09)	
Homogeneous, with disclosure	0.01	0.00	0.02	-0.02	1.000
	(0.08)	(0.07)	(0.06)	(0.10)	

Notes: Counterfactual means for both panels appear on the continuation page of this table. The four cooperation-norm outcomes range from 1 (strongly disagree) to 5 (strongly agree). Non-substantive responses are coded missing. Each estimate is relative to no leader in the same composition-by-disclosure environment. Meeting 1 is elicited immediately after the first accountability meeting, Meeting 2 at the second meeting, and Endline after the recognition period. Separate regressions for each survey control for the same outcome's baseline response, gender, and assignment wealth stratum and absorb neighborhood × disclosure × composition fixed effects. The pooled regressions combine the three surveys, include survey-wave × neighborhood × disclosure × composition fixed effects, and allow the baseline-response, gender, and wealth-stratum slopes to vary by wave. Standard errors in parentheses are clustered by group-neighborhood using the main paper's finite-sample correction. Family p-values apply the Bonferroni minimum-p test across the four displayed outcomes within each leader-environment and survey wave. Depending on item availability and don't-know responses, observations range from 983 at Meeting 1, 943–946 at Meeting 2, 965 at Endline, and 2891–2894 in pooled regressions. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (unadjusted).

Table E-1 – LEADER EFFECTS ON COOPERATION NORMS AND DIVINE MONITORING BELIEFS (continued)

Panel C. No-leader counterfactual means for the evolution in Panel A

	Homogeneous, with disclosure				Mixed, without disclosure			
Outcome	Meeting 1	Meeting 2	Endline	Pooled	Meeting 1	Meeting 2	Endline	Pooled
Well regarded	4.45	4.48	4.45	4.46	4.43	4.42	4.41	4.42
	(0.06)	(0.06)	(0.07)	(0.05)	(0.06)	(0.06)	(0.07)	(0.04)
Moral obligation	4.59	4.52	4.46	4.52	4.41	4.51	4.56	4.49
	(0.05)	(0.05)	(0.07)	(0.05)	(0.06)	(0.06)	(0.07)	(0.05)
God rewards	4.64	4.55	4.63	4.61	4.57	4.55	4.57	4.56
	(0.04)	(0.06)	(0.05)	(0.04)	(0.04)	(0.05)	(0.06)	(0.04)
God punishes	4.14	4.05	4.22	4.14	4.19	4.29	4.30	4.26
	(0.11)	(0.10)	(0.07)	(0.07)	(0.08)	(0.09)	(0.08)	(0.06)

Panel D. Pooled no-leader counterfactual means for Panel B

Environment	Well regarded	Moral obligation	God rewards	God punishes
Mixed, without disclosure	4.42	4.49	4.56	4.26
	(0.04)	(0.05)	(0.04)	(0.06)
Mixed, with disclosure	4.39	4.41	4.58	4.05
	(0.06)	(0.05)	(0.04)	(0.08)
Homogeneous, without disclosure	4.52	4.35	4.54	4.13
	(0.04)	(0.05)	(0.03)	(0.05)
Homogeneous, with disclosure	4.46	4.52	4.61	4.14
	(0.05)	(0.05)	(0.04)	(0.07)

Notes: The means use the same outcomes, estimation samples, controls, and fixed effects as the corresponding treatment estimates on the preceding page. They average model predictions over no-leader observations in each displayed environment and survey window, retaining those observations' covariates. The pooled means use the stacked survey sample. Both leader comparisons in Panel B use the same environment-specific reference. Parentheses report group-clustered standard errors.

Table E-2 – LEADER EFFECTS ON IDENTITY SALIENCE

Panel A. Evolution in the two environments where leaders increased participation

	Traditional leader: homogeneous, with disclosure				Religious leader: mixed, without disclosure			
Identity rank	Meeting 1	Meeting 2	Endline	Pooled	Meeting 1	Meeting 2	Endline	Pooled
Congolese	0.45*	0.20	0.05	0.23	-0.09	-0.26	0.06	-0.10
	(0.25)	(0.23)	(0.18)	(0.14)	(0.19)	(0.23)	(0.22)	(0.18)
Kasaien	0.09	-0.20	0.02	-0.03	0.06	0.18	0.21	0.15
	(0.23)	(0.20)	(0.21)	(0.14)	(0.19)	(0.24)	(0.20)	(0.15)
Christian	0.02	0.44**	-0.05	0.13	0.28	0.28	0.59**	0.38*
	(0.24)	(0.22)	(0.24)	(0.18)	(0.26)	(0.26)	(0.26)	(0.20)
Own territory	0.26	-0.05	0.45***	0.23*	-0.12	-0.15	-0.25	-0.17
	(0.18)	(0.20)	(0.17)	(0.13)	(0.23)	(0.22)	(0.21)	(0.16)
Own denomination	-0.31	-0.26	-0.14	-0.23	-0.27	-0.30	-0.20	-0.26*
	(0.24)	(0.24)	(0.22)	(0.19)	(0.21)	(0.20)	(0.18)	(0.14)
Own tribe	-0.49*	-0.12	-0.34	-0.32*	0.13	0.25	-0.42*	-0.01
	(0.26)	(0.23)	(0.21)	(0.18)	(0.19)	(0.21)	(0.23)	(0.16)
Family p-value	0.354	0.274	0.045	0.487	1.000	0.870	0.140	0.331

Panel B. Pooled effects in every leader environment

	Congolese	Kasaien	Christian	Own territory	Own denomination	Own tribe	Family p-value
Traditional leader							
Mixed, without disclosure	0.00	0.14	-0.11	-0.21	0.11	0.07	0.872
	(0.16)	(0.15)	(0.16)	(0.14)	(0.15)	(0.15)	
Mixed, with disclosure	0.17	0.20	-0.19	-0.02	-0.31**	0.16	0.258
	(0.12)	(0.14)	(0.15)	(0.16)	(0.15)	(0.13)	
Homogeneous, without disclosure	0.11	-0.05	0.07	-0.03	-0.19	0.13	1.000
	(0.16)	(0.11)	(0.18)	(0.12)	(0.15)	(0.13)	
Homogeneous, with disclosure	0.23	-0.03	0.13	0.23*	-0.23	-0.32*	0.487
	(0.14)	(0.14)	(0.18)	(0.13)	(0.19)	(0.18)	
Religious leader							
Mixed, without disclosure	-0.10	0.15	0.38*	-0.17	-0.26*	-0.01	0.331
	(0.18)	(0.15)	(0.20)	(0.16)	(0.14)	(0.16)	
Mixed, with disclosure	-0.05	-0.11	-0.02	0.18	0.02	-0.03	1.000
	(0.13)	(0.15)	(0.15)	(0.14)	(0.16)	(0.14)	
Homogeneous, without disclosure	0.06	0.04	0.15	-0.21*	-0.33**	0.25**	0.208
	(0.14)	(0.12)	(0.17)	(0.12)	(0.15)	(0.12)	
Homogeneous, with disclosure	0.07	0.10	0.30	0.06	-0.19	-0.34**	0.242
	(0.15)	(0.14)	(0.20)	(0.14)	(0.17)	(0.17)	

Notes: Counterfactual means for both panels appear on the continuation page of this table. Outcomes are reverse-coded salience ranks from a forced ranking of six identities; higher values indicate greater salience. Each estimate is relative to no leader in the same composition-by-disclosure environment. Meeting 1 is elicited immediately after the first accountability meeting, Meeting 2 at the second meeting, and Endline after the recognition period. Separate regressions for each survey control for the same identity's baseline rank, gender, and assignment wealth stratum and absorb neighborhood × disclosure × composition fixed effects. The pooled regressions combine the three surveys, include survey-wave × neighborhood × disclosure × composition fixed effects, and allow the baseline-rank, gender, and wealth-stratum slopes to vary by wave. Standard errors in parentheses are clustered by group-neighborhood using the main paper's finite-sample correction. Family p-values apply the Bonferroni minimum-p test across the six ranks within the displayed leader-environment and survey wave; the ranks sum mechanically to 21, so no aggregate identity index is reported. Observations are 983 at Meeting 1, 947 at Meeting 2, 961 at Endline, and 2,891 in the pooled regressions. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (unadjusted).

Table E-2 – LEADER EFFECTS ON IDENTITY SALIENCE (continued)

Panel C. No-leader counterfactual means for the evolution in Panel A

	Homogeneous, with disclosure				Mixed, without disclosure			
Outcome	Meeting 1	Meeting 2	Endline	Pooled	Meeting 1	Meeting 2	Endline	Pooled
Congolese	3.80	4.25	4.21	4.09	4.27	4.38	4.35	4.33
	(0.17)	(0.15)	(0.13)	(0.10)	(0.14)	(0.15)	(0.14)	(0.12)
Kasaien	3.25	3.89	3.89	3.68	3.53	3.65	3.93	3.70
	(0.13)	(0.13)	(0.13)	(0.09)	(0.13)	(0.13)	(0.14)	(0.11)
Christian	4.04	3.58	3.66	3.75	3.74	3.71	3.28	3.58
	(0.17)	(0.16)	(0.16)	(0.14)	(0.16)	(0.18)	(0.17)	(0.12)
Own territory	3.19	3.18	2.90	3.09	3.18	3.50	3.57	3.41
	(0.11)	(0.12)	(0.12)	(0.08)	(0.14)	(0.15)	(0.14)	(0.11)
Own denomination	3.08	2.95	2.93	2.99	3.12	2.84	2.64	2.87
	(0.17)	(0.17)	(0.14)	(0.13)	(0.16)	(0.14)	(0.12)	(0.10)
Own tribe	3.64	3.15	3.40	3.40	3.15	2.92	3.23	3.10
	(0.18)	(0.14)	(0.16)	(0.13)	(0.13)	(0.12)	(0.15)	(0.10)

Panel D. Pooled no-leader counterfactual means for Panel B

Environment	Congolese	Kasaien	Christian	Own territory	Own denomination	Own tribe
Mixed, without disclosure	4.33	3.70	3.58	3.41	2.87	3.10
	(0.12)	(0.11)	(0.12)	(0.11)	(0.10)	(0.10)
Mixed, with disclosure	4.23	3.71	3.90	3.25	2.82	3.10
	(0.08)	(0.09)	(0.11)	(0.10)	(0.11)	(0.09)
Homogeneous, without disclosure	4.29	3.78	3.76	3.47	2.86	2.83
	(0.09)	(0.07)	(0.13)	(0.07)	(0.11)	(0.09)
Homogeneous, with disclosure	4.09	3.68	3.75	3.09	2.99	3.40
	(0.10)	(0.09)	(0.14)	(0.08)	(0.13)	(0.13)

Notes: The means use the same outcomes, estimation samples, controls, and fixed effects as the corresponding treatment estimates on the preceding page. They average model predictions over no-leader observations in each displayed environment and survey window, retaining those observations' covariates. The pooled means use the stacked survey sample. Both leader comparisons in Panel B use the same environment-specific reference. Parentheses report group-clustered standard errors.

Table E-3 – GOVERNANCE AND INSTITUTIONAL OUTCOMES: TESTS BY OUTCOME FAMILY

		Smallest adjusted q in the two mobilizing environments		Rejections at $q < 0.10$	
Outcome family	Outcomes	Chief: homogeneous, disclosure	Pastor: mixed, no disclosure	All leader effects	All contrasts
Government service performance	5	0.858	0.858	0/60	0/40
Government accountability and knowledge	4	0.849	0.608	0/48	0/32
Performance: public and community authorities	4	0.901	0.901	0/48	0/32
Performance: other authorities	4	0.638	0.638	0/48	0/32
Expected assistance from the locality chief	3	0.812	0.686	0/36	0/24
Local chief service performance	5	0.561	0.999	0/60	0/40
Local chief accountability and capture	2	0.664	0.664	0/24	0/16
Trust in public and local authorities	5	0.304	0.304	0/60	0/40
Trust in religious and civil-society actors	4	0.641	0.641	0/48	0/32
All governance families	36	0.304	0.304	0/432	0/288

Notes: We summarize effects on 36 outcomes measured at baseline and endline, grouped into nine families. Each ANCOVA controls for the corresponding baseline outcome, gender, and wealth stratum and absorbs neighborhood × disclosure × composition fixed effects; standard errors are clustered at the randomized group–neighborhood level using the main-paper finite-sample correction. The table reports the smallest Benjamini–Hochberg adjusted q-value within each mobilizing treatment cell and outcome family. Adjustment is performed separately within each thematic outcome family for all leader effects and all treatment-effect contrasts; we keep the nine families separate and adjust effects and contrasts separately. The last two columns report adjusted rejections divided by tests across all leader effects and all treatment-effect contrasts, respectively. We exclude predetermined ties to chiefs, complementary allocations in the diversion questions, and provider categories that have no natural ordering. Appeal frequency is recoded monotonically, institutional-performance items are oriented so higher values indicate better performance, and refusals on name-knowledge items are treated as missing. None of the 720 tests remains significant after adjustment at $q < 0.10$.

E.2 Relationships with Peers and Leaders

Table E-4 – ENDPOINT CHANGES IN OPINIONS ABOUT ORDINARY PEERS

	Inspires respondent	Moral example	Enforcement authority	Trusts responsibility	Would help target	Expected help from target
Panel A. Leader effects by composition						
Traditional leader						
Mixed	0.04	0.11	0.09	0.14	0.12	0.02
	(0.09)	(0.09)	(0.08)	(0.10)	(0.09)	(0.09)
Homogeneous	0.14	0.08	0.04	-0.09	0.00	0.11
	(0.08)	(0.08)	(0.08)	(0.09)	(0.07)	(0.09)
Religious leader						
Mixed	0.20*	0.05	0.08	0.12	0.24**	0.05
	(0.09)	(0.08)	(0.08)	(0.09)	(0.09)	(0.10)
Homogeneous	0.05	0.08	0.09	-0.07	0.05	-0.00
	(0.09)	(0.08)	(0.08)	(0.10)	(0.07)	(0.08)
Panel B. Environment-specific leader effects						
Traditional leader						
Mixed, without disclosure	0.14	0.27	0.14	0.08	0.18	0.13
	(0.12)	(0.12)	(0.10)	(0.14)	(0.14)	(0.12)
Mixed, with disclosure	-0.07	-0.06	0.04	0.20	0.04	-0.10
	(0.12)	(0.12)	(0.14)	(0.14)	(0.12)	(0.13)
Homogeneous, without disclosure	0.10	0.05	0.07	-0.16	-0.21	-0.11
	(0.12)	(0.11)	(0.12)	(0.13)	(0.10)	(0.12)
Homogeneous, with disclosure	0.17	0.10	0.01	-0.02	0.21**	0.32**
	(0.11)	(0.11)	(0.10)	(0.12)	(0.08)	(0.13)
Religious leader						
Mixed, without disclosure	0.25	0.06	0.16	0.20	0.43**	0.15
	(0.12)	(0.10)	(0.10)	(0.13)	(0.15)	(0.14)
Mixed, with disclosure	0.15	0.04	0.01	0.03	0.06	-0.04
	(0.13)	(0.13)	(0.14)	(0.13)	(0.09)	(0.14)
Homogeneous, without disclosure	0.13	0.11	0.12	-0.11	-0.02	-0.13
	(0.12)	(0.11)	(0.11)	(0.14)	(0.10)	(0.12)
Homogeneous, with disclosure	-0.04	0.05	0.06	-0.03	0.11	0.13
	(0.12)	(0.11)	(0.12)	(0.14)	(0.10)	(0.11)
Panel C. No-leader counterfactual means						
Mixed, pooled across disclosure	0.21	0.21	0.01	0.00	-0.14	-0.08
	(0.06)	(0.05)	(0.05)	(0.05)	(0.07)	(0.06)
Homogeneous, pooled across disclosure	0.11	0.13	-0.04	-0.02	-0.09	-0.08
	(0.05)	(0.04)	(0.05)	(0.06)	(0.04)	(0.05)
Mixed, without disclosure	0.12	0.13	0.03	-0.06	-0.21	-0.13
	(0.07)	(0.06)	(0.06)	(0.08)	(0.11)	(0.09)
Mixed, with disclosure	0.29	0.29	-0.02	0.07	-0.06	-0.02
	(0.08)	(0.07)	(0.07)	(0.07)	(0.07)	(0.09)
Homogeneous, without disclosure	0.08	0.12	-0.07	0.03	-0.01	-0.00
	(0.07)	(0.06)	(0.07)	(0.08)	(0.06)	(0.07)
Homogeneous, with disclosure	0.15	0.15	-0.01	-0.07	-0.18	-0.17
	(0.07)	(0.06)	(0.07)	(0.08)	(0.06)	(0.08)
Respondent–target pairs	2,632	2,631	2,633	2,633	2,633	2,633
Ordinary-target pairs	2,071	2,071	2,072	2,072	2,072	2,072
Assigned-leader-target pairs	561	560	561	561	561	561
Respondents	885	885	885	885	885	885
Group-neighborhood clusters	323	323	323	323	323	323
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes	Yes	Yes	Yes
Baseline outcome control	No	No	No	No	No	No

Notes: This table shows how residents' opinions of ordinary teammates changed between the start of Meeting 1, before any discussion, and the start of Meeting 2. Each outcome is the later answer minus the earlier answer for the same respondent and teammate, expressed in baseline standard deviations. Both help amounts are inverse-hyperbolic-sine transformed before calculating changes and standardized by their baseline standard deviation. To illustrate magnitudes in CDF in the main text, we apply the inverse transformation at the baseline mean. These conversions do not estimate the average treatment effect in CDF. Panel A pools the disclosure conditions and compares mixed with homogeneous groups. Panel B estimates all four environments separately. Figure 9 instead pools the other three environments separately for each leader type; the resulting coefficients therefore differ. Every effect is relative to opinions of ordinary teammates in no-leader groups under the same experimental conditions. Panel C shows predicted changes for ordinary teammates in no-leader groups, in baseline-standard-deviation units. We average predictions over matched respondent–teammate pairs in each composition or environment, using the same reference for ordinary-peer and assigned-leader effects. The companion Table E-5 uses the same model to report changes in opinions of assigned leaders. Standard errors in parentheses are clustered by randomized group. Willingness-to-help estimates include the 25-meter spillover indicator; outcome-specific LASSO selected no additional baseline controls. The table is discussed in Section 6; Figure 9 highlights the two help outcomes. Stars use Benjamini–Hochberg adjusted p-values across the six ordinary-peer outcomes for each treatment comparison: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E-5 – ENDPOINT CHANGES IN OPINIONS ABOUT ASSIGNED LEADERS

	Inspires respondent	Moral example	Enforcement authority	Trusts responsibility	Would help target	Expected help from target
Panel A. Leader effects by composition						
Traditional leader						
Mixed	0.05	0.02	-0.02	0.19	0.08	0.07
	(0.11)	(0.10)	(0.08)	(0.10)	(0.12)	(0.10)
Homogeneous	0.03	-0.02	-0.04	0.12	0.16	0.07
	(0.08)	(0.08)	(0.08)	(0.10)	(0.08)	(0.08)
Religious leader						
Mixed	0.07	0.08	0.09	0.23*	0.32***	0.17
	(0.09)	(0.08)	(0.09)	(0.10)	(0.10)	(0.10)
Homogeneous	0.11	0.00	0.08	0.16	0.04	0.02
	(0.09)	(0.08)	(0.09)	(0.11)	(0.06)	(0.08)
Panel B. Environment-specific leader effects						
Traditional leader						
Mixed, without disclosure	0.18	0.09	0.03	0.32	0.28	0.12
	(0.15)	(0.16)	(0.12)	(0.15)	(0.16)	(0.14)
Mixed, with disclosure	-0.09	-0.05	-0.09	0.06	-0.14	0.01
	(0.14)	(0.13)	(0.10)	(0.13)	(0.17)	(0.14)
Homogeneous, without disclosure	0.15	-0.04	0.13	0.14	0.18	-0.02
	(0.11)	(0.11)	(0.11)	(0.14)	(0.14)	(0.11)
Homogeneous, with disclosure	-0.08	-0.01	-0.19	0.11	0.15	0.16
	(0.11)	(0.11)	(0.12)	(0.15)	(0.10)	(0.13)
Religious leader						
Mixed, without disclosure	0.15	0.10	0.14	0.22	0.42**	0.21
	(0.11)	(0.09)	(0.13)	(0.13)	(0.16)	(0.15)
Mixed, with disclosure	-0.01	0.06	0.03	0.24	0.22	0.14
	(0.15)	(0.13)	(0.13)	(0.16)	(0.12)	(0.13)
Homogeneous, without disclosure	0.16	0.05	0.14	0.03	-0.06	-0.10
	(0.12)	(0.10)	(0.14)	(0.15)	(0.10)	(0.10)
Homogeneous, with disclosure	0.06	-0.04	0.01	0.29	0.15	0.14
	(0.13)	(0.12)	(0.13)	(0.15)	(0.07)	(0.14)
Panel C. No-leader counterfactual means						
Mixed, pooled across disclosure	0.21	0.21	0.01	0.00	-0.14	-0.08
	(0.06)	(0.05)	(0.05)	(0.05)	(0.07)	(0.06)
Homogeneous, pooled across disclosure	0.11	0.13	-0.04	-0.02	-0.09	-0.08
	(0.05)	(0.04)	(0.05)	(0.06)	(0.04)	(0.05)
Mixed, without disclosure	0.12	0.13	0.03	-0.06	-0.21	-0.13
	(0.07)	(0.06)	(0.06)	(0.08)	(0.11)	(0.09)
Mixed, with disclosure	0.29	0.29	-0.02	0.07	-0.06	-0.02
	(0.08)	(0.07)	(0.07)	(0.07)	(0.07)	(0.09)
Homogeneous, without disclosure	0.08	0.12	-0.07	0.03	-0.01	-0.00
	(0.07)	(0.06)	(0.07)	(0.08)	(0.06)	(0.07)
Homogeneous, with disclosure	0.15	0.15	-0.01	-0.07	-0.18	-0.17
	(0.07)	(0.06)	(0.07)	(0.08)	(0.06)	(0.08)
Respondent–target pairs	2,632	2,631	2,633	2,633	2,633	2,633
Ordinary-target pairs	2,071	2,071	2,072	2,072	2,072	2,072
Assigned-leader-target pairs	561	560	561	561	561	561
Respondents	885	885	885	885	885	885
Group-neighborhood clusters	323	323	323	323	323	323
Neighborhood × disclosure × composition FEs	Yes	Yes	Yes	Yes	Yes	Yes
Baseline outcome control	No	No	No	No	No	No

Notes: This table shows how residents' opinions of their assigned chief or pastor changed between the start of Meeting 1, before any discussion, and the start of Meeting 2. Each outcome is the later answer minus the earlier answer for the same respondent and leader, expressed in baseline standard deviations. Both help amounts are inverse-hyperbolic-sine transformed before calculating changes and standardized by their baseline standard deviation. To illustrate magnitudes in CDF in the main text, we apply the inverse transformation at the baseline mean. These conversions do not estimate the average treatment effect in CDF. Panel A pools the disclosure conditions and compares mixed with homogeneous groups. Panel B estimates all four environments separately. Figure 9 instead pools the other three environments separately for each leader type; the resulting coefficients therefore differ. Every effect is measured relative to opinions of ordinary teammates in no-leader groups under the same experimental conditions. Panel C shows predicted changes for ordinary teammates in no-leader groups, in baseline-standard-deviation units. We average predictions over matched respondent–teammate pairs in each composition or environment, using the same reference for ordinary-peer and assigned-leader effects. The companion Table E-4 uses the same model to report changes in opinions of ordinary teammates. Standard errors in parentheses are clustered by randomized group. Willingness-to-help estimates include the 25-meter spillover indicator; outcome-specific LASSO selected no additional baseline controls. The table is discussed in Section 6; Figure 9 highlights the two help outcomes. Stars use Benjamini–Hochberg adjusted p-values across the six assigned-leader outcomes for each treatment comparison: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.